

THINKING WITH CASES:

An Introduction to Bioethics



Brendan Shea, PhD

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Brendan Shea, PhD

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This book is a work of human–artificial intelligence collaboration; the introduction explains how it was made and who is accountable for it.

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About This Book

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Thinking with Cases: An Introduction to Bioethics

Brendan Shea, PhD

Real medical dilemmas, taken apart — so you can decide what you think.

This book is an open introduction to bioethics built around true stories of medicine at its hardest — moments where thoughtful people, acting in good faith, reached the edge of what their training could settle and had to decide anyway. The opening chapter equips a reader coming to the field fresh with the essential toolkit: the four principles of biomedical ethics, the moral theories that sit behind them, the short and brutal history that produced our rules of consent and research, and the simple skill of taking an argument apart to find the premise in dispute. The chapters that follow put those tools to work, one case at a time. Each gives every side its strongest voice and ends not with a verdict but with an open question — because the aim is not to tell you what to think, but to make you a better thinker.

Begin with **the introduction**, which explains the ideas and how to use the cases. After that, the cases can be read in any order.

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This book is a work of human–artificial intelligence collaboration; the introduction explains how it was made and who is accountable for it.

The collection is meant to grow as medicine raises new dilemmas. This edition was generated from the source materials on the date shown.

Thinking with Cases: An Introduction to Bioethics

How to read the case studies in this book

Starting in the Middle

It is the middle of the night, and a man is dying. He has end-stage cancer and a day or two left, and the pain has broken through the dose he is on. His doctor has left a standing order for morphine — not a single amount, but a *range*, to be given as the nurse judges the patient needs it. She can give a little or a lot. A larger dose would finally settle his pain. It might also slow his breathing enough to bring the end sooner, tonight instead of tomorrow.

His daughter is at the bedside, holding his hand, and she looks up at the nurse and says she cannot bear to watch him suffer anymore. The nurse draws morphine into the syringe and pauses, her thumb on the plunger. How much should she give?

No rule in the binder answers this. The order permits the whole range and leaves the choice to her. Her training tells her what each dose will do to his breathing; it does not tell her which one she ought to push. If her aim is to ease his pain and the shortened life is a side effect she foresees but does not intend, most ethicists — and most laws — say she has done nothing wrong. But she is the one holding the syringe, and the line between relieving death's pain and hastening it runs straight through her hands.

What should she do?

Notice what the question is *not*. It is not a question about pharmacology — she knows exactly what each dose does. It is not a question about policy — the order and the law permit every choice in front of her. It is a question about what is *right*, and it cannot be answered by knowing more facts alone. Two nurses could agree on every detail of the man's condition and still disagree about the syringe. That gap, between the medical facts and the right action, is where bioethics lives.

This book is a collection of such moments. Each chapter tells a true story in which thoughtful people, acting in good faith, reached the edge of what their training could settle and had to decide anyway. The stories are not here to be solved and filed away. They are here to be *thought with*. By the end of each one you should not have “the answer.” You should have a sharpened, more honest version of the question — and a clearer sense of what you yourself believe and why.

That is a strange promise for a book to make, so this opening chapter explains the tools. It introduces the core ideas of bioethics for a reader coming to the field fresh: what the discipline is, the four principles that organize most of its arguments, the ethical theories that sit behind those principles, and the basic skill of taking an argument apart to see how it works. It then explains how the cases in this book are built, and how they were written. None of this requires a background in medicine or philosophy. It requires only the willingness to sit with a hard question a little longer than feels comfortable.

What Bioethics Is

Bioethics is the study of the ethical questions raised by medicine, health care, and the life sciences. It asks what we owe one another when bodies, minds, and lives are at stake: when to treat and when to stop, who decides, what may be researched and on whom, how scarce help should be shared, and where the limits of medicine's power should fall.

The field is young. It took its modern shape only in the second half of the twentieth century, when three forces collided. Medicine grew suddenly powerful — ventilators, dialysis, organ transplants, intensive care — and could now keep bodies alive in conditions no one had faced before. A series of research scandals came to light, in which doctors had treated patients as material rather than as people. And the old, unquestioned authority of the physician began to give way to a new insistence that patients have a say in what happens to them. Bioethics emerged to think clearly about problems that were genuinely new.

It helps, early on, to mark off what ethics is *not*, because three reflexes tend to short-circuit the conversation.

- **Ethics is not the same as law.** Laws can be unjust, and many things that are legal are still wrong (and a few illegal things may be right). “It’s allowed” and “it’s permitted here” do not answer the moral question; they change the subject.
- **Ethics is not the same as religion.** Religious traditions offer some of the deepest moral thinking we have, but a bioethical argument has to be one that people of different faiths, or none, can follow and weigh. The reasons have to be shareable.
- **Ethics is not the same as personal preference.** “I just feel it’s wrong” reports a fact about you, not a reason others can examine. Ethics deals in reasons that can be stated, challenged, and defended.

It also helps to know that ethical questions come in different kinds. Some are **descriptive**: what do people actually believe, and how do they in fact behave? Those are questions for psychology and sociology. Some are **metaethical**: what *is* goodness, and can moral claims even be true? Those are questions for philosophers. This book lives almost entirely in a third kind, **normative** ethics, which asks the practical question the nurse at the bedside is facing: *what should be done here, and why?*

The Four Principles

If bioethics has a common language, it is the language of four principles. In the 1970s the philosophers Tom Beauchamp and James Childress proposed that most moral disputes in medicine could be framed using four basic commitments that nearly everyone already shares, whatever their religion or politics (Beauchamp and Childress 2019). Their book made these four principles the working vocabulary of the field.

- **Autonomy** — respect for a person's right to make decisions about their own life and body. This is the principle behind informed consent, and behind the patient's right to say no.
- **Beneficence** — the duty to act for the good of others, to help, to promote well-being. It is the oldest impulse in medicine: do good.
- **Non-maleficence** — the duty to avoid causing harm. *First, do no harm.* It often pulls hardest exactly when beneficence is most tempting.
- **Justice** — the duty to treat people fairly and to share benefits and burdens equitably. It governs who gets the scarce organ, the ICU bed, the place in the trial.

Beauchamp and Childress argued that these four are part of what they called the **common morality** — the set of basic moral commitments shared across cultures and traditions. You do not have to be persuaded of a grand theory to accept them; you already live by them. Critics have sometimes called this “fourth-grade morality,” as if it were too simple to be serious. The reply is that the four principles are simple to *state* and very hard to *apply* — and that their simplicity is a strength, because it gives people who disagree about everything else a place to begin.

The principles do real work precisely because they **conflict**. The nurse with the syringe feels beneficence (the relief the morphine would bring) pulling against non-maleficence (the breathing it might suppress). A competent patient who refuses a life-saving transfusion sets autonomy against beneficence. A transplant committee feels one patient's need against justice to all the others. In an easy case the principles agree, and there is no dilemma. In a hard case — the only kind worth a chapter — they point in different directions, and something has to give.

This is why bioethics is not a matter of memorizing rules. There is no fixed ranking that puts autonomy permanently above beneficence, or justice above both. The work is in the application: deciding what each principle *demand*s in this particular situation (philosophers call this **specification**) and how much weight each should carry when they collide (**balancing**).

Consider a brief example. An elderly woman with early dementia still wants to live alone, though her doctors think she is no longer safe there. Autonomy says respect her wish; non-maleficence warns of the fall waiting to happen; beneficence wants her cared for. There is no formula that solves this. The careful move is to *specify*: how much capacity does she still have for this particular choice, and what is the real size of the risk? And then to *balance*: a small risk freely chosen may weigh less than the dignity of her own home, while a grave and certain danger may tip the other way. Two thoughtful people can run that process honestly and still land in different places. That disagreement is not a failure of the method. It is the honest shape of a hard problem.

Four Lenses for a Hard Case

Behind the four principles sit older and more ambitious ideas — the moral theories that have occupied philosophers for centuries. You do not need to master them to use this book. But it helps to know the main ones, because each works like a *lens*: a way of looking at a case that brings certain features into focus and pushes others to the edge. Faced with the same story, the lenses often ask different questions and sometimes reach different answers.

Consequentialism judges an action by its results. The most famous version, **utilitarianism**, says the right act is the one that produces the most good — the greatest well-being, the least suffering — counting everyone equally. Jeremy Bentham and John Stuart Mill gave the view its classic form: the question is not who you are or what rule you followed, but how much better or worse the world becomes (Mill 1863; Bentham 1789). A modern utilitarian, Peter Singer, has followed the logic to conclusions many find startling, arguing that the suffering of animals counts as fully as our own and that the affluent are obligated to give far more than they do (Singer 1975). Utilitarianism's strength is its clarity and its refusal to play favorites. Its danger is that it can, in principle, justify sacrificing an individual for the greater good — which is exactly what some of the worst cases in this book involve.

Kantian ethics, named for Immanuel Kant, judges actions by duty rather than results. Its central command is to treat every person as an *end in themselves*, never merely as a means to someone else's purpose (Kant 1785). Some things may not be done to a person even if doing them would help many others, because persons are not tools. This is the deepest objection to using patients as unwitting research material, and it underwrites the modern insistence on consent. Where the utilitarian asks "what produces the most good?", the Kantian asks "what do we owe this person, regardless of the good?"

Virtue ethics, descending from Aristotle, shifts the question from the act to the *agent*. Instead of "what should I do?" it asks "what would a good person — honest, compassionate, courageous, just — do here, and what kind of person do I become by choosing this?" (Aristotle 340 BC). It captures something the other theories can miss: that character and habit, not just isolated decisions, are the substance of a moral life.

The ethics of care, developed by Carol Gilligan and others, starts from relationships rather than rules (Gilligan 1982). It notices that real moral life is lived among people who are dependent on one another — patient and nurse, parent and child — and that abstract principles can obscure the concrete obligations of those bonds. It is a natural lens for medicine, where so much of the work is relationship.

Put the lenses to work on a single case and the difference shows at once. Suppose a hospital has one ICU bed and two patients who need it. The utilitarian asks which choice saves more life and relieves more suffering overall. The Kantian asks whether either patient is being treated as a mere means — whether each is given the respect owed to a person, not weighed like cargo. The virtue ethicist asks what a just and compassionate clinician would do, and how she should carry the loss of whoever is not chosen. The care ethicist asks about the relationships and dependencies in the room — who is already in this clinician's care, and what that bond demands. None of these questions is foolish, and they do not always agree.

These theories, then, are not rival machines that stamp out answers. They are ways of seeing, and a skilled reader learns to switch among them. Often the most useful question about a case is not “which theory is correct?” but “what does each lens reveal that the others hide?” Each one, in the end, can be read as a different account of *why* the four principles matter.

How to Read a Moral Argument

The cases in this book are built around arguments, and the single most useful skill for reading them is also the simplest: learning to take an argument apart.

An **argument**, in this sense, is not a quarrel. It is a set of reasons — called **premises** — offered in support of a conclusion. Laid out plainly, an argument looks like this:

1. A competent adult has the right to refuse any medical treatment.
2. This patient is a competent adult who refuses the treatment.
3. Therefore, his refusal should be honored.

Setting an argument out in this **standard form** — premises first, conclusion last — is powerful because it makes disagreement *locatable*. Once the steps are visible, you can ask two distinct questions. First: does the conclusion actually follow from the premises, if the premises are true? (This is **validity**, a matter of logic.) Second: are the premises in fact true? (A valid argument with true premises is called **sound**.) An argument can be perfectly logical and still go wrong because one of its premises is false.

This gives the reader a precise tool. When two thoughtful people disagree about a case, they rarely disagree about *everything*. Usually they accept most of the steps and part ways over exactly one — a single premise that one believes and the other denies. Find that premise and you have found the real dispute. In the refusal argument above, almost everyone grants the first and third steps; the whole question turns on the second, on whether the patient is truly *competent* to refuse. That is where the argument should be pressed.

Throughout this book you will see arguments set out in this numbered form, in boxes, with the contested premise named. The point is never to win. It is to see, with as little fog as possible, where reasonable people actually divide — and to make you state which premise *you* would defend.

How the Rules Were Written

Many of the safeguards we now take for granted — that a patient must consent, that research must be reviewed, that subjects can walk away — are recent, and most were written in response to disaster. The history is worth knowing, because the cases in this book are the latest chapters in a long argument, not the first.

For most of medicine’s history, the doctor decided and the patient complied. The physician’s duty was beneficence, and consent as we understand it barely existed. The change began, brutally, at

Nuremberg. After the Second World War, the trial of Nazi doctors who had performed lethal experiments on prisoners produced the **Nuremberg Code** of 1947, the first clear statement that the *voluntary consent* of the research subject is essential ([Nuremberg Military Tribunals 1947](#)). The world medical community followed, eventually, with the **Declaration of Helsinki**, the international standard for ethical research that is still revised today ([World Medical Association 2024](#)).

The codes did not end the abuses. In the United States, the **Tuskegee Syphilis Study** ran for forty years, into the 1970s, withholding effective treatment from hundreds of Black men so that researchers could observe the untreated course of the disease — without their informed consent, and after a cure existed. When it came to light, the outrage led to the **Belmont Report** of 1979, which distilled the ethics of human research into three principles that still govern it: **respect for persons, beneficence, and justice** ([National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979](#)). Belmont, in turn, gave us the Institutional Review Boards that must approve research with human subjects, and the modern apparatus of consent forms, risk assessment, and oversight ([Jonsen 1998](#)).

Other cases sharpened the same lessons. Henrietta Lacks, a Black woman who died of cancer in 1951, had cells taken without her knowledge that became one of the most important tools in the history of medicine — grown, sold, and studied for decades while her own family knew nothing and shared in none of it. Her story folds together every theme of this history at once: consent, justice, race, and the question of what we owe the people whose bodies make medical progress possible.

The same shift remade the ordinary clinic. The physician's authority gave way to a partnership in which the patient's values, not only the doctor's judgment, direct the care. One useful way to see the change comes from Ezekiel and Linda Emanuel, who described four models of the relationship between doctor and patient ([Emanuel and Emanuel 1992](#)). In the **paternalistic** model, the physician decides what is best and acts on it — the old default. In the **informative** model, the doctor supplies facts and the patient alone chooses, a kind of vending-machine medicine. In the **interpretive** model, the doctor helps the patient understand what they truly want. In the **deliberative** model, doctor and patient reason together toward the best choice. The history of modern medical ethics is, in large part, a movement away from the first model — though which of the others should replace it is itself a live question, and several of the cases in this book turn on it.

At the center of that shift is **informed consent** — understood not as a signature on a form but as a *process* of disclosure and choice ([Faden and Beauchamp 1986](#)). For consent to be genuine, four things must hold: the patient must receive adequate **disclosure** of the relevant facts; must have real **understanding** of them; must choose **voluntarily**, free of coercion; and must have the **capacity** to decide ([Appelbaum 2007](#)). Capacity is more subtle than it looks — it is not about whether a patient agrees with the doctor, but about whether they can take in information, weigh it, and express a choice. A patient may have the capacity to refuse a flu shot but not to manage a complex cancer decision; capacity is specific to the choice at hand. Alongside consent stands **confidentiality**, the old promise that what a patient reveals will be protected — a promise now strained by databases, families, and public-health duties.

Return now to the nurse with the syringe. Her dilemma is not an accident of one bad shift. The same advances that gave medicine its new power over death — the drugs, the machines, the ability to hold dying open for hours or days — are what placed that choice in her hands at all; a century ago there would have been no range to titrate. And the rules built since Nuremberg, for all they reshaped, never resolved the oldest tension of the bedside: the duty to relieve suffering pulling against the duty never to be the cause of a patient's death. She stands inside both the new powers and the ancient conflict. The cases that follow are, again and again, places where commitments like these collide.

Why Reason from Cases?

It would be possible to write an ethics book made only of principles and theories, with cases tucked in as examples. This book does the reverse, and the choice is deliberate.

Reasoning from cases has a long and respectable history. In medieval and early-modern ethics it was called **casuistry** — the practice of working out what is right by comparing the case at hand to clear examples whose rightness or wrongness is not in doubt. Casuistry fell into disrepute when it was abused to rationalize whatever a clever arguer wanted. But in the 1980s the philosophers Albert Jonsen and Stephen Toulmin revived it for bioethics, arguing that committees facing real medical dilemmas reached agreement not by deducing answers from grand theories, but by reasoning carefully from cases (Jonsen 1998). People who disagreed sharply about *why* something was wrong could still agree *that* it was — and could reason outward from there.

This matches how moral understanding actually grows. We rarely learn ethics by memorizing a rule and applying it. We learn by encountering a hard situation, feeling the pull of competing goods, watching how others reason, and refining our judgment against new cases. A principle learned in the abstract is inert until you have used it on something that resists. The cases in this book are that resistance — chosen because each one strains the principles in a different direction, so that working through the whole set builds a flexible, tested judgment rather than a list of memorized verdicts.

The Anatomy of a Case

Each case in this book follows the same shape, so that once you learn to read one, you can read them all.

Every chapter opens with an **at-a-glance** card — who, what, where, and why it matters — so you know the stakes within fifteen seconds. Then comes **the story**, told as carefully and concretely as the facts allow, with real names, dates, and details, because ethics done in the abstract is ethics done badly. Technical terms are explained where they first appear; you are never expected to arrive already knowing them.

The heart of each chapter is **the argument the case started** — a connected exchange of positions, laid out as a sequence of moves. One side states its case; the strongest objection answers it; a

reply answers the objection; and so on, with the key arguments set out in standard form and the contested premise named at each step. The exchange is built to be *fair*: each position is given in its strongest version, in the voice of someone who actually holds it. A case in which one side is obviously right is not worth your time.

Each chapter ends with **discussion questions** and, deliberately, without a verdict. This is the book's one firm rule. The cases are designed to *open* your judgment, not to close it. The final move of each argument leaves a question still live — because that is the truth of these problems, and because the point is to make you a better thinker, not to tell you what to think. You can read these chapters alone, pausing to argue with the page, or in a group, where the disagreement is the whole point.

A Note on How This Book Was Made

This book is a work of human–artificial intelligence collaboration, and it would be strange — in a book about honesty, consent, and the proper use of powerful tools — not to say so plainly.

The ideas, the structure, the selection of cases, the arguments, and the final judgment about every sentence are mine. Artificial-intelligence tools were used as an aid in drafting and in synthesizing material, much as earlier writers used research assistants and editors. I directed that work, supplied the expertise that shaped it, checked the facts and the sources, and take full responsibility for what remains — including any errors. The technology can assemble and rephrase; it cannot decide what is true or what matters, and I have not asked it to.

I disclose this for two reasons. The first is simple honesty about how the work was produced. The second is that this book argues, repeatedly, that powerful tools in medicine and science demand transparency about how they are used and who remains accountable for the results. It would be a poor argument if the book did not hold itself to the same standard.

About the Author, and the Aim

I have taught bioethics to students for about sixteen years — most of them nursing and health-science students, along with others meeting these questions for the first time. I also serve on the Institutional Biosafety Committee at Mayo Clinic, where some of the abstractions in this book become concrete decisions about real research, and I have published on a number of the issues the cases raise. Outside of academic philosophy, I also spent around a decade working with developmentally disabled adults. These experiences, along with my training as a academic philosopher, have deeply impacted the ways in which I approach the subject.

Using and Sharing This Book

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The collection is also meant to grow. New cases will be added over time as medicine raises new dilemmas — and medicine always does. Whichever cases you find here, the method is the same, and so is the invitation: read the story, weigh the arguments, find the premise you would defend, and decide what you think. Then hold that conclusion loosely enough to keep thinking. That is what it means to think with cases.

Dax Cowart and the Right to Refuse Life-Saving Treatment: Autonomy, Beneficence, and Paternalism

“Please Let Me Die” — a bioethics case study

At a Glance

- **Who:** Donald Herbert “Dax” Cowart (1947–2019), a 25-year-old former Air Force pilot from Henderson, Texas.
- **What:** Burned over roughly two-thirds of his body in a 1973 propane explosion. He repeatedly, and competently, demanded that treatment stop, and was overruled for more than a year.
- **Where / When:** Parkland Memorial Hospital (Dallas), a rehabilitation hospital in Houston, and John Sealy Hospital at the University of Texas Medical Branch (Galveston), 1973–1974.
- **Why it matters:** Cowart’s case became the most-discussed American example of a clash between patient **autonomy** (a competent person’s right to refuse treatment) and clinical **beneficence** (the duty to act for the patient’s good). Even after a full recovery and a successful career, he insisted his refusal should have been honored. The argument he started has run for fifty years and is not over.
- **Pair with:** History & the Clinical Relationship

The Case

On the afternoon of July 25, 1973, Dax Cowart and his father, Ray, drove out to look at a piece of rural land in east Texas. Dax — then named Donald, a name he would later change — was twenty-five. He was a former high-school football captain, a rodeo rider, and an Air Force pilot recently back from service. By every account, including his own, he was a young man who took his strong, fast, capable body for granted.

They did not know that an underground propane pipeline near where they parked had been leaking. The gas had pooled, invisible and odorless, in the low ground around the car. When Ray turned the ignition to leave, a spark from the engine ignited it. The explosion threw Dax clear of a wall of fire. He ran through pine woods until he could not run anymore, beating at flames with hands that were already destroyed.

A farmer reached him first. Cowart's first recorded words, to that stranger, were a request for a weapon: "Get me a gun. Can't you see I'm a dead man. I'm going to die anyway." Both men were taken to a small hospital in Kilgore, then transferred together — 140 miles, in the same ambulance — to the burn unit at Parkland Memorial Hospital in Dallas. Ray Cowart died on the way. Dax survived. The damage was severe:

- Burns covered about 65% of his body.
- The deepest, third-degree burns destroyed the skin of his face, ears, and hands.
- He was blinded in both eyes.
- Most of his fingers were later amputated.
- He was conscious, and he stayed conscious.

What followed was fourteen months of treatment that Cowart would later describe, without exaggeration, as "pure hell." The medical necessity was real.

Background

Why the treatment was unavoidable. Burn depth is graded by layer:

- *First-degree* burns affect only the outer skin (an ordinary sunburn).
- *Second-degree* burns blister and are intensely painful.
- *Third-degree* (full-thickness) burns destroy all skin layers.

The amount burned is reported as percent of **total body surface area** (TBSA). At ~65% TBSA, much of it third-degree, Cowart was at the edge of what was survivable in 1973. Survival required **debridement** — the physical removal of dead tissue, because new skin cannot grow over it and dead tissue breeds lethal infection. This was often done in a *Hubbard tank*, a steel pool of warm antiseptic water that softened the dead skin so it could be scrubbed and cut away. The procedure works. Before modern pain control, it was also one of the most agonizing routine treatments in medicine.

From the start — at Parkland, and at every stage afterward — Cowart asked the people treating him to stop. Not in passing, and not only when the pain was worst. He asked his doctors, his nurses, and his mother, again and again, to end treatment and let him die. He understood what he was asking for. He understood that without the tankings and skin grafts he would die of infection within weeks. He wanted that outcome over the one being forced on him. He felt, he said, that he was "being skinned alive."

A psychiatrist, Dr. Robert B. White, was brought in to evaluate him at Galveston. The implied hope was that White would find Cowart too depressed or too overwhelmed to be making a real decision. That would have authorized treating him against his will with a clear conscience. White concluded the opposite. Cowart, he found, was "not in the least incompetent or mentally ill." It was during this period that White recorded the interview that became the 1974 film *Please Let Me Die* — a competent man, on camera, lucidly asking to be allowed to die, and being treated anyway (White 1974).

The most revealing turn in the story is one the standard retellings often skip. After about eight

months at Parkland, Cowart was moved to a rehabilitation hospital in Houston. There he again refused further treatment — and the rehabilitation physician *accepted the refusal*. For a moment, the system did what Cowart asked. Then his mother, Ada, his legal next of kin, and his attorney intervened. They had him transferred to the burn unit at John Sealy Hospital in Galveston, where treatment — and the tankings — resumed under the unit’s director, Dr. Duane Larson. The decisive fact of the case is not that no one would listen. Someone did. He was moved away from the person who listened.

He survived, was discharged blind and severely disabled, and was, by his own description, deeply depressed for years. He attempted suicide more than once. Then his life changed. He earned a law degree from Texas Tech University in 1986, married, practiced law, and became one of the most articulate patient-rights advocates in the country. He died on April 28, 2019, at seventy-one, of complications from leukemia and liver cancer.

For the rest of that life he said two things that many listeners could not hold together. He was glad to be alive. And his physicians had been wrong to keep him alive against his will. He did not think these statements contradicted each other. The good outcome, he insisted, was never the question. The question was *whose decision it had been* — and that question became one of the longest-running arguments in modern bioethics.

The Argument Dax Started

Cowart’s case became the test the modern right to refuse treatment had to pass. The principle was already stirring in 1970s bioethics — that patients, not physicians, should rule over their own bodies — but it had not yet met a patient who was fully competent, in agony, and demanding to be allowed to die from injuries he was likely to survive. For fifty years Cowart, his physicians, and a line of philosophers and lawyers fought over the one question his case posed without escape: may a competent person refuse life-saving care precisely because the life saved would be his to live?

The argument from autonomy

Cowart’s own position is the cleanest. He insisted the issue was control over his body, not the value of the result: “the right to control your own body is a right you’re born with, not something that you have to ask anyone else for, not the government, not your treating physician, not your next-of-kin” (Cowart and Burt 1998).

Argument

The Autonomy Argument

1. A competent adult has a moral right to refuse any medical treatment.
2. Dax Cowart was a competent adult who refused treatment.
3. Therefore, treating Cowart over his objection violated his moral rights.

Premise 1 is now close to settled doctrine. Courts and ethicists broadly accept that bodily self-determination includes the right to decline care. So the argument's whole weight falls on **premise 2** — and that is exactly where his physicians attacked it.

Background

Capacity, competence, and the surrogate. **Decision-making capacity** is a clinical judgment that a patient can do four things:

1. Understand the relevant information.
2. Appreciate how it applies to them.
3. Reason about the options.
4. Communicate a choice.

Capacity is *decision-specific* and *time-specific*. "Competence" is the parallel legal term. When a patient lacks capacity, a **surrogate** (often next of kin) decides for them — but a surrogate has **no authority to override a patient who still has capacity**. Two psychiatrists found Cowart had capacity. His mother's authorization was nonetheless treated as controlling. That is the legal pressure point premise 2 exposes.

The beneficence reply: pain, shock, and "they almost always change their minds"

The physicians did not deny premise 1. They denied premise 2. Their claim was not that Cowart could not *talk* coherently. It was that a person in catastrophic pain, in shock, facing a future he could not yet imagine, is not making a *settled, autonomous* choice. He is crying out from an unbearable present. Dr. Larson, on the film *Dax's Case*, put both the conviction and its limit in one breath: "No other burn patient has made it so clear ... that they really do desire to die. ... This is an ongoing battle that we have to individualize."

Argument

The Beneficence / Temporary-Incapacity Reply

1. If a patient is in extreme pain and shock, their capacity to refuse treatment is undermined.
2. Dax Cowart was in extreme pain and shock.
3. Therefore, Dax Cowart lacked the capacity to refuse treatment.

This is essentially the position the philosopher H. Tristram Engelhardt later defended in the *Dax's Case* essays. He called it a genuine conflict "at the roots of health care" between freedom and best interest (Kliever 1989). The reply also has empirical backing that did not exist when commentators first took Cowart's side:

- A 1978 NIH consensus statement held that shock makes early decision-making nearly impossible for burn victims.

- In a later study, burn survivors interviewed years afterward said almost uniformly that informed consent was unrealistic at the moment of injury.
- They said the capacity to consent *developed over time*.
- None thought withdrawing support would have been right for them (Brewster et al. 2006).

Background

Informed consent vs. informed refusal — and how new it was. **Informed consent** — disclosure plus free agreement before treatment — was barely a decade old in law in 1973. *Canterbury v. Spence*, which set the modern “reasonable patient” disclosure standard, had been decided only the year before (United States Court of Appeals, District of Columbia Circuit 1972). The harder idea, **informed refusal** — a competent patient’s right to *decline* life-saving care — was almost unrecognized. Many physicians trained in an older **paternalistic** model treated refusal of life-saving treatment as itself evidence of impaired judgment. That is precisely the reasoning premise 1 of the reply formalizes (see Faden and Beauchamp (1986)).

But the reply has a soft spot. Premise 2 is a *statistical generalization* — most patients are glad later — used to override *this* patient. Applying an average to the individual in front of you is a recognizably **utilitarian** move: maximize good outcomes across the class of burn patients. It invites the objection Cowart spent his life pressing.

Cowart’s counter: a good outcome doesn’t make the choice yours

Cowart’s reply is not “you were wrong about the outcome.” He concedes the outcome. His reply is that a good result does not retroactively transfer the decision to the people who made it.

Argument

The Retrospective-Endorsement Objection

1. The right to make a decision does not depend on the outcome of that decision.
2. Pointing to a patient’s later gladness is an appeal to the outcome.
3. Therefore, a patient’s later gladness does not prove that the decision belonged to the medical team.

He sharpened it with an analogy: a surgeon who operates on the wrong patient is not vindicated because the patient happens to recover.

This is where a concept from ethical theory does real work. Cowart’s two claims — *I am glad to be alive* and *I was wronged* — are not a contradiction. They are a textbook case of **moral residue**: even when overriding a duty is, on balance, defensible, the overridden duty does not vanish. Something wrongful remains to be acknowledged. As he said: “I do not feel grateful to anyone for fighting my request to die ... no, I’m not glad they forcibly treated me because the pain that I went through was pure hell” (Cowart and Burt 1998). From a **Kantian** angle, the same point is

that treating a competent person merely as a means to a future *others* judge worthwhile fails to respect him as a person who may author his own life — even badly.

Burt's relational counter: "I won't abandon you"

The legal scholar Robert Burt, who debated Cowart publicly for years, accepted almost all of this and still resisted the conclusion that the team should simply have complied. His move changes the frame from a transaction (consent given or withheld) to a relationship.

Argument

The Relational Counter

1. Desperate demands to die are often cries for sustained relationship rather than pure refusals.
2. If a demand is a cry for relationship, then immediately complying with it is a form of abandonment.
3. Therefore, immediately honoring a patient's refusal to live can be a form of abandonment.

This is recognizably an **ethics of care** position: obligations grow out of relationship and dependency, not only out of rights. It is also where the dialogue gets its most famous phrase. Burt kept asking *when comes the end of the day?* He conceded that staying cannot be indefinite, because a relationship that never lets the patient's "no" be final has quietly become the old paternalism in kinder language. Cowart's answer was blunt: "an hour was an eternity, with the burns and the pain I was going through." Burt's honest reply was that he did not always know. That unresolved gap is what the next move tries to close.

The Arnold–Menzel procedure: turning principles into a process

Denis Arnold and Paul Menzel took the Burt–Cowart standoff seriously as an *unanswered question*: when, exactly, does resistance to a competent refusal have to yield? They proposed a step-by-step procedure to answer it without smuggling paternalism back in. Their guiding question reframes the whole debate away from absolute rights. Not "who holds the trump card?" but "**Whose pain is it, anyway?**" — a question, they write, about who may decide that agonizing pain *now* is worth a possible good life *later* (Arnold and Menzel 1998).

The procedure is, in the vocabulary of principlism, an exercise in **specification** and **balancing**. It takes two abstract principles that conflict in the abstract — respect for autonomy and beneficence — and turns them into a concrete sequence a real clinical team can follow at the bedside. The decisive split is whether the patient's pain is being adequately controlled. Uncontrolled pain both clouds judgment *and* is itself a harm the patient is being forced to endure. Read the diagram as a single question that forks: *first fix the pain or honor a short waiting period. Either way, the patient — not the team and not the family — ends up holding the decision.*

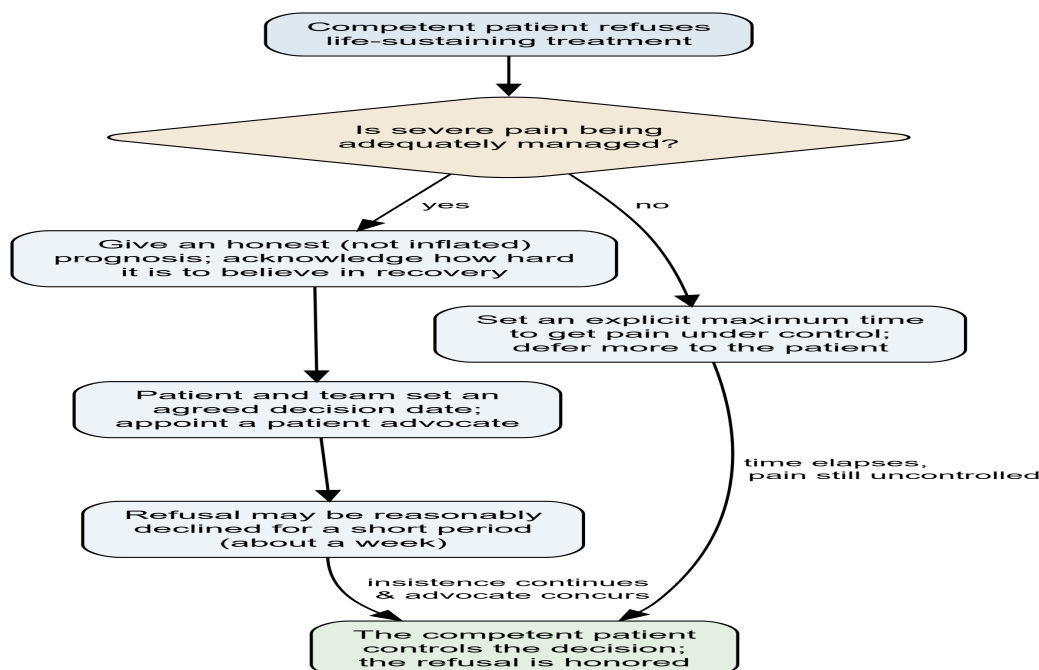


Figure 1: The Arnold–Menzel “end of the day” procedure. Burt asked when a care team must finally stop resisting a competent patient’s refusal; Arnold and Menzel answer with a process rather than a fixed deadline. The pivotal question (gold diamond) is whether severe pain is controlled. If it is, the team owes the patient an honest prognosis, an agreed decision date, and an advocate, and may decline the refusal only for a short period (about a week). If pain is *not* controlled, the team may hold only long enough to set and pursue an explicit pain-control deadline. On both paths the endpoint (green) is the same: the competent patient controls the decision. Adapted from Arnold & Menzel (1998).

The argument that hasn’t ended: the empirical turn

Cowart’s case itself never produced a court ruling. The legal architecture grew up around it. Two cases were decisive:

- *In re Quinlan* (1976) recognized that life-sustaining treatment could be withdrawn through a surrogate ([Supreme Court of New Jersey 1976](#)).
- *Cruzan v. Director* (1990) treated the refusal of such treatment as a protected liberty, while letting states demand clear evidence of the patient’s wishes ([Supreme Court of the United States 1990](#)).

Out of that lineage came living wills, durable powers of attorney, and the federal Patient Self-Determination Act. The law moved Cowart’s way. The bedside did not always follow. In a 2019 remembrance, the bioethicist Greg Pence reported a burn physician saying, of patients who ask to die, “They all want to die, and I just ignore them. I don’t let them give up on themselves” ([Parsi and Winslade 2019](#)). Advance directives, when they exist, are quietly overridden a large share of the time. The doctrine settled. The disagreement did not.

If the story stopped with Arnold and Menzel, the lesson would still look tidy: respect autonomy, control pain, hand the patient the decision. The most important recent move complicates exactly that lesson — and it comes not from law but from data. The bioethicist Monica Gerrek argues that building burn ethics almost entirely on Cowart has spread a belief — that burn units are inherently paternalistic — that the wider evidence does not support (Gerrek 2018). She sets beside Dax the case of Andrea Rubin, severely burned in 2014 with comparable injuries. The two cases ran in parallel:

- Rubin’s friends, like Dax’s, pleaded that she would not want to live this way.
- Her father, like Dax’s mother, consented to treatment.
- She was sedated for weeks and could not decide.
- After recovery, Rubin did *not* feel betrayed. She believes she genuinely lacked capacity early on, that the team acted in her interest, and that honoring her friends’ pleas would have been a fatal mistake.

Rubin’s case does not refute Cowart. Two psychiatrists found *he* had capacity, and hers is also just one story. What it does is reopen premise 2 of the original autonomy argument as a genuinely empirical question that cannot be settled from a single famous case. Capacity in catastrophic injury is harder to read, and more variable across people, than either “burn units are tyrannical” or “patients always come around” allows. Cowart himself, asked how a physician could ever be *sure* a patient truly wants to die, conceded: “I doubt that there is any way a physician can be absolutely sure.” Fifty years on, that is roughly where the argument rests — not at an answer, but at a harder question, with Cowart’s autonomy claim intact and the certainty once used against it gone on both sides.

Discussion Questions

1. Write out **The Autonomy Argument** and **The Beneficence Reply** in your own words — premises first, then the conclusion. The two arguments disagree about one thing: whether Cowart was *competent* to refuse during his treatment. Then say what kind of evidence — clinical, philosophical, or statistical — could actually settle that question.
2. After he recovered, Cowart said two things: he was glad to be alive, and his rights had been violated. The idea of **moral residue** says a decision can turn out fine and still leave a moral debt behind. Use that idea to explain why his two statements are not a contradiction. What does the idea tell us about paternalistic decisions that “turned out fine”?
3. The Beneficence Reply uses a generalization about a group — “most patients in Cowart’s situation are later glad they were treated” — to override one person’s choice. When, if ever, is it fair to apply a true generalization about a group to the specific person in front of you? What principle are you relying on?
4. You are the nurse assigned to Cowart’s daily tank treatment. He has just told you again that he wants it to stop and wants to die. What do you owe him in that moment that the institution does not? What, concretely, do you do?

5. Apply the Arnold–Menzel procedure to Andrea Rubin’s case. Does the procedure give the answer Rubin would endorse? If not, what does the mismatch tell you — is the procedure flawed, is Rubin’s case importantly different from Cowart’s, or is the argument Cowart started still open?

Further Reading

- *Please Let Me Die* (1974) and *Dax’s Case* (1984) — the two documentaries; the second includes interviews with Cowart, his physicians, his nurse, and his mother.
- Lonnie D. Kliever, ed., *Dax’s Case: Essays in Medical Ethics and Human Meaning* (Southern Methodist University Press, 1989) ([Kliever 1989](#)).
- Keith Burton, “A Chronicle: Dax’s Case as It Happened,” in Kliever, ed. (1989) ([Burton 1989](#)).
- Daniel E. Slotnik, “Dax Cowart, Who Fought for Patients’ Rights, Dies at 71,” *The New York Times*, May 15, 2019 ([Slotnik 2019](#)).

The Tuskegee and Guatemala Syphilis Studies: Research Ethics, Whistleblowing, and Medical Distrust

Deception and the Long Half-Life of Distrust — a bioethics case study

At a Glance

- **Who:** About 600 Black men of Macon County, Alabama — 399 with latent syphilis, 201 uninfected controls — and, later, more than 1,300 Guatemalans deliberately infected by the same U.S. agency.
- **What:** The U.S. Public Health Service watched untreated syphilis progress to autopsy for **40 years (1932–1972)**. The agency withheld penicillin and deceived the men into believing they were being treated. A parallel PHS project in Guatemala (1946–1948) infected prisoners, soldiers, and psychiatric patients on purpose.
- **Where / When:** Tuskegee/Macon County, Alabama, 1932–1972; Guatemala, 1946–1948. Exposed by a whistleblower and the press in 1972. The Guatemala records were found in 2003.
- **Why it matters:** Tuskegee is widely called the single most important event in the rise of American **bioethics**. It produced the modern rules — **informed consent**, the **IRB**, the Belmont principles — and a debate, still unsettled, about whether rules were ever the thing that was missing, and about the medical distrust the study left behind.
- **Pair with:** History & the Making of Research Ethics

The Case

In 1932, the United States Public Health Service (PHS) — the federal agency that would become the CDC — began what its own scientists titled the *Tuskegee Study of Untreated Syphilis in the Negro Male*. The premise was simple. To its designers, it was scientifically attractive: find people with untreated **syphilis** and observe what the disease does to a human body if nothing is done, all the way to autopsy.

They chose Macon County, Alabama. Most residents were Black. Most were poor. Almost none had access to a doctor. That was the point. The men recruited — about 600 in all — were sharecroppers working white-owned land in a system of debt peonage. Of the 600:

- 399 had latent syphilis.
- 201 served as healthy controls.

The PHS enlisted the respected Tuskegee Institute and a trusted Black public-health nurse, Eunice Rivers, whose constancy kept the men coming back for decades. The men were never told they had syphilis. They were told they had “**bad blood**,” a vague local phrase, and that the government doctors were treating them for it.

Background

What untreated syphilis does. Syphilis is a bacterial infection. After an early stage it can go *latent* — silent for years — and then, in roughly a third of untreated cases, attack the heart, blood vessels, brain, and spinal cord. The consequences the PHS’s own venereal-disease workers printed on a matchbook to scare people into testing were blunt and accurate: “Blindness, heart injury, insanity, death.” None of that was named to the Tuskegee men.

The deception was not passive. To collect the data they wanted, investigators did several specific things:

- They performed painful diagnostic spinal taps and called them a free “spinal shot” — a *treatment*.
- One PHS officer wrote that the men might refuse the procedure if they understood it was only diagnostic. Its real purpose, he said, should “be kept from them as far as possible.”
- Because the study’s final data point was the autopsy, the PHS offered burial insurance. This was a powerful inducement in a poor rural community where a proper funeral mattered. Families would surrender the bodies in return.
- One investigator wrote that the team had “no further interest in these patients until they die.”
- When a mobile treatment unit later swept the county offering free penicillin to everyone, PHS staff quietly flagged their enrollees to local doctors with the instruction that the man was under study and *not to be treated*.



Figure 1: A Public Health Service worker draws blood from a study participant during an annual “roundup,” Milstead, Macon County, 1953. Nurse Eunice Rivers stands at left.

National Archives and Records Administration. Public domain. Reproduced in ([Tobin 2022](#)).

Penicillin became the standard cure for syphilis in the mid-1940s. The men were not given it. PHS scientists argued that the arrival of a cure made the study *more* valuable, not less. It was a never-to-be-repeated chance to see the full natural course of a disease that would soon no longer run its course anywhere else. The study generated about fifteen papers in reputable medical journals over thirty-seven years. The titles alone announced what it was — one 1964 article was called “The Tuskegee Study of Untreated Syphilis; the 30th Year of Observation.” A 1955 paper reported that more than thirty percent of the autopsied men had died directly of syphilis. No physician, anywhere, published a letter of objection.

It ended only because one PHS employee would not let it go. By the most-cited later reckonings:

- Dozens of the men had died of syphilis or its complications.
- Dozens of their wives had been infected.
- Children had been born with congenital syphilis.

In 1973, after the story broke, a class-action suit produced an out-of-court settlement. The government eventually provided lifetime medical care to survivors and affected families. In 1997 — a quarter century after the study stopped — President Clinton apologized: what the government did “was shameful,” a study “so clearly racist.” No scientist was ever prosecuted.

The case is usually taught as a failure of consent. The questions it actually raises are harder than that, and they are still open.

The Argument Tuskegee Started

Few cases have produced as much agreement *that* something monstrous happened and as much disagreement about *what the monstrous thing was*. That second question is not academic. Nearly every reform that followed — federal consent rules, the Belmont Report, the very existence of review boards — rests on some answer to it, and the answers do not all point the same way.

“We were missing rules”

The official lesson is institutional. Tuskegee, on this view, happened because the United States had no system requiring that subjects consent and no body to review research before it ran. So the country built one:

- The 1974 National Research Act.
- **Institutional review boards** (IRBs) to approve and monitor studies.
- Federal informed-consent regulations.
- In 1979, the Belmont Report’s three principles — respect for persons, beneficence, and justice.

Background

Informed consent, the IRB, and Belmont. **Informed consent** means a competent person is told the truth about a study’s risks, benefits, and purpose and freely agrees to take part. An **IRB** is a committee that must approve human research before it begins and can stop it. The **Belmont Report** distilled the post-Tuskegee consensus into three duties:

1. Respect persons as autonomous agents.
2. Do not harm and maximize benefit.
3. Distribute research’s burdens and benefits fairly.

These are the spine of every research-ethics rulebook in use today ([National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research 1979](#)).

Argument

The Regulatory-Gap Argument

1. If a system lacks independent review and consent regulations, then research abuses like Tuskegee are inevitable.
2. Modern research systems now have independent review and consent regulations.
3. Therefore, research abuses like Tuskegee are no longer possible in modern research systems.

Premise 1 is where this breaks. It assumes the gap was the *absence* of rules. The next move denies that.

“The rules existed — and were ignored”

The Nuremberg Code was written in 1947 after Nazi doctors were tried for experimenting on prisoners. It opens with the sentence “The voluntary consent of the human subject is absolutely essential.” It existed for the last twenty-five years of Tuskegee. American researchers of the era reportedly dismissed it as “a good code for barbarians” — something for Nazis, not for them. Rules were not missing. They were not felt to apply.

Then there is Guatemala. While researching Surgeon General Thomas Parran’s papers in 2003, the historian Susan Reverby opened the donated files of a PHS doctor named John Cutler and found something worse than Tuskegee ([Reverby 2011](#)). Between 1946 and 1948, the same Public Health Service had **deliberately infected** more than 1,300 Guatemalans — prisoners, soldiers, and patients in a psychiatric hospital — with syphilis, gonorrhea, and chancroid, to test prevention. When infection through arranged contact proved unreliable, investigators inoculated subjects directly. At least eighty-three deaths were recorded. One psychiatric patient, after months of induced disease, was dying when a researcher infected her further. She died days later.

The decisive fact about Guatemala is not its cruelty but its respectability. Consider the chain of approval:

- The principal investigator, John Mahoney, was arguably the most honored American physician-scientist of his day. He won a Lasker Award the year the experiments began.
- The proposal was reviewed and funded by an NIH study section staffed by physicians from the country’s leading medical schools.
- The Surgeon General was kept informed and enthusiastic. He is recorded remarking that “we couldn’t do such an experiment in this country.”

Nothing failed in the chain of command. Everyone with authority approved.

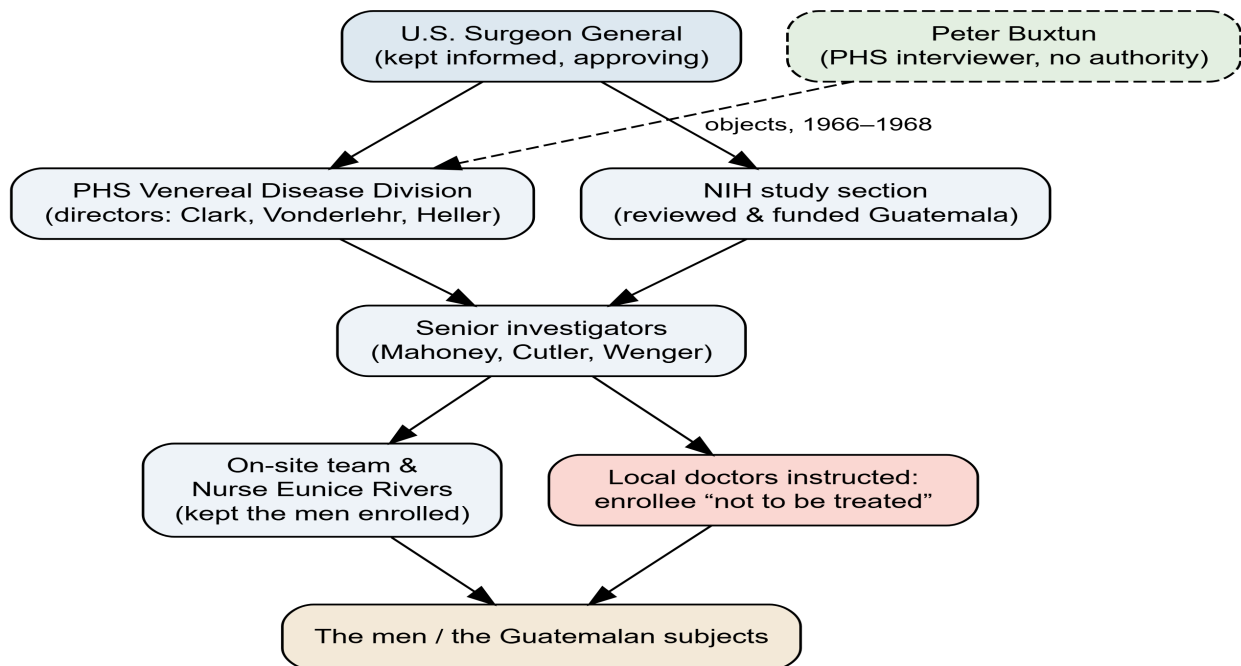


Figure 2: The PHS chain of command in the Tuskegee and Guatemala studies. The point of the diagram is what it does *not* contain: a broken link. Approval and knowledge flowed through every legitimate level of authority — the Surgeon General, the funding review, the senior investigators, the trusted on-site nurse — down to instructing local doctors that enrolled men were ‘not to be treated.’ The one person who finally stopped Tuskegee, Peter Buxtun, sits *outside* this structure entirely (dashed). A rule is enforced by the people inside the boxes; that is why ‘add a rule’ did not, by itself, answer the problem.

Argument

The Conscience Argument

1. Rules only prevent abuse if researchers believe the rules apply to them and their subjects.
2. The PHS researchers did not believe existing ethical rules applied to their subjects.
3. Therefore, the failure was a failure of institutional and individual conscience, not a lack of rules.

That individual was Peter Buxtun. The PHS hired him in 1965 as a venereal-disease interviewer in San Francisco. He was a Czech-born social worker whose family had fled to America in 1937. He learned of the study from a coworker, obtained a folder of internal “roundup” reports, and recognized the shape of what he was reading. He had read the Nuremberg trial record. His path inside the agency went like this:

1. He filed formal objections in 1966 and again in 1968.
2. The CDC summoned him to Atlanta, and a senior official berated him.

3. A blue-ribbon panel was convened in 1969. Its dominant voice argued, “You will never have another study like this; take advantage of it.” Only one member, Gene Stollerman, urged that the men be treated as patients rather than as a data set.
4. The panel voted to continue. Buxtun asked the CDC, in writing, “What is the ethical thing to do?” He received no answer.
5. He took the documents to the press. The story broke in July 1972.



Figure 3: Peter Buxtun, the PHS social worker whose complaints, ignored inside the agency for years, finally reached the press in 1972.

Public domain. Reproduced in ([Tobin 2022](#)).

Why did it take one outsider seven years, when no insider acted in forty? Buxtun’s own answer was wry: “It’s tough being a whistleblower when you don’t even know you’re a whistleblower.” The stronger pressure to weave in here is from social psychology. Stanley Milgram’s obedience experiments — and a later study modeled on them, in which university students were asked to recruit others into an obviously dangerous sham experiment — found that most people, given a legitimate-seeming authority, comply and look away. They do this even when dissent is safe, anonymous, and easy ([Elliott 2018](#)). Milgram called this surrender the “agentic state”: we recast ourselves as instruments of the authority and let our conscience go quiet. The Tuskegee physicians were not monsters by Milgram’s lights. They were ordinary people inside a structure that made silence feel normal.

“But heroes are not a policy”

If the Conscience Argument is right, it leaves a problem. You cannot run a research system on the hope that each abuse will produce its own Buxton — especially since the system punishes the ones it gets.

Argument

The Institutional-Design Reply

1. Relying on individual conscience only works if whistleblowers can safely come forward.
2. Under current institutional designs, whistleblowers who come forward are routinely punished.
3. Therefore, preventing future research abuses requires structural institutional reform, not just appeals to personal courage.

The constructive half of this comes, again, from Milgram. Obedience dropped sharply under three specific conditions:

- When the authority figure was made less imposing.
- When the victim was made more present.
- And — most powerfully — when *even one* other person in the room dissented.

The design implication is concrete:

1. Deflate the prestige of the lead researcher.
2. Make subjects visible rather than abstract.
3. Protect real (not cosmetic) channels for objection.
4. Make it possible for objectors to find each other.

The recurring real-world successes — the New Zealand “unfortunate experiment,” for instance — came from *teams* of clinicians acting together, not solitary martyrs. An anonymous “ethics hotline,” the usual institutional answer, is the thing the evidence trusts least.

Who was chosen, and why

There is a wrong here that none of the moves above quite names. The Tuskegee men were not random. They were selected *because* they were poor, Black, unschooled, and without doctors. They were chosen for the very powerlessness that made them easy to deceive and unlikely to be believed. A PHS internal phrasing reduced them to a single compound label, a string of demographic adjectives ending in “men.” That is not a consent failure or an oversight failure. It is a **justice** failure: a vulnerable group made to carry a burden for a science that would never have been done to the powerful.

Background

Vulnerability and fair subject selection. Kenneth Kipnis argued that “vulnerable” is not a list of groups but a set of *pressures* a participant is under — economic, social, deferential, institutional — that compromise a genuinely free choice (Kipnis 2001). The Belmont principle of **justice** demands *fair subject selection*. Research must not load its risks onto those least able to refuse, nor (the mirror error) exclude groups from research’s benefits. Tuskegee is the textbook case of wrongful **exploitation** of the vulnerable. Modern ethics also watches for wrongful **exclusion**.

Argument

The Justice Reframing

1. In Tuskegee, subjects were selected specifically because their poverty and race made them unable to refuse or complain.
2. Selecting subjects because of their vulnerability is a form of exploitation that violates the principle of justice.
3. Therefore, the deepest wrong of Tuskegee was a violation of justice, which cannot be fixed merely by improving consent procedures.

This is not safely historical. Research today still gravitates toward people under exactly Kipnis’s pressures:

- The institutionalized.
- The incarcerated.
- The cognitively impaired.
- The poor.

That is why “fair subject selection” remains an active demand and not a settled box to check.

The argument that hasn’t ended

There is a final move, and it is the one with the most current stakes. The standard account says: Tuskegee is *why* many Black Americans distrust medicine and research, so the remedy is outreach, apology, and education to repair the trust the study broke. After 1972, the empirical legacy is, in fact, measurable. Marcella Alsan and Marianne Wanamaker found that disclosure of the study was followed by reduced medical contact among older Black men and worse health outcomes — a real and lasting cost (Alsan and Wanamaker 2018).

But Vanessa Northington Gamble and Susan Reverby press a sharp objection to the standard account. Black medical distrust did not begin with Tuskegee and is not exhausted by it. It tracks a long history and, more importantly, *present* mistreatment — documented disparities in pain treatment and outcomes, structural barriers in care now (Gamble 1997). To explain today’s distrust by pointing only to a study that ended in 1972 can become a way of locating the problem safely in the past. It can cast Black patients as irrationally haunted rather than accurately

observant. (Reverby also notes a precise irony: the Tuskegee men were not distrustful refusers. They *trusted* the government doctors, which is exactly how they were used.) The COVID-19 vaccination gap became the live test of this dispute: was the right response to recite Tuskegee and reassure, or to address the present-day inequities that make wariness a reasonable inference?

What actually moved the gap is instructive. Early 2021 messaging that leaned on history-and-reassurance — *that was then, the rules are different now, the vaccine is safe* — tended to land poorly. It answered a question many Black patients were not asking. A review of equity strategies found that the interventions that raised uptake were structural and relational rather than rhetorical:

- Vaccine sites placed inside trusted Black churches and barbershops.
- Outreach delivered by community health workers, clergy, and Black clinicians rather than by the institutions themselves (Dada et al. 2022).

The recurring mechanism is *concordance* — patients are more likely to accept advice, including vaccination, from clinicians who share their background or have earned standing in their community. A randomized field experiment in Oakland found that Black men assigned to a Black physician agreed to substantially more preventive services, including invasive screening, than those assigned to a non-Black physician — a gap that no recitation of history opens or closes (Alsan et al. 2019). The lesson tracks Reverby's point precisely: the operative variable was not how vividly the past was explained but how much present trustworthiness the patient could actually observe in the room.

That gives a clinician something to do rather than something to recite. It does not, however, settle the argument. It arguably concedes the critique's premise — trust follows present conditions, not historical correction — while leaving the standard view's strongest reply intact: history is *part* of why those present conditions are read the way they are, and a patient who invokes Tuskegee is often naming a pattern, not a date — distrust as a response to everyday racism, not only to one archived study (Bajaj and Stanford 2021). The dispute is which description should drive the response.

Argument

The Distrust Dispute

1. If historical events like Tuskegee are the primary cause of medical distrust, then distrust should be solved by historical education and apology.
2. Medical distrust is primarily caused by ongoing, present-day systemic medical racism and disparities.
3. Therefore, medical distrust can only be solved by changing present-day medical practices, not just explaining the past.

Fifty years on, that is where the argument rests — not at a verdict, but at a question that a nurse, a researcher, and a public-health officer each have to answer the next time someone says, reasonably, that they are not sure they trust the people offering to help them.

Discussion Questions

1. Write out **The Regulatory-Gap Argument** and **The Conscience Argument** in your own words — premises first, then the conclusion. They disagree about one premise. Identify it. Then explain what the Guatemala study — run by a respected lead researcher, reviewed by NIH, approved by the U.S. Surgeon General — tells us about which side has the better case.
2. Nurse Eunice Rivers was trusted by the men in the study because she was caring, consistent, and one of their own. That trust is part of what kept them enrolled for decades. Imagine you are a nurse who has just realized that the “treatment” you have been giving is a deception. What do you owe these particular patients that the institution does not? What, concretely, do you do — knowing that nurses who reported similar abuses often lost their jobs?
3. **The Justice Reframing** says the deepest wrong of Tuskegee was *who* the researchers chose as subjects, not the missing consent forms. Imagine a future study with perfect informed consent — but where the volunteers are poor, uninsured, and badly need the money. Would the consent fix Tuskegee’s wrong, or just move it somewhere else? Use the idea of **fair subject selection** to defend your answer.
4. **The Institutional-Design Reply** says hospitals can do specific things to keep individual dissent from being silenced. Pick a hospital you know. Name one concrete change — not an ethics hotline — that would make a dissenting nurse or resident less alone. Drawing on Milgram’s obedience studies, explain why your change would actually work.
5. A patient eligible for a new vaccine tells you she does not trust it. She mentions Tuskegee. The case describes two ways to respond: **reassure her and explain the science**, or **acknowledge that present conditions still give some patients real reasons to distrust**. Give the strongest version of each response. Then say which you would actually give her, and why.

Further Reading

- James H. Jones, *Bad Blood: The Tuskegee Syphilis Experiment* (Free Press, 1981; rev. 1993) ([J. H. Jones 1981](#)) — the definitive history.
- Susan M. Reverby, *Examining Tuskegee: The Infamous Syphilis Study and Its Legacy* (University of North Carolina Press, 2009) ([Reverby 2009](#)).
- Harriet A. Washington, *Medical Apartheid* (Doubleday, 2006) ([Washington 2006](#)) — the longer history of which Tuskegee is one chapter.
- Carl Elliott, “Tuskegee Truth Teller,” *The American Scholar* 87, no. 1 (2018) ([Elliott 2018](#)) — the Buxtun profile and the whistleblowing evidence.
- Presidential Commission for the Study of Bioethical Issues, *“Ethically Impossible”: STD Research in Guatemala from 1946 to 1948* (2011) ([Presidential Commission for the Study of Bioethical Issues 2011](#)).

Sylvain Lesné and the A β *56 Affair: Research Misconduct, Replication, and the Amyloid Hypothesis

“Blots on a field” — a bioethics case study

At a Glance

- **Who:** Sylvain Lesné (b. 1974), neuroscientist at the University of Minnesota; Matthew Schrag (b. ~1984), Vanderbilt neurologist and image sleuth; Karen Ashe, Lesné’s senior collaborator and a founding figure in mouse models of Alzheimer’s disease.
- **What:** A 2006 paper in *Nature* claimed to have isolated a form of amyloid-beta — “A β 56” — *that caused memory loss in rats. The paper anchored the leading theory of Alzheimer’s for sixteen years. In 2022, a whistleblower published evidence that dozens of Western blot images in Lesné’s papers had been digitally faked. Nature* retracted the paper in 2024.*
- **Where / When:** University of Minnesota; Vanderbilt University; published in *Science* July 2022; retraction June 2024; Lesné resigned February 2025.
- **Why it matters:** The case sits at the crossroads of **research misconduct**, the **replication crisis**, and **subject protection**. If a basic preclinical result is faked, every drug trial built on it enrolls human volunteers under false **clinical equipoise** — and the field’s billions in public funding rest on a mirage.
- **Pair with:** Research Ethics (Protecting Research Subjects)

The Case

In August 2021, Matthew Schrag got a phone call. He was a junior neurologist at Vanderbilt, thirty-seven years old, soft-spoken, with a small sign on his desk reading “Everything is figureoutable.” A colleague wanted to connect him with an attorney. The attorney represented two short-sellers — investors betting that a small biotech, Cassava Sciences, would lose value. Cassava was developing an Alzheimer’s drug called Simufilam. The short-sellers thought the underlying research was fake. Would Schrag, who had already criticized the FDA’s approval of another anti-amyloid drug, agree to examine the published images? He said yes. Over the next months, the attorney paid him about \$18,000 (Piller 2022).

What began as a paid side job became something larger. Schrag's method was unglamorous. He downloaded published figures, zoomed in, and looked for telltale signs that an image had been built in Photoshop rather than captured from a real experiment:

- Patches of pixels repeated in places they should not be.
- Bands that looked identical when they should be unique.
- Edges too clean to come from a working gel.

His targets were **Western blots**. A Western blot uses electricity to drag protein samples through a gel, separating them by size into stacked bands that are stained and photographed. Two blots from the same true experiment can never look pixel-identical. If they do, someone copied and pasted.

Background

Western blot. A Western blot detects a specific protein in a tissue or fluid sample. The sample runs through a gel that separates proteins by size. The proteins are then moved to a membrane and tagged with antibodies that bind only to the target protein. The result is a photograph of dark bands at certain positions — a visual fingerprint. The exact pattern of background noise and band shape is random, so two real blots should never match.

While studying Cassava, Schrag noticed something far bigger. The same patterns of image trouble appeared in an older and much more famous paper: a 2006 article in *Nature*, first-authored by Sylvain Lesné. Lesné was then a postdoc at the University of Minnesota, working with the well-known Alzheimer's scientist Karen Ashe ([Lesné et al. 2006](#)). The paper reported the isolation of a specific form of amyloid-beta. The authors called it A β 56 (*pronounced "A-beta star fifty-six"*). *They claimed that injecting purified A β 56 into young rats made them fail memory tests they had passed before.* In the language of biomedical research, this was a smoking gun. It was a direct claim that a particular amyloid species *causes* dementia, not merely tracks it.

Background

The amyloid hypothesis. Since the early 1990s, the leading theory of Alzheimer's has held that sticky protein deposits called amyloid-beta plaques in the brain are a main cause of memory loss. The hypothesis has shaped drug development for thirty years. Hundreds of clinical trials have tested antibodies and small molecules meant to prevent, dissolve, or clear amyloid. Most have failed. The few approved drugs — aducanumab (Aduhelm), lecanemab, donanemab — produce small, disputed benefits and can cause brain swelling or bleeding.

The 2006 paper was cited more than 2,300 times. It launched a generation of follow-up studies. It helped Lesné win his own NIH-funded lab at Minnesota in 2009. And, according to Schrag, dozens of the Western blots in it and in Lesné's later papers showed signs of digital editing: bands repeated where they should be unique, backgrounds spliced together, edges that looked drawn rather than photographed ([Piller 2022](#)).

Schrag was careful. He refused to use the word "fraud" in his public critiques. He could not

prove intent. He could not see the original, unedited image files. “I focus on what we can see in the published images, and describe them as red flags, not final conclusions,” he told the journalist Charles Piller. “The data should speak for itself.” But when *Science* sent his findings to top independent image analysts and senior Alzheimer’s researchers, they agreed with him. The reviewers included:

- **Elisabeth Bik**, a microbiologist and the world’s best-known forensic image consultant.
- **Donna Wilcock**, an Alzheimer’s researcher at the University of Kentucky, who called some of the images “shockingly blatant.”
- **Thomas Südhof**, a Nobel laureate at Stanford.

Bik concluded that the authors “appeared to have composed figures by piecing together parts of photos from different experiments... The obtained experimental results might not have been the desired results, and that data might have been changed to better fit a hypothesis” (Piller 2022).

Piller’s investigation appeared in *Science* in July 2022 under the title “Blots on a Field?”. It named Schrag publicly for the first time. The fallout came in stages:

1. *Nature* issued an expression of concern within days of the article.
2. In June 2024, the editors and most of the original authors agreed to retract the 2006 paper.
3. In February 2025, the University of Minnesota concluded its inquiry. It found that Lesné had committed research misconduct, and he resigned his faculty post.

The deeper damage is harder to count. The U.S. National Institutes of Health spent roughly \$1.6 billion on amyloid-related Alzheimer’s research in fiscal year 2022 alone — about half the federal Alzheimer’s budget. Südhof’s verdict was blunt: “The immediate, obvious damage is wasted NIH funding and wasted thinking in the field because people are using these results as a starting point for their own experiments” (Piller 2022).

The A β *56 affair turned out not to be alone. Over the next three years, Piller and his collaborators documented apparent image manipulation in the work of several other senior figures:

- **Eliezer Masliah**, hired in 2016 to lead the National Institute on Aging’s expanded Alzheimer’s program.
- **Berislav Zlokovic**, a stroke researcher at USC whose work supported a federally funded trial that was later suspended.
- **Marc Tessier-Lavigne**, the president of Stanford University, who resigned in 2023 after a student journalist found altered images in papers from his lab.

Across forty-six leading Alzheimer’s researchers, imaging experts flagged nearly six hundred suspect papers. Those papers had collectively been cited some 80,000 times (Piller 2025).

The question the A β *56 affair raises is not only whether Sylvain Lesné faked data. That question is now largely settled. The harder question is what it means for a field, and for the human volunteers downstream of it, when a basic result turns out to have been an “elaborate mirage.”

The Argument A β *56 Started

Long before Lesné, two quarrels were already running in Alzheimer's research: whether the amyloid hypothesis had become a paradigm too dominant to question, and whether fraud that fakes no patient data can still harm patients. The Lesné affair forced the two together. The argument it reignited is about what counts as harm, what counts as evidence, and what a research enterprise owes the people who enroll in its trials.

Schrag's argument: fraud in preclinical research is a research-ethics violation

Schrag's public framing is careful, but it contains a sharper ethical claim than his prose makes obvious. He worried from the start that volunteers in Cassava's Simufilam trials "faced risks of side effects with no chance of benefit." That worry generalizes. If the preclinical foundation of a drug-development program is faked, the trials built on it cannot meet **clinical equipoise**. Equipoise is the rule that a community of skilled clinicians must be in honest, good-faith doubt about whether the experimental treatment beats the alternative. Equipoise is what makes it acceptable to expose trial subjects to risk. Without it, the trial is not research. It is human experimentation in the older, condemned sense.

Argument

Schrag's Argument

1. A clinical trial is ethically permissible only if there is genuine community equipoise about whether the experimental intervention will benefit participants.
2. Equipoise requires that the preclinical evidence supporting the intervention be honestly produced.
3. Several anti-amyloid drug programs rely, directly or by citation chain, on preclinical results that appear to have been faked.
4. Therefore, those trials enroll participants under a false equipoise and violate the conditions for ethical research.

The disputed premise is the second one. Does equipoise depend on the honesty of the preclinical evidence in particular, or on the honesty of the broader scientific community at any given moment? If the field still has other good reasons to believe amyloid matters, perhaps a single faked study does not poison the well. That is where the main reply starts.

The principal reply: science is robust to individual fraud

A defender of the amyloid hypothesis can grant most of Schrag's facts and still resist his conclusion. The hypothesis does not stand or fall with A β *56. The link between amyloid and Alzheimer's was established long before 2006. Three independent lines of evidence pointed at it:

- **Pathology.** Amyloid plaques were identified in Alois Alzheimer's original 1906 case.

- **Genetics.** The amyloid-precursor-protein gene was tied to inherited Alzheimer's in 1991.
- **Down syndrome.** People with an extra copy of that gene develop early Alzheimer's pathology with striking reliability.

A β *56 was one proposed mechanism among many. Losing it does not collapse the hypothesis.

Argument

The "Robustness" Reply

1. The ethics of a clinical trial depends on the *overall* state of evidence supporting the intervention, not the integrity of any single paper.
2. The amyloid hypothesis is supported by multiple independent lines of evidence (genetics, pathology, imaging) that do not depend on Lesné's work.
3. Therefore, even if Lesné's specific findings are faked, current anti-amyloid trials need not violate equipoise.

The reply's soft spot is empirical. How much of the field's confidence in amyloid actually traveled — by citation and by influence on grant decisions — through the A β *56 result and the broader network of suspect papers? If the answer is "most of it," the reply is weak. If "very little," it is strong. Piller's team estimates that roughly six hundred questionable Alzheimer's papers have been cited around 80,000 times (Piller 2025). That is not noise. It also does not, on its own, prove that the field has been bent.

The counter-reply: structural harm and the therapeutic misconception

The counter-reply does not argue that the field has been bent. It argues that the *participants* have been wronged regardless. Trial volunteers — especially in early-stage Alzheimer's, where cognitive decline is the condition being treated — are vulnerable subjects. They suffer from what the bioethicist Paul Appelbaum named the **therapeutic misconception**. This is the tendency of seriously ill research participants to believe that what is being done to them is being chosen for their personal benefit, the way a doctor's prescription would be.

Background

Therapeutic misconception. The therapeutic misconception is the false but natural belief that a research protocol is a form of personalized care. Patients in trials commonly do three things at once:

- They underestimate the chance of being randomized to placebo.
- They overestimate the chance of personal benefit.
- They fail to register that the trial's main purpose is general knowledge, not their own treatment.

The misconception is strongest in conditions like Alzheimer's, where standard care offers little hope.

The structural harm, in this telling, is not that the trials produced bad medical outcomes. Many anti-amyloid trials are honestly run by investigators who had no idea anything was wrong with the preclinical literature. The harm is that the *consent* the participants gave was undermined upstream. They were told, in effect, that skilled independent researchers had reason to believe the drug might help. That claim turned out, in part, to be false. The wrong is a wrong of disclosure and of respect for persons. It is not erased by the fact that the trialists were themselves deceived. Here is where Kant's idea of treating persons as ends, never merely as means, earns its keep. A research subject used to test a drug whose preclinical basis was faked has been used.

A reframing move: Stegenga and medical nihilism

A different kind of critic accepts most of the exchange and then shifts the frame. The philosopher of medicine Jacob Stegenga argues, in his book *Medical Nihilism*, that we should “have low confidence in the effectiveness of medical interventions” in general (Stegenga 2018). The reason is not that researchers cheat. It is that even honest clinical evidence is shaped by structural problems:

- Small effect sizes.
- Selective publication of positive results.
- *p*-hacking — running analyses until something looks significant.
- Surrogate endpoints that may not track real benefit.
- Conflicts of interest from industry funding.
- Career pressure to find positive results.

On Stegenga's view, the Lesné case is not an outlier in an otherwise healthy science. It is a dramatic version of the field's normal pathology.

Argument

The Medical Nihilism Argument

1. The honest evidence base for most medical interventions is distorted by small effect sizes, surrogate endpoints, selective publication, industry funding, and pressure to find positive results.
2. Once those distortions are properly discounted, the residual evidence shows that most medical interventions deliver small benefits relative to their harms and costs.
3. Therefore, we should have low confidence in the effectiveness of most medical interventions.

The disputed premise is the second. Skeptics of Stegenga grant that the distortions are real but deny that discounting for them leaves most interventions looking marginal.

The two anti-amyloid drugs to reach the U.S. market are a test case. The first was Biogen's aducanumab, sold as Aduhelm. The FDA approved it in June 2021 over the near-unanimous objection of its own advisory committee. Three committee members resigned in protest. Key facts:

- The approval used the **accelerated approval** pathway. This pathway lets a drug reach patients based on a **surrogate endpoint** — here, the clearance of amyloid plaques from PET scans — rather than direct evidence that patients live longer or think more clearly.
- The two pivotal trials had been stopped early for futility.
- A post hoc re-analysis of a subset of patients given the highest dose produced a slim positive signal. The FDA accepted it.
- Biogen priced the drug at \$56,000 per patient per year. The Medicare program raised Part B premiums for every beneficiary in the country to cover the projected cost.
- By early 2024, Biogen had withdrawn Aduhelm from the market for commercial failure.

Background

Accelerated approval and surrogate endpoints. Accelerated approval is an FDA pathway that lets a drug for a serious illness reach patients based on a *surrogate endpoint* — a measurable proxy (a blood test, a scan finding, a biomarker) “reasonably likely” to predict real clinical benefit. The maker must complete confirmatory trials after approval. The pathway speeds access in conditions like cancer and HIV. Its risk is that the surrogate sometimes fails to track what patients actually care about: surviving longer, thinking more clearly, walking without help.

Lecanemab (sold as Leqembi) and donanemab (Kisunla) are the successors. The FDA approved lecanemab in January 2023 and donanemab in July 2024. Their trials were not stopped early. Their data are real. The headline numbers:

- They slow cognitive decline by about half a point on the eighteen-point Clinical Dementia Rating Sum of Boxes scale over eighteen months. Many neurologists openly doubt whether that difference matters in daily life.
- They cause **amyloid-related imaging abnormalities** (ARIA) — brain swelling or small bleeds visible on MRI — in roughly one in five treated patients, with rare but documented deaths.
- Annual list price runs to roughly \$26,500 per patient. That figure does not include infusion, monitoring MRI, or clinic fees. Medicare Part B now covers the drug for patients meeting its criteria, leaving a 20% coinsurance of about \$5,300 a year for anyone without supplemental coverage — a reminder that “covered” and “affordable” are different claims.

If Stegenga is right, the bioethical question is not only “how do we catch the next Lesné?” It is “what should we do when even our non-fraudulent medicine routinely overpromises?” The amyloid program looks, on this view, like a field that absorbed faked data so smoothly because the rest of its evidence base was, even at its best, weaker than its advocates claimed. Aduhelm did not require fraud. It required an honest field willing to count amyloid clearance as good enough.

Stegenga’s prescription is what he calls **gentle medicine**. The default should lean toward less intervention. Aggressive treatment should be reserved for cases where the evidence base clears a high bar:

Argument

The Gentle Medicine Argument

1. When confidence in an intervention's benefit is low and its harms are non-trivial, prescribing it does more to violate non-maleficence than withholding it does to violate beneficence.
2. For most medical interventions in their current evidence base, confidence in benefit is low and harms are non-trivial.
3. Therefore, medicine should default to less intervention, reserving aggressive treatment for cases where the evidence is unusually strong.

A defender of standard practice will press on the second premise. “Most” is doing a great deal of work. And the alternative to prescribing a marginal drug is sometimes prescribing nothing — itself a choice with costs. Some patients would have benefited. Family caregivers carry a heavy burden the drug was meant, however slightly, to ease. The argument does not, on its own, tell a memory-clinic nurse what to say to a frightened sixty-eight-year-old. It does suggest the conversation should start somewhere other than the brochure.

This is not a comforting frame. It implies that better misconduct policing — sharper image-forensics software, mandatory raw-data deposits, stronger journal retraction rules — addresses a real problem but not the deepest one. The deepest problem is the gap between what biomedical research can reliably deliver and what patients, regulators, and journalists are induced to expect from it.

The move that hasn't ended

The institutional fallout has been steady:

1. **June 2024.** The original authors of the A β *56 paper agreed to retract it.
2. **September 2024.** The NIH found that Eliezer Masliah — until that month the leader of the agency's Alzheimer's program — had committed research misconduct. He was removed from his post ([Piller 2025](#)).
3. **February 2025.** The University of Minnesota concluded its inquiry. Lesné resigned.

None of these moves answers the live question.

The live question is what biomedical research should do now that the tools of image faking and the tools of image forensics are both becoming much more powerful. Generative AI can produce Western blots that look real to the eye. Forensic AI can scan an entire journal's back catalog for duplications in minutes — Elisabeth Bik did her early work by hand. The arms race is real. It is being run on top of an incentive structure — publish-or-perish, novelty-or-irrelevance, single-laboratory secrecy — that the field has not seriously revised since the 1970s.

The parallel to Tuskegee is instructive but limited. Tuskegee produced direct, identifiable victims and a clear set of failures: no consent, no treatment, no public oversight. The A β *56 affair produces diffuse, statistical victims:

- Alzheimer’s patients who enrolled in trials that should have been redesigned.
- Families who paid out of pocket for drugs that should have been studied differently.
- A generation of postdocs whose careers were built on a hypothesis that was overinvested in.

There is no single wronged person to apologize to. There is, instead, a field-level question about what the public is owed when it funds science. Not perfect results, but a system honest enough that its mistakes are recoverable. The question is whether we now have one.

Discussion Questions

1. Write out **Schrag’s Argument** and the **Robustness Reply** in your own words — premises first, then the conclusion. The two arguments share one premise about what makes a clinical trial ethically OK to run. Identify it. Then say which side has the better read on what that premise requires.
2. The counter-reply says that patients in anti-amyloid trials were *wronged* even when their own trials were honestly run — because the science behind those trials had been corrupted upstream by Lesné’s fraud. Is this idea of “being wronged” too broad? If you accept it, where does it stop? If you reject it, why?
3. Stegenga’s **medical nihilism** says the Lesné case is just an extreme version of how biomedical research normally fails. Push back on this claim. Name one part of biomedical research that you think *does* work well, and explain why it survives Stegenga’s critique.
4. You are a nurse in a memory clinic. A patient with mild cognitive impairment asks you whether she should enroll in a new Alzheimer’s drug trial. She mentions news coverage of the “new Alzheimer’s drugs.” She has not read the *Science* article on the fraud. What do you say to her? What do you not say?
5. The 2006 *Nature* paper was retracted in 2024 — eighteen years after it was published, and only after a whistleblower (paid in part by stock-market short-sellers) brought it to public attention. Design one specific reform — at the journal, the university, the NIH, or the FDA — that would have caught the problem earlier. What trade-off would your reform force on the system?

Further Reading

- Charles Piller, *Doctored: Fraud, Arrogance, and Tragedy in the Quest to Cure Alzheimer’s* (Atria, 2025) — book-length investigation.
- Charles Piller, “Blots on a Field?”, *Science*, 21 July 2022 — the original exposé.
- Jacob Stegenga, *Medical Nihilism* (Oxford, 2018), chapters 1–3 — the philosophical reframing.

- The PubPeer entries on Lesné's papers (pubpeer.com) — the actual flagged images, with public commentary.
- John Ioannidis, "Why Most Published Research Findings Are False", *PLoS Medicine* (2005) — the foundational replication-crisis paper.

Kate Cox, Damla Karsan, and Conscience in Post-Dobbs Medicine: Refusal, Provision, and Professional Integrity

“I don’t want to wait until I’m bleeding” — a bioethics case study

At a Glance

- **Who:** Kate Cox, a thirty-one-year-old mother of two from suburban Dallas; Dr. Damla Karsan, her OB-GYN; Ken Paxton, the Attorney General of Texas.
- **What:** In December 2023, Cox became the first pregnant woman since *Roe v. Wade* was overturned to sue in open court for the right to an abortion. A trial judge granted her request. Within hours, Paxton wrote to three Texas hospitals warning that anyone performing the procedure could still face prosecution. The Texas Supreme Court ruled against her. Karsan, the named physician-plaintiff, would have faced up to life in prison if she had gone ahead.
- **Where / When:** Travis County and the Texas Supreme Court, December 2023. The broader question is national and ongoing.
- **Why it matters:** The case forces a hard question about **conscientious refusal** and its mirror image, **conscientious provision**. When a clinician’s medical and moral judgment conflicts with the law of the state in which she practices, what does professional **integrity** require? *Dobbs* did not invent that question, but it inverted it.
- **Pair with:** Abortion (The Law and Reproductive Rights)

The Case

In August 2023, Kate Cox saw a positive home pregnancy test and ran into the playroom where her husband Justin was wrestling on the floor with their two toddlers. The Coxes had always wanted a large family — three or four children. They lived in a Dallas-area bungalow with two child-sized lawn chairs in the backyard, two sippy cups in the sink, and an Olaf the snowman in the living room. Neither Kate nor Justin was particularly political. When the Supreme Court issued its 2022 decision in *Dobbs v. Jackson Women’s Health Organization*, returning the regulation of abortion to the states, the news barely registered with them. Texas’s strict ban, they assumed, would have medical exceptions that would cover anything serious.

At eighteen weeks, Cox's obstetrician called while she was driving and asked her to pull over. Early screening tests suggested Trisomy 18. This is a chromosomal condition in which a third copy of chromosome 18 produces severe developmental abnormalities. Over the next several weeks of ultrasounds and maternal-fetal-medicine consultations, the diagnosis was confirmed. The fetus had malformations of:

- The brain.
- The spine.
- The neural tube.
- The heart.
- The skull.
- The limbs.

"Each time we had an ultrasound, there was more bad news," Cox later told *Time*. The expected outcome was either death in utero or, if the baby survived birth, hospice care and death within hours or days.

Background

Trisomy 18 (Edwards syndrome). Trisomy 18 occurs when a fetus has three copies of chromosome 18 instead of two. Roughly half of fetuses with the condition are stillborn. Among live births, most die within the first week. A small minority of children with mosaic forms live longer, sometimes into adolescence, generally with profound disabilities. There is no curative treatment. Standard care is "comfort care" — keeping the infant warm and free of pain — rather than aggressive intervention.

The diagnosis also created acute risks for Cox herself. Her two previous deliveries had been by Cesarean section. A third C-section, or an induced labor at term following a fetal demise, carried elevated risks:

- Uterine rupture.
- Hemorrhage.
- Hysterectomy — meaning she might never be able to bear another child.

She had already been to the emergency room several times for cramping and elevated vital signs. Her physician, Dr. Damla Karsan, judged that termination of the pregnancy was the medically indicated course.

Texas law at the time forbade nearly all abortions. The statutes permitted a narrow exception when, in the "reasonable medical judgment" of the attending physician, the pregnancy placed the woman at risk of "a serious risk of substantial impairment of a major bodily function." The exception was poorly defined. Karsan believed Cox qualified. She also believed that if she was wrong — or if a prosecutor disagreed — she could face severe penalties:

- Loss of her medical license.
- Fines of at least \$100,000 per violation.
- Up to ninety-nine years in prison.

Background

The Texas medical exception. Texas’s overlapping abortion statutes (the Human Life Protection Act and Senate Bill 8) permit abortion only when, in the physician’s “reasonable medical judgment,” the patient faces a “life-threatening physical condition” or “a serious risk of substantial impairment of a major bodily function.” The statutes:

- Do not list qualifying diagnoses.
- Do not provide a pre-clearance mechanism.
- Require the physician to make the call alone.

A prosecutor may second-guess that call afterward. The physician bears the burden of proving the judgment was reasonable.

On December 5, 2023, with Cox now twenty weeks pregnant, the Center for Reproductive Rights filed *Cox v. Texas* on behalf of Cox, her husband, and Dr. Karsan. They asked a Travis County district judge for a **temporary restraining order** — a court order blocking the state from prosecuting Karsan or her hospital if she performed the abortion. Two days later, Judge Maya Guerra Gamble granted it. From the bench, she said: “The idea that Ms. Cox wants desperately to be a parent, and this law might actually cause her to lose that ability, is shocking and would be a genuine miscarriage of justice.”

Within hours, the Texas Attorney General, Ken Paxton, sent letters to three Houston-area hospitals where Karsan held privileges. The letters made three points:

- The trial court’s order did not insulate anyone from civil liability under SB 8, which authorizes private lawsuits for at least \$100,000 per abortion.
- The order also did not insulate anyone from criminal prosecution after the order was lifted, which Paxton was actively seeking.
- The hospitals should consider their own exposure.

Paxton then petitioned the Texas Supreme Court for an emergency stay.

By the time the Texas Supreme Court ruled, on December 11, 2023, Kate Cox had already left the state. The court held that Karsan had not met the statutory standard. Her sworn declaration that Cox faced a serious risk of impairment of a major bodily function was, in the court’s view, an opinion — not the demonstration that the statute required. “Only a doctor can exercise ‘reasonable medical judgment’ to decide whether a pregnant woman ‘has a life-threatening physical condition,’ ” the court wrote. “A woman who meets the medical-necessity exception need not seek a court order. ... But the law does not permit a court to grant a pre-authorization” ([Supreme Court of Texas 2023](#)). Cox obtained the abortion in New Mexico.

What follows turns on the physician rather than the patient. Damla Karsan had not been compelled to take Cox’s case. She could have referred Cox out of state — many Texas OB-GYNs did — and avoided the criminal exposure entirely. She chose instead to do three risky things:

1. Put her name on the lawsuit.
2. Swear under oath that the medical-exception standard was met.

3. Publicly disagree with a state law she had to live under.

She was not refusing to participate in a procedure she found morally objectionable. She was insisting on participating in a procedure she believed she was morally obligated to provide. The question her case raises is whether the profession of medicine has room for that kind of conscience too.

The Argument Karsan Started

The classical debate about conscience in medicine was built around the figure of the *refuser*. Three examples:

- A Catholic pharmacist who would not dispense Plan B.
- An OB-GYN who would not perform elective abortions.
- A nurse who would not assist with physician-assisted dying.

The debate asked when, if ever, the law should protect such refusals. Karsan's case raises a stranger version of the question: when, if ever, should the law protect *provision*? The philosophical materials built for the first question travel awkwardly into the second.

The integrity argument for conscientious refusal

The foundational defense of conscience clauses, given its fullest form by the bioethicist Mark Wicclair, holds that being asked to perform an act one believes to be deeply wrong is being asked to damage one's own moral integrity. Moral integrity is itself a basic human good that medicine should respect (Wicclair 2000). A pharmacist who believes emergency contraception ends a human life is not, on this view, merely expressing a preference. She is being asked to act against her core identity.

Argument

The Integrity Argument for Refusal

1. Moral integrity — the consistency between a person's deeply held moral commitments and her actions — is a basic component of human well-being.
2. Compelling a clinician to perform an act she sincerely believes to be seriously wrong damages that integrity.
3. The state and the profession should not damage a clinician's integrity unless doing so is necessary to prevent serious harm to others.
4. Therefore, clinicians have a defeasible right to refuse to participate in legal procedures they find morally abhorrent.

The disputed premise is the third, which depends on what "serious harm to others" amounts to. If a refusal merely inconveniences a patient who has other options, the right to refuse holds. If a

refusal denies a patient timely access to legal, indicated care — particularly in a region with no alternative provider — the right runs out. That is where the principal reply lives.

Stahl and Emanuel's reply: physicians are not conscripts

The bioethicists Ronit Stahl and Ezekiel Emanuel argue that the comparison to military conscientious objection breaks down at the point of voluntariness. Soldiers are drafted. Physicians are licensed. Medicine is a profession one enters voluntarily. The entry deal goes like this: a clinician accepts fiduciary obligations to patients in exchange for state-granted privileges. To accept the license and then refuse the obligations is to want the benefit without the burden (Stahl and Emanuel 2017).

Argument

The Stahl-Emanuel Reply

1. Medicine is a public profession whose members voluntarily accept fiduciary obligations to patients in exchange for state-granted privileges.
2. Those obligations include the duty to provide legal, professionally indicated care that the patient requests, regardless of the clinician's personal moral views.
3. Clinicians who cannot in conscience meet those obligations should leave the relevant specialty, not import their personal morality into the patient encounter.
4. Therefore, conscience clauses that exempt clinicians from providing legal, indicated care are illegitimate.

The soft spot here is the third premise. It works smoothly for a young medical student choosing a specialty. It works clumsily for a thirty-year veteran of obstetrics whose state legislature changes the rules underneath her. The argument also threatens to swallow more than its authors may have intended. The same logic would seem to compel participation in:

- MAID (medical aid in dying).
- Capital punishment medical clearances.
- Any other lawful practice the state designates as professionally indicated.

Brock's counter-reply: the conventional compromise

The philosopher Dan Brock proposed, well before *Dobbs*, what he called the “conventional compromise.” A clinician may refuse personally, but must meet three conditions:

1. Disclose the refusal in advance.
2. Refer the patient to another qualified provider.
3. Not abandon the patient mid-care.

The refusing pharmacist hands the prescription to a colleague. The refusing OB-GYN provides the name of a clinic across town. Brock then anticipated and rejected the “complicity” objection

— that referring is itself moral participation — by arguing that the moral weight of a referral is too thin to constitute the kind of agency that integrity protects ([Brock 2008](#)).

The conventional compromise was, for two decades, the working consensus of American medical ethics. It held because it assumed a background condition: that a willing provider was *somewhere*. In a country where abortion was legal everywhere and contraception was available at any pharmacy, the refusing clinician could always point to a door. *Dobbs* removed the assumption. In a state with no in-state legal provider, the obligation to “refer” collapses into a duty to direct the patient to break the law, leave the state, or wait.

The reframing: conscientious provision

Here the question shifts from refusal to its mirror image, provision. The classical defenders of conscientious refusal — Wicclair, Brock, even the dissenters Stahl and Emanuel — were arguing about a different setting. In their setting, the contested care was legal, and the contested clinician was the one declining to provide it. Lisa Harris and her co-authors at the *Hastings Center Report* describe the post-*Dobbs* clinician as living under a “sword of Damocles.” The contested care is now criminalized. The contested clinician is the one who provides it ([Harris et al. 2024](#)). The integrity argument has not gone away. It has changed direction.

The strongest version of the inversion comes from Isa Ryan, Ashish Premkumar, and Katie Watson in the *AMA Journal of Ethics* ([Ryan et al. 2022](#)). They draw a distinction the classical literature mostly elides:

- **Personal morality** — the convictions one holds as a private individual.
- **Role morality** — the ethical obligations one takes on by accepting a professional role.

The pre-*Dobbs* conscience clause is a personal-morality protection. The pharmacist’s private view of emergency contraception is shielded from the role’s demands. Ryan and her co-authors argue that the post-*Dobbs* landscape now demands the *role-morality* analogue. Codes of professional ethics should protect clinicians who provide medically indicated care that their state has criminalized. That provision is what the role itself requires. In their words: “Codes of medical ethics should be updated to state that, where abortion is illegal, breaking the law to provide care for reasons of conscience is not unethical.”

Argument

The Argument for Conscientious Provision

1. Personal morality grounds conscientious refusal: a clinician may decline care she sincerely believes is wrong, on the same integrity grounds the refusal tradition recognizes.
2. Role morality grounds professional obligation: by accepting the role of obstetrician or emergency physician, a clinician takes on duties to her patients that are not reducible to her private moral views.
3. When state law criminalizes care that role morality requires, role-based conscience generates the same kind of integrity claim as personal-based conscience does — pointed

toward provision rather than refusal.

4. Therefore, the profession should protect conscientious provision on par with conscientious refusal, including, where necessary, by recognizing that providing standard-of-care medicine in defiance of an unjust law is itself an ethical act.

A defender of the asymmetry will press on the third premise. Refusal, the objection runs, is a negative claim — *leave me out of it* — that can be honored without anyone breaking the law. Provision is a positive claim that, where care is illegal, *requires* law-breaking. That is too much to ask the profession to underwrite. Ryan and her co-authors anticipate the objection. They argue that the asymmetry is itself an artifact of the prior legal regime, not a deep feature of conscience. What the profession underwrites is not law-breaking as such. It is the role's commitment to patient welfare when the law fails it.

One case from outside the United States shows what is at stake. In October 2012, Savita Halappanavar, a thirty-one-year-old dentist living in Galway, was admitted to University Hospital Galway at seventeen weeks of pregnancy with an inevitable miscarriage. Her membranes had ruptured. The fetus was not viable. Irish law at the time prohibited abortion unless the woman's life was at risk. The medical team waited several days for the fetal heartbeat to stop on its own. Halappanavar developed sepsis. By the time the pregnancy was terminated, the infection was uncontrollable. She died on October 28, 2012. The case galvanized Irish public opinion and led to the 2018 referendum that repealed the constitutional ban. Damla Karsan, in Houston in December 2023, was practicing in a legal environment that posed the same demand Halappanavar's physicians had faced in 2012: to wait until a patient was sick enough to qualify.

The move that hasn't ended

What follows from the inversion is structural, not symmetrical. Pre-*Dobbs* conscience clauses protected individual clinicians from the demands of the profession. The new conscience question runs through *institutions* and *jurisdictions* — entities that classical bioethics did not theorize as conscience-bearers at all. Three current pressure points show the shape of the problem:

- **Catholic hospital systems.** Roughly one in six U.S. acute-care hospital beds is now in a Catholic system, and in some states the share is much higher. Catholic hospital directives prohibit sterilization, most forms of contraception, and abortion even in cases like Cox's. Clinicians who work in those systems are bound by institutional conscience whether or not they share its commitments.
- **State shield laws.** A growing patchwork of state shield laws — New York, Massachusetts, California, and others — now protect clinicians who mail mifepristone into states that ban it. This poses the constitutional question of whether a state may reach across its borders to prosecute a doctor whose conduct is legal where she stands. It is no longer hypothetical: Dr. Margaret Carpenter, a New York physician, has been criminally indicted in Louisiana and hit with a six-figure civil judgment in Texas, while New York has refused extradition and, in October 2025, declined to enforce the Texas judgment. One physician is

simultaneously a felon, a debtor, and a protected practitioner, depending on which state you ask.

- **EMTALA conflicts.** The federal Emergency Medical Treatment and Labor Act, which requires hospitals to stabilize patients in emergencies, came into direct conflict with state bans — and the conflict was never resolved. The Supreme Court dismissed *Moyle v. United States* as improvidently granted in 2024, the federal government dropped the Idaho suit in March 2025, and HHS rescinded its EMTALA emergency-abortion guidance in May 2025. The question was not answered so much as withdrawn, leaving clinicians in ban states with less federal clarity than they had in 2023.

These are not problems Wicclair or Brock anticipated. Their framework was built for the individual clinician with a private moral commitment. It is now being asked to mediate between four very different parties:

- State legislatures.
- Federal statutes.
- Religious health systems.
- Clinicians whose oaths now conflict with their licenses.

The question that remains is no longer whether conscience matters in medicine. It is whose conscience the law should recognize when the conscience of a state, a hospital, and a physician all point in different directions.

Discussion Questions

1. Write out the **Integrity Argument for Refusal** and the **Argument for Conscientious Provision** in your own words — premises first, then the conclusion. The two arguments share one premise. Identify it. Then say whether the post-*Dobbs* legal environment changes which side that shared premise now supports.
2. Stahl and Emanuel argue that clinicians who cannot meet a profession's obligations should leave the specialty. Does that response work for Damla Karsan — who completed her training years before Texas changed its abortion law? Give the strongest version of Stahl and Emanuel's reply to Karsan. Then give the strongest version of Karsan's reply back.
3. Brock's **conventional compromise** says a clinician may refuse to provide care she objects to, as long as she refers the patient to someone who will. Brock's rule depends on one assumption about the world that has to be true for the rule to work. What is that assumption? Describe a present-day case where the assumption no longer holds. Does the compromise still work in some modified form, or does it have to be abandoned?
4. You are a labor-and-delivery nurse in a Catholic hospital in a state with a near-total abortion ban. An ER patient is admitted at seventeen weeks. Her membranes have ruptured. The fetus still has a heartbeat, likely to stop on its own in a day or two. Her white-blood-cell count is rising — an early sign of infection. Hospital policy and state law both require

waiting. Your professional judgment tells you the patient is heading toward sepsis. What do you do? Which argument from the case is doing the work in your answer?

5. Catholic hospital systems describe themselves as institutions with a conscience. Many state legislatures describe their abortion statutes as expressions of the moral conscience of the people who voted for them. The integrity argument was built for *individual* people with consciences. Does it carry over to an institution or to a state? Pick one and defend your answer.

Further Reading

- Charlotte Alter, “How Kate Cox Became a Reluctant Face of the Abortion-Rights Movement,” *Time*, March 27, 2024 — the long-form profile.
- Lisa H. Harris et al., “Beneath the Sword of Damocles: Moral Obligations of Physicians in a Post-Dobbs Landscape,” *Hastings Center Report*, May–June 2024.
- *In re State of Texas*, 23-0994 (Texas Supreme Court, Dec. 11, 2023) — the per curiam ruling, brief and worth reading in the original.
- Kitty Holland, *Savita: The Tragedy That Shook a Nation* (Transworld Ireland, 2013) — the investigative account of the Halappanavar case.
- Ronit Y. Stahl and Ezekiel J. Emanuel, “Physicians, Not Conscripts,” *New England Journal of Medicine*, April 6, 2017.

Adriana Smith: Brain Death, Pregnancy, and Bodies the Law Has No Box For

“The decision should have been left to us” — a bioethics case study

At a Glance

- **Who:** Adriana Smith (1994–2025), a thirty-year-old registered nurse in Atlanta. Her mother, April Newkirk, became her decision-maker. Her partner, Adrian Harden, was the baby’s father.
- **What:** In February 2025, Smith went to a hospital with severe headaches and was sent home. The next morning she was found gasping for air. Blood clots had flooded her brain. She was declared brain-dead — legally dead — at nine weeks pregnant. The hospital kept her body on machines for about four months, over the family’s objection, citing Georgia’s abortion law. Her son Chance was delivered prematurely in June. Smith’s body was removed from support four days later.
- **Where / When:** Emory University Hospital Midtown, Atlanta, Georgia, February–June 2025. A custody fight over Chance ran into December 2025.
- **Why it matters:** This looks like an abortion case but is not one. It forces three questions a clinician can meet on any shift:
 - When a patient is already dead, who decides what happens to her body — and to the pregnancy inside it?
 - Why does this situation fit *none* of medicine’s familiar categories — organ donation, withdrawal of care, or research?
 - When the law is unclear, *whose* body ends up absorbing the doubt?
- **Pair with:** Autonomy, Paternalism, and Reproduction (Part A — Paternalism and Its Limits)

The Case

Adriana Smith was thirty, a nurse, and nine weeks pregnant. In early February 2025 she went to Northside Hospital in Atlanta with a severe headache. She was given medicine and sent home. No scan of her brain was done. The next morning her partner, Adrian Harden, woke to the sound of her gasping for air and called 911. At Emory, a scan showed several blood clots in her brain.

Her mother gave permission for a procedure to relieve the pressure, but doctors said it could not help. The injury was too far gone. Within a short time, Smith met the criteria for brain death.

Background

What brain death is. A person is brain-dead when the *entire* brain, including the brainstem, has permanently and completely stopped working. Under the Uniform Determination of Death Act — the model law in Georgia and almost every other state — brain death is legal death. It is the same, in law and in medicine, as the heart stopping for good. A brain-dead patient is not in a coma, not unconscious, and not “on life support” in the usual sense. The patient has died. A ventilator and drugs can keep blood moving through the organs of a body whose owner is gone.

So the hospital was not keeping Adriana Smith alive. She was dead. The machines kept her organs working for the sake of the nine-week fetus she carried. Her mother, April Newkirk, later told reporters what the staff had said: Georgia law required them to continue support because the fetus had a detectable heartbeat. The family was given no choice. The plan was to maintain Smith’s body until roughly thirty-two weeks, when the baby would have a fair chance outside the womb. “We didn’t have a choice or a say about it,” Newkirk said. “We want the baby. That’s a part of my daughter. But the decision should have been left to us — not the state” (Lewis et al. 2025).

The law the hospital pointed to was the LIFE Act.

Background

The LIFE Act (O.C.G.A. § 16-12-141). Georgia’s 2019 statute bans most abortions once a “human heartbeat” can be detected — about six weeks. It defines an “unborn child” as a member of the human species carried in the womb, and it treats that child as a legal person for more than abortion. Under Georgia law, a fetus with a detectable heartbeat can be claimed as a tax dependent and can be the subject of a wrongful-death lawsuit. Legal scholars describe the LIFE Act as, at its core, a *fetal-personhood* law, with the abortion ban as one branch of it (State of Georgia 2019).

For months the machines ran while the fetus grew. Reports later said scans had found possible problems with the fetus. The family, by their account, was never offered a way to stop. In mid-May, with the pregnancy at about five months, Newkirk went to the local press. The story was national within two days.

It drew a coalition you would expect — abortion-rights groups, reproductive-justice organizers, civil-liberties lawyers. It also drew one you might not. Several abortion opponents said the law did **not** require what Emory was doing. The state’s Republican Attorney General, Chris Carr, put it in writing.

Background

Carr’s statement, May 2025. “There is nothing in the LIFE Act that requires medical professionals to keep a woman on life support after brain death. Removing life support is not an action ‘with the purpose to terminate a pregnancy.’” Georgia House Republican leaders agreed, calling the law “completely irrelevant” to Smith’s case ([Office of the Attorney General of Georgia 2025](#)).

Emory did not change course. The hospital said only that it used clinical experts, the medical literature, and legal guidance to make decisions “in compliance with Georgia’s abortion laws and all other applicable laws” ([Lewis et al. 2025](#)). In other words: the Attorney General’s opinion was not the hospital lawyers’ opinion, and the lawyers were the ones in charge.

On June 13, 2025, surgeons delivered Chance by emergency cesarean. He was born at about six months of pregnancy — *extremely premature* by World Health Organization terms — weighing one pound, thirteen ounces. (Accounts of his exact gestational age vary.) Four days later, Emory removed Adriana Smith’s body from support. A funeral followed that month.

The question her case raises is not whether anyone broke the law. The Attorney General said the hospital was not forced to do this. The question is harder: *what kind of medical act was Emory performing for four months, and what rules should govern it?*

Eight People in One Room

Before taking sides, it helps to slow down and ask a plain question that nurses and ethics committees ask all the time: **who has a stake here, and what does each one want?** This is called a **stakeholder analysis**. You list everyone the decision touches, name their interest, and note where those interests collide. It does not tell you the answer. It keeps you from missing someone — which is exactly what the family says happened here.

The stakeholders in Adriana Smith’s case:

- **Adriana Smith herself.** She was dead, but the dead still have interests we honor — in dignity, and in not having their bodies used in ways they would have refused. She left no advance directive, so no one knew her wishes.
- **The fetus, later the baby Chance.** An interest in being born and in a decent life. But scans suggested possible harm, and his odds were uncertain.
- **April Newkirk, the mother.** Adriana’s legal decision-maker. She wanted the baby — but wanted the *choice* to be the family’s. She called the months of waiting “torture.”
- **Adrian Harden, the father.** Deeply invested, present from the first 911 call. He had almost no legal standing the whole time.
- **The bedside nurses and doctors.** Many were Smith’s own former colleagues. They gave months of intensive care to a body that could not get better, with no clear goal and real moral distress.
- **The hospital as an institution.** Worried about prison time for its doctors and uncapped lawsuits under a personhood statute.

- **The state of Georgia.** A stated interest in protecting fetal life — voiced loudly by some officials and waved off by others.
- **The public.** Watching a case that would shape how the next family is treated.

The list points to a recurring tension in the case. The person with the strongest legal voice — the hospital’s lawyer — had the least personal stake. The people with the deepest personal stake — the mother and father — had little or no legal voice.

Why This Is Not an Abortion

The first thing to get straight is the one most headlines got wrong. This was not an abortion case, and that is not a technicality. It turns on a distinction that runs through all of ethics: the difference between what you **intend** and what you merely **foresee**.

The LIFE Act defines abortion as an act done “with the *purpose* to terminate a pregnancy.” That word, *purpose*, is the hinge. When doctors stop machines on a brain-dead body, what is their purpose? Standard medical practice says: to stop futile treatment on a patient who has died. The death of the fetus is a foreseen side effect of that — sad, but not the *aim*. On this reading, the act is simply not an abortion, because the purpose is different (Watson 2025).

This is a real-world use of a principle called the **Doctrine of Double Effect (DDE)**.

Background

The Doctrine of Double Effect. An old idea, often traced to Thomas Aquinas, that an act with two effects — one good, one bad — can be permissible even if the bad effect is certain, *provided* you aim at the good one and merely tolerate the bad one. A doctor may give a dying patient enough morphine to ease pain even if it shortens life, because the aim is comfort, not death. Same outcome, different intention, different moral (and legal) verdict.

DDE explains the strange coalition. It let abortion opponents say the ban did not apply *without giving up the belief that the fetus is a person*. You can hold that the fetus is a full person and still say: stopping a dead woman’s machines is not an act *aimed* at ending the pregnancy. That is why even some pro-life voices agreed disconnection was permissible.

Argument

The “Not an Abortion” Argument (the Attorney General’s position)

1. The LIFE Act bans only acts done *with the purpose* to terminate a pregnancy.
2. Stopping support for a brain-dead patient aims at ending futile treatment on a dead body; the fetus’s death is foreseen, not intended.
3. Therefore the LIFE Act does not forbid — and does not even speak to — stopping support, and the family’s wishes should govern.

Here is the catch, and it is the heart of the hospital's fear. **DDE is controversial, and "intent" is slippery.** Critics of the doctrine have long argued that the line between intending a harm and foreseeing it is easy to blur and easy to game. From the outside, the two acts look identical: in both, machines come off and the fetus dies. The only difference lives inside someone's head — and a courtroom is exactly the place where someone else gets to second-guess what was really in your head.

That is precisely what Emory's lawyers seem to have worried about. It is one thing for the Attorney General to *announce* that the purpose is innocent. It is another to bet a doctor's freedom on a future prosecutor — or a wrongful-death jury — agreeing about what that doctor "really meant." Under the LIFE Act, a physician can face up to ten years in prison, and the institution faces lawsuits with no dollar cap. So the soft spot in DDE — that intent is hard to prove — was not a seminar puzzle for Emory. It *was* the legal risk. The hospital read every ambiguity in the most defensive direction, because the doctrine that should have protected it was one a hostile reader could attack.

The contrast with Kate Cox's case in Texas is instructive. There, a single physician, Damla Karsan, put her own name on a lawsuit and accepted the personal legal risk. Emory, by contrast, was an institution weighing liability rather than an individual acting on conscience. A similar kind of statute produced opposite choices.

Almost, But Not Quite

If this is not an abortion, what is it? The honest answer is that it does not fit any box medicine already has. It looks like several familiar things and turns out to be none of them. That is what makes it genuinely hard.

It is almost like organ donation — but the analogy breaks at both ends. We routinely use a dead person's body to help others. The reason that works, ethically and legally, is that organs are treated almost like *property*: things that can be given, with consent, under gift law. We have a whole system of donor cards, registries, and family authorization. But a fetus is not property. Under the LIFE Act it is a *person*, with standing to sue and to be claimed on taxes. You cannot "donate" it, and you cannot discard it, the way you would a kidney. So the consent-and-property machinery that settles organ cases has nothing to grab onto. And it breaks from the other side too: organ donation needs the dead person's prior *yes*. Adriana Smith never gave one. No one used her like a registered donor; they used her without any donation at all.

It is almost like withdrawing care from a dying patient — but she was not dying. She was dead. Normally a family can refuse life-sustaining treatment, and the law backs them. Georgia even has an Advance Directive Act for exactly this. But that law only applies to a patient with "a terminal condition or state of permanent unconsciousness." Brain death is neither. It is *death*. A dead person cannot have treatment "withdrawn," and an advance directive does not apply to her, because directives are instructions from the living ([Watson 2025](#)). The familiar end-of-life framework simply switches off.

It is almost like research — and here the experts openly disagree. The bioethicist Arthur Caplan argued that keeping a brain-dead body pregnant for months has almost no evidence base, no settled protocol, and no proven track record. A medical act like that, he said, is not *treatment* at all — it is closer to an *experiment*. And experiments on human subjects require informed consent, which no one here could give and no one sought.

Argument

The Experimentation Argument (Caplan)

1. A medical act with essentially no evidence base, no established protocol, and no prior outcomes is *research*, not ordinary care.
2. Research on a human subject requires informed consent — from the subject or a legal surrogate.
3. Long-term support of a brain-dead pregnant body meets the test for research, and no such consent was sought for Smith.
4. Therefore Emory performed *unconsented human experimentation*, which research ethics forbids no matter what an abortion law says (Caplan 2025).

But this claim is disputed. A 2025 review in *The American Journal of Bioethics* by Ariane Lewis and colleagues pushes back hard. They count **at least forty-seven reported cases** of brain death in pregnancy since 1982, and they point to detailed international guidance — the World Brain Death Project even publishes a management checklist for it. On their account, this is “not novel or experimental.” It is rare and difficult, but known (Lewis et al. 2025).

So who is right? The two sides may not fully contradict each other. Caplan could mean that the *four-month length*, and the post-*Dobbs* legal pressure, were unprecedented even if brain death in pregnancy is not. Lewis could be right that the basic practice has decades of cases behind it. Underneath the disagreement sits a deeper question: **what exactly makes a medical act count as “experimental”?** It could be any of three things:

- That it has never been done before.
- That it has no settled protocol.
- That there is no good evidence it works.

These come apart. Many accepted treatments are short on evidence, yet no one calls them experiments. Where the line falls decides whether Caplan’s powerful argument lands.

Even declaring brain death is done differently in pregnancy. One more sign this case sits outside the normal categories: the usual brain-death exam includes an “apnea test,” which briefly stops the ventilator to see if the patient breathes. Recent work suggests this test may be risky for a fetus, so some experts say it should be skipped in pregnancy in favor of blood-flow scans. Some hospital policies have even barred declaring brain death in a pregnant patient at all until the fetus is delivered (Lewis et al. 2025). Here, the pregnancy affects how death itself is determined.

When the Baby Was Born, the Problem Moved

The trouble did not end when Chance was born. It just changed shape — and exposed a second gap in the same law.

The LIFE Act had treated the *fetus* as a person with powerful standing: enough to keep a dead woman on machines for months, enough for tax and wrongful-death purposes. But when Chance was actually born, his biological father, Adrian Harden, discovered he had **no automatic legal rights to his own son at all**. Because Harden and Smith had never married, Georgia law gave an unwed father no parental standing by default — even with a positive DNA test, and even though the baby’s mother was dead. His attorneys warned that Chance could be placed in foster care without a court order ([Dukes 2025](#)).

So Harden had to sue for custody of his own child. The fight stretched across most of the year:

- **August 2025** — he filed an emergency petition for legitimation and custody.
- **September 2025** — a court granted temporary custody, letting Chance go home with him.
- **December 2, 2025** — a judge issued the final order granting sole custody.

That last date was nearly six months after his son was born, and ten months after Adriana collapsed.

These two outcomes come from different bodies of law. The LIFE Act is fetal-protection law; an unwed father’s rights are governed by Georgia’s family and legitimation statutes, which long predate it. There is no logical contradiction between them. But placed next to each other in a single family’s story, they show how unevenly the law assigns standing. The fetus had standing the moment a heartbeat was detected. The biological father, with a positive DNA test and the mother deceased, had none until a court granted it months later.

Whose Body Absorbs the Doubt

A final factor is harder to measure but well documented. Georgia’s LIFE Act is, on its face, race-neutral; it says nothing about race. Yet the way ambiguous cases actually get resolved often tracks *who the patient is* — something that, in principle, should make no difference to the outcome.

The pattern is documented, not merely suspected. Black women in the United States are three to four times more likely to die from pregnancy-related causes than white women, and studies find racial gaps in how quickly pregnant patients are even seen in emergency rooms ([Lewis et al. 2025](#)). Adriana Smith was sent home with a headache and no brain scan; whether bias played a role in that first visit is something her own mother raised.

The deeper worry is structural. When a law is ambiguous and the penalties are severe, *someone* has to absorb the doubt — and it is rarely distributed evenly. Hospitals with thin budgets and many low-income patients have the least room to risk a lawsuit, so their lawyers read every gray area in the most defensive way. Wealthier hospitals have more freedom to read the same gray

area in the patient's favor. The same statute can land very differently depending on whose body it lands on. The reproductive-justice group SisterSong, a longtime opponent of the LIFE Act, made Smith's case a rallying point for exactly this reason ([SisterSong Women of Color Reproductive Justice Collective 2025](#); [Roberts 1997](#)).

None of this decides whether Emory was right. It adds a further consideration: when the rules are unclear, the cost of that uncertainty is not shared equally. It tends to fall hardest on the hospitals — and the patients — with the least room to absorb legal risk.

Discussion Questions

1. Write out **The “Not an Abortion” Argument** in your own words, premises first. Which single premise was Emory's legal team most afraid to rely on, and why?
2. The case turns on the gap between *intending* a death and *foreseeing* it. Describe one other medical situation where that gap matters. Is the distinction real, or just a way to feel better about the same act?
3. Caplan calls the four months “experimentation”; Lewis calls it rare-but-known care. Pick the one feature you think best decides whether a medical act is “experimental,” and use it to judge Smith's case.
4. You are the night-shift charge nurse in a Georgia ICU. A patient is declared brain-dead at eleven weeks pregnant. Her sister, the legal decision-maker, asks that support be stopped; the hospital lawyer has not called back. What do you say in the meeting, and which idea from this case is doing the work?
5. The law gave a nine-week fetus more standing than it gave the baby's living father. Propose one change to fix that mismatch — then name one thing your change would *not* fix.

Further Reading

- Ariane Lewis, Gwendolyn Quinn, and Kimberly Mutcherson, “Ethical Controversies in the Adriana Smith Case in Georgia,” *The American Journal of Bioethics* (2025). The clearest medical-ethical overview; the source for the claim that brain death in pregnancy is rare but not new.
- Arthur Caplan, “The Adriana Smith Case Unfolding in Atlanta Raises Many Questions,” *Bioethics Today* (May 2025). Short, sharp, and the origin of the “experimentation” framing.
- Katie Watson, “Brain Death in Pregnancy — Abortion, Advance-Directive, or End-of-Life Law?” *New England Journal of Medicine* 393:4 (2025). A three-page tour of why the case fits no legal box.
- Hilde Lindemann Nelson, “The Architect and the Bee: Some Reflections on Postmortem Pregnancy,” *Bioethics* 8:3 (1994). The philosophical roots of the worry about using the dead.

- The Wikipedia article “Adriana Smith pregnancy case” — unusually well-sourced for a factual timeline.

He Jiankui and the CRISPR Twins: Germline Editing, Consent, and the Limits of Scientific Self-Regulation

‘Lulu and Nana’ — a bioethics case study

At a Glance

- **Who:** He Jiankui (b. 1984), a Chinese biophysicist; “Lulu” and “Nana,” twin girls he claimed to have edited as embryos; a third child, sometimes called “Amy,” born in 2019.
- **What:** He used CRISPR-Cas9 to disable the *CCR5* gene in human embryos and implanted them, producing the first known live births of gene-edited babies.
- **Where / When:** Shenzhen, China. The experiment ran from 2017 to 2018. He revealed it on November 25, 2018, twenty-four hours before the Second International Summit on Human Gene Editing opened in Hong Kong. He was sentenced to three years in prison in December 2019 and released in 2022.
- **Why it matters:** This is the first known clinical use of human **germline editing** — changes to DNA that pass to every cell of the resulting person and to their descendants. The case is a study in what a careful, decade-long international **near-consensus** can and cannot do when one ambitious actor decides not to wait.
- **Pair with:** Genomics & Enhancement

The Case

The world found out on a Sunday night. On November 25, 2018, the technology reporter Antonio Regalado published an article in *MIT Technology Review* describing a Chinese clinical trial that was, the title said, “creating CRISPR babies.” Within hours the Associated Press confirmed the story: a researcher named He Jiankui had used gene editing on human embryos, implanted them, and a pair of twin girls — he called them Lulu and Nana — had been born about a month earlier. He posted five short YouTube videos the same night. In one of them, against a clean white wall, he explained calmly that he had “edited a gene” to give two healthy newborn girls “the ability to resist HIV” (Greely 2019).

The Second International Summit on Human Gene Editing was scheduled to open in Hong Kong the next morning. He was on the program. He had not told the organizers what he was about to say.

Background

Some vocabulary the story turns on. **CRISPR-Cas9** is a programmable tool that scientists use to cut DNA at a chosen spot. A short guide molecule finds the target sequence; the Cas9 enzyme makes the cut; the cell repairs the cut, and the edit takes effect.

A change made to one tissue in one person — say, to a patient's bone marrow — is a **somatic edit**. It is not inherited. A change made to an early embryo (or to sperm or egg cells) is a **germline edit**. It ends up in every cell of the resulting person, including her own eggs or sperm, and so passes to her children, her grandchildren, and onward.

Somatic editing is now standard medicine. The FDA approved the first somatic CRISPR therapy, **Casgevy** for sickle-cell disease, in December 2023. Germline editing in human embryos meant for live birth has been banned for clinical use in nearly every country with the relevant law. The He experiment is the only known exception.

He Jiankui was 34. He had a physics doctorate from Rice University and a postdoctoral fellowship at Stanford with Stephen Quake, where he had worked on DNA sequencing technology. In 2012 he had returned to China under the Thousand Talents recruitment program and joined the new Southern University of Science and Technology in Shenzhen. He started a sequencing company, Direct Genomics, in 2012, with the Nobel laureate Craig Mello on its scientific advisory board. By 2017 he had decided to do something his sequencing background did not qualify him for. He would use CRISPR to make HIV-resistant children.

His recruitment plan ran through an AIDS support organization in Beijing called Baihualin. Eight Chinese couples agreed to take part. In every case the father was HIV-positive and the mother was not. He paid each couple about 280,000 yuan, roughly \$40,000, to cover the procedures. The consent form they signed was twenty-three pages long. Its first sentence described what they were joining as “an AIDS vaccine development project.” It said the goal was to help produce “HIV-resistant infants” (Greely 2019). The form spent more pages on the financial liabilities that would attach if the parents talked to anyone about the project than it spent on what gene editing actually is.

He's target was the **CCR5** gene, which encodes a protein on the surface of certain immune cells. Most strains of HIV use the CCR5 protein as the door through which they enter the cell. About 1% of people of European ancestry are born with two copies of a natural variant of the gene called CCR5Δ32, in which a 32-letter stretch is deleted. People with two copies of CCR5Δ32 are largely resistant to those strains of HIV. He's plan was to use CRISPR to make a similar deletion in human embryos.

Background

The medical problem was already solved. Couples in which one partner is HIV-positive and the other is not have safely had biological children for years. The standard methods are sperm washing followed by intrauterine insemination or IVF, with antiretroviral therapy for the positive parent. Transmission rates are effectively zero. There was no medical need for any of these eight couples to have their embryos edited (Greely 2019).

The risks were real. CRISPR can cut DNA at unintended sites — called **off-target effects** —

with unknown consequences. Edits in a single-cell embryo can fail to copy to every later cell, producing a **mosaic** child whose tissues carry different edits. And the protein He was disabling does many things besides serve as a doorway for HIV; CCR5 plays roles in cognition, in resistance to West Nile virus, and possibly in lifespan. The edits would be present in the children's eggs, and pass to any children they had.

Five of the eight women became pregnant. One miscarried. One withdrew. One carried twins to term. The twins — He called them Lulu and Nana, and the names were almost certainly pseudonyms — were born by emergency C-section in October 2018, premature but healthy. He paid the medical costs and arranged for the family to disappear from public view. The DNA tests he ran on the babies before implantation showed that he had not actually reproduced the natural CCR5 Δ 32 deletion. He had produced novel variants, with effects no one had ever studied, in two living children. He implanted the embryos anyway.

He flew to Hong Kong on November 26 and walked onto the Summit stage on November 28. The room had spent the previous twenty-four hours absorbing Regalado's article. He was visibly calm. He apologized for not telling the organizers in advance. He said he was proud of what he had done. He said the parents were "well-educated." He took questions and answered most of them with confidence that surprised his critics more than his content did.

By that evening the Chinese vice-minister of science and technology had announced an investigation. Within a week, 122 Chinese scientists had signed an open letter calling He's experiment "madness." His university said it knew nothing about the work. The hospital ethics committee whose approval He had cited said its signatures had been forged. He was placed under what looked like loose detention at his apartment in Shenzhen for most of 2019.

A second woman in the trial had been pregnant when He spoke in Hong Kong. She gave birth, also in secret, in 2019. That third child — sometimes called Amy in the limited reporting since — appears to be alive and healthy. Almost nothing else about her, Lulu, or Nana has been disclosed.

In December 2019 a Chinese court convicted He of "illegal medical practice" and sentenced him to three years in prison and a three-million-yuan fine. Two collaborators received shorter terms. He was banned for life from human reproductive research. He served his sentence and was released in 2022. He has since opened a small lab in Beijing, given interviews defending the original experiment, and announced plans for "rare disease" gene therapy work ([Cyranoski 2020](#)). The international consensus against clinical germline editing has held for now. The question is what, exactly, the consensus is *for*.

The Argument He Started

He's experiment broke no clear American law, because it was not done in America. It broke Chinese laws ambiguously enough that a court had to reach for "illegal medical practice." What it broke unambiguously was a long, careful, near-unanimous statement from the world's leading scientific bodies that this exact step should not yet be taken. The argument the case started is not

just about He. It is about what scientific self-regulation can do when one person decides not to listen.

His Defense

In Hong Kong, He defended what he had done as a serious response to a serious disease. The argument he made, paraphrased, ran something like this.

Argument

He Jiankui's Defense

1. HIV is a serious disease, especially in families where one parent is already infected.
2. CRISPR editing of *CCR5* can confer resistance to the strains of HIV that use that protein.
3. The parents understood the procedure and consented to it.
4. Therefore, this experiment was an ethically acceptable use of gene editing.

Premise 2 is where the defense falls. The edits He actually made were not the natural *CCR5*Δ32 variant. They were novel deletions and substitutions, never studied in any organism, with unknown effects on immune function and brain development. He knew this before he implanted the embryos. Lulu and Nana may or may not have meaningful HIV resistance. They will, for the rest of their lives, carry edits whose downstream effects no one can predict.

The Risk-Benefit Reply

Henry Greely, in the most thorough early analysis of the case, called it a “reckless ethical disaster” (Greely 2019). The objection is not philosophical at first. It is empirical: there was no benefit on offer, and the risks were real and lifelong.

Argument

The Risk-Benefit Reply

1. Standard reproductive medicine already prevents HIV transmission from infected fathers to their children, at very low cost and near-zero risk.
2. The *CCR5* edits actually made by He did not match the natural protective variant and had never been studied in any organism.
3. Any unintended edits, off-target effects, or mosaicism in Lulu and Nana are permanent and heritable.
4. Therefore, the risk-benefit balance of this experiment was unjustifiable on standard research-ethics criteria.

This reply does most of the work needed to condemn what He did. It does not yet say anything special about *germline* editing. The next move asks whether the case would still have been wrong

if the risks had somehow been low — and finds that consent itself does not stretch the way He needed it to stretch.

The Future Person's Consent

The parents signed a form. Lulu and Nana did not — and could not. Once they exist, they cannot un-consent to the edits they were born with. Any children they have will inherit those edits in turn. The German philosopher Jürgen Habermas had argued, fifteen years before the case, that an asymmetry runs between children who are *grown* under nature's contingency and children who are *made* under someone else's specific plan: the made child cannot, in principle, retroactively endorse the design choices that produced her (Habermas 2003).

Argument

The Future Person's Consent Reply

1. Germline edits affect every cell of the resulting person and pass to her descendants.
2. The persons most affected by a germline edit are the resulting child and her future descendants.
3. Those persons cannot consent before they exist, and cannot reverse the edit after they exist.
4. Therefore, parental consent alone is structurally insufficient to authorize a germline edit.

This is where Derek Parfit's **non-identity problem** earns its place in the case. Lulu and Nana cannot complain that they were harmed by the edits, because the only alternative was that they not exist. A different embryo would have meant a different person, or no person at all. The wrong does not sit cleanly in "harm to specific children." It sits in something subtler: in the kind of relationship He set up between himself, the parents, and two future adults who were, in Habermas's phrase, made into the project of someone else's life.

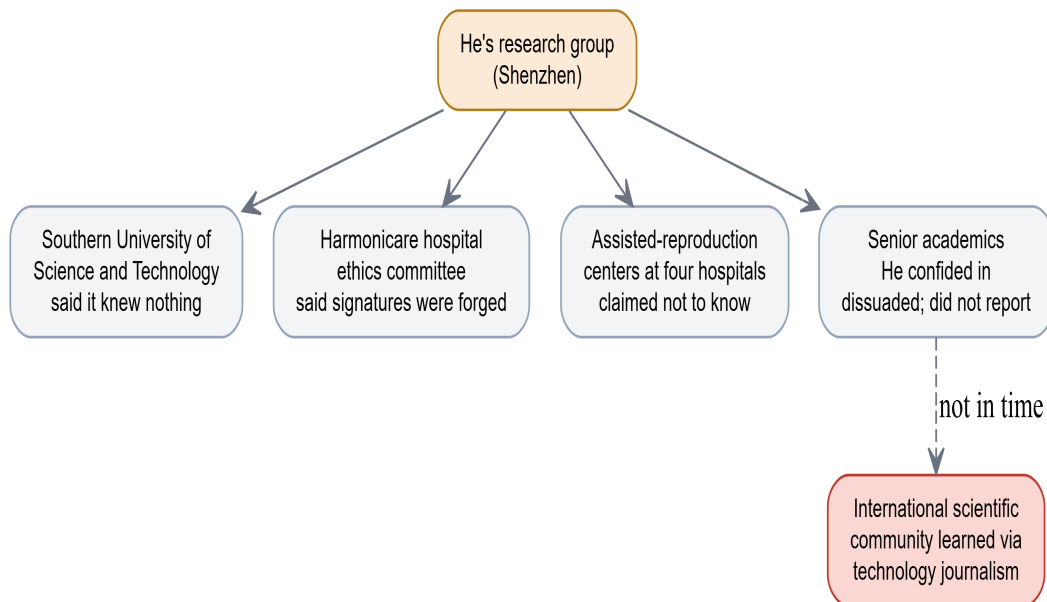
The Limits of Self-Regulation

A natural reply to all of this is: well, then, change the rules. But the rules were not the problem. The rules had already been written. They were not law. They were the carefully considered judgment of the relevant scientific community, repeated in detail.

In 2015, an international group of scientists and ethicists meeting in Napa Valley — modeled on the 1975 Asilomar conference on recombinant DNA — published a statement in *Science* calling for a moratorium on clinical germline editing while the science and the ethics matured. The First International Summit, in Washington in December of that year, concluded that clinical germline editing should not proceed "unless and until (i) the relevant safety and efficacy issues have been resolved ... and (ii) there is broad societal consensus." In February 2017, the U.S. National Academies of Sciences and Medicine issued a 328-page report, *Human Genome Editing: Science, Ethics, and Governance*, listing ten specific criteria that would need to be met before any heritable

germline edit could proceed ([National Academies of Sciences, Engineering, and Medicine 2017](#)). The Nuffield Council on Bioethics, in July 2018, set out a similar framework for the United Kingdom.

He's experiment, judged against these documents, failed almost every criterion. He had no clinical justification: the families could already have unaffected children. He had no pre-clinical safety data of the kind the Academies required. He had no oversight: his university said it knew nothing. He had no transparency: he confided in a handful of senior academics — including his Stanford postdoctoral supervisor Stephen Quake, the Nobel laureate Craig Mello, and the bioethicist William Hurlbut — and most of them either tried to dissuade him or simply did not inform anyone else when they could not ([Greely 2019](#)).



The cascade of oversight that should have caught He's experiment, and didn't. Each box is a body that, on standard accounts of research ethics, would have been expected to know, ask, or stop the work.

The diagram shows the structure of the failure. Self-regulation works when the relevant community shares norms and enforces them on each other. It assumes the deviant case does not happen, or if it does, that someone in the chain will speak. In He's case, every box failed in its own way. His university did not look. His hospital ethics committee may have been deceived or simply ignored. The senior academics he confided in tried, individually, to talk him out of it, but did not raise alarms outside their conversations. By the time the international community knew, two children had been born.

What Has and Has Not Followed

The case is now eight years old. The consensus it tested has, so far, held.

In July 2021 the World Health Organization issued a framework for global governance of human genome editing, including recommendations for a global registry of clinical trials and a mechanism for whistleblowers. The Third International Summit at the Francis Crick Institute in March 2023 reaffirmed the moratorium on clinical germline editing while explicitly endorsing the development of somatic CRISPR therapies. In December 2023 the FDA approved Casgevy, the first somatic CRISPR therapy for sickle-cell disease, at a list price of about \$2.2 million per patient. The same molecular tool that put He Jiankui in prison is now the technology behind a celebrated treatment.

And He himself, freed from prison in April 2022, has reopened a small Beijing laboratory. His public statements continue to defend the original experiment, framed now around “rare disease” gene therapy rather than HIV resistance (Cyranski 2020). More recently he has proposed editing **discarded human embryos** to test whether a particular variant protects against Alzheimer’s disease — research he says would stop short of establishing a pregnancy, and would comply with domestic and international rules. He states he has no intention of producing more gene-edited babies. Whether that reassures you may depend on how much weight his assurances carried in 2018. He has not been able to publish in any major journal, but he is not silent. He is not the only researcher in the world who could try this again. The 2021 WHO framework is a set of recommendations. The 2023 Summit was a meeting. The Academies’ criteria are guidance. None of these documents prevent a second He Jiankui. They only describe what one would, in retrospect, be guilty of.

The question that has not closed is this: is the international consensus holding because it is sound, or because no one has yet decided to break it again? Self-regulation produced eighteen months of horror in 2018 and one prison sentence in 2019. It has not, since, produced a structure that could stop the next attempt before another child is born.

Discussion Questions

1. Reconstruct **He Jiankui’s Defense** and the **Risk-Benefit Reply** in standard form. They share one premise. Identify it and write the strongest version of that premise you can give.
2. Parfit’s non-identity problem suggests Lulu and Nana cannot have been harmed by the edits, because their alternative was nonexistence. Yet most readers feel strongly that the experiment was wrong. Where, exactly, does the wrong lie if not in harm to the twins? Use one ethical concept from the dialogue to ground your answer.
3. He’s consent form described the experiment as “an AIDS vaccine development project.” Suppose he had instead used the phrase “the first heritable human gene-editing experiment.” Would proper informed consent have been possible from the eight couples? Defend your view.
4. You are a research nurse at a fertility clinic. A patient mentions, in passing, that her husband is a biologist who has been “doing some CRISPR experiments on our IVF embryos at home before transfer.” What do you do, in what order, and what authority — clinical, legal, or professional — backs each step?

5. The FDA approved Casgevy in December 2023. It uses CRISPR to treat sickle-cell disease through *somatic* editing of a patient's own stem cells. He Jiankui used the same molecular tool to alter the germline of twin girls. Drawing on the case, identify the single feature — not “intent,” not “regulation” — that does the moral work of separating these two uses of CRISPR.

Further Reading

- Henry T. Greely, “CRISPR’d babies: human germline genome editing in the ‘He Jiankui affair,’” *Journal of Law and the Biosciences* 6 (2019): 111–183. The fullest and most accessible single account, with extensive primary documents.
- *Make People Better* (Cody Sheehy, dir.), 2022. A documentary built around Antonio Regalado’s reporting, with interviews from several principals.
- National Academies of Sciences, Engineering, and Medicine, *Human Genome Editing: Science, Ethics, and Governance* (National Academies Press, 2017). The pre-He report whose criteria the experiment violated.
- David Cyranoski, “What CRISPR-baby prison sentences mean for research,” *Nature* 577 (2020): 154–155. Short, useful summary of the legal outcome and what it did and didn’t settle.

Noelia Castillo: Psychiatric Suffering, Decisional Capacity, and the Limits of Euthanasia Law

“I just want to go peacefully” — a bioethics case study

At a Glance

- **Who:** Noelia Castillo Ramos (2000–2026), a young woman from Barcelona, Catalonia, Spain.
- **What:** After a 2022 suicide attempt left her paraplegic and in constant pain, she asked to die under Spain’s euthanasia law. Her request was approved, then tied up for 601 days by her father’s legal appeals before she died at age 25.
- **Where / When:** Catalonia, Spain, 2024–2026, under the 2021 Organic Law Regulating Euthanasia.
- **Why it matters:** Castillo’s case forces the hardest question in end-of-life ethics: can a person whose suffering is bound up with **mental illness** make a genuinely autonomous, irreversible choice to die — and who gets to decide whether she can? It also reverses the usual family fight: here a **competent** patient fought *for* death while her family fought to keep her alive.
- **Pair with:** Euthanasia, Assisted Death, and Their Critics

The Case

Noelia Castillo was born in Barcelona in November 2000. Her childhood was unstable. When she was about thirteen, her parents lost their home and separated, and her mental health began to suffer. She later said she sometimes had to follow her father to bars and wait there until the early hours of the morning. She was diagnosed with two conditions that would shadow the rest of her story: **obsessive–compulsive disorder** and **borderline personality disorder**.

Background

Two diagnoses, in plain terms.

- **Obsessive–compulsive disorder** (OCD) involves intrusive, unwanted thoughts and repeated behaviors a person feels driven to perform to ease the anxiety those thoughts cause.

- **Borderline personality disorder** (BPD) brings intense, unstable emotions, a shaky sense of self, and stormy relationships. It also raises the risk of self-harm and suicide, which matters here.

Neither condition, by itself, removes a person's ability to make decisions. But BPD can drive suicidal feelings. So it sits at the center of the dispute that follows: when such a person asks to die, is that a free choice or a symptom of the illness?

From 2015 to 2019, Castillo lived in social care. She left at eighteen, as a legal adult. In interviews she said she had survived several sexual assaults that she never reported, and so they were never investigated or confirmed in court. In October 2022, days after one of these assaults, she attempted suicide by jumping from a building. She was twenty-one.

She survived the fall, but it changed everything. The injuries left her with **irreversible paraplegia** — permanent paralysis of the lower body — along with chronic physical pain, neurological damage, and deep psychological suffering. She now used a wheelchair and lived, in her own words, in constant pain.

Eighteen months later, in April 2024, Castillo formally asked to die under Spain's euthanasia law.

Background

Spain's euthanasia law. The 2021 Organic Law Regulating Euthanasia lets a capable, conscious adult request an assisted death. They must have a "serious and incurable illness" or a "serious, chronic, and disabling condition." It must cause "constant and unbearable physical or mental suffering" with no relief the person finds tolerable ([Cortes Generales \(Spain\) 2021](#)). Spain does **not** require that the patient be dying. A request must be made twice in writing, reviewed by independent doctors, and signed off by a regional committee. A doctor may then either give the lethal medication directly (**euthanasia**) or supply it for the patient to take (**assisted death**).

In July 2024, Catalonia's Guarantee and Evaluation Commission approved her request unanimously. The procedure was set for the following month. It did not happen for another 601 days.

The obstacle was her father. With the support of the Christian Lawyers Association, an ultra-conservative advocacy group, he launched a series of appeals to stop the procedure. His central argument was about her mind. He claimed that her personality disorder meant she could not make a sound judgment about ending her life. The state, he said, had a duty to protect her from a decision driven by illness.

The courts disagreed, again and again. In March 2025, a Barcelona court rejected the appeal and found Castillo capable of deciding. The High Court of Justice of Catalonia upheld her right in September 2025. Spain's Supreme Court rejected the father's appeal in January 2026, and the Constitutional Court dismissed his claim. On March 10, 2026, the European Court of Human

Rights rejected his final appeal. At every level, the courts found that Castillo had the capacity to choose.

In a television interview recorded the day before she died, Castillo defended her decision and made a point of not recommending it to anyone else. “I just want to go peacefully now and to stop suffering,” she said. “I don’t want to be an example for anyone — it’s simply my life.” On March 26, 2026, at age twenty-five, she received euthanasia at a healthcare center in Sant Pere de Ribes. Her mother asked to be in the room. Castillo chose to die alone.

Her case became a national argument before and after her death. It was debated in Spain’s Congress. Several parties spoke in support. The People’s Party and Vox criticized the outcome. Catholic bishops condemned it, with one calling it “another step toward a culture of death.” A great deal of false information also spread. Political figures claimed she had been attacked by migrant children, and many reports said she had been granted euthanasia “for depression.” Fact-checkers found both untrue (S. Jones 2026). The official basis for her euthanasia was the chronic, irreversible **physical** pain from her injuries.

That last fact is the knot at the center of the case. On paper, her euthanasia was approved for a physical, disabling condition — suffering the law plainly covers. But the reason it was fought all the way to Strasbourg was psychiatric. The claim was that a mind shaped by trauma and personality disorder cannot truly consent to its own death. The law treated her as a disability-and-pain case. Her father treated her as a psychiatric one. Both were describing the same woman.

The Argument Castillo Started

For decades, assisted-death laws drew their hardest line around a single assumption: that the suffering being relieved was *physical* and the patient’s judgment was intact. Castillo’s case removes both supports at once. Her suffering was psychiatric, and the very condition that made her eligible — long-standing depression — is one that can distort the will it torments. The dispute is whether a request to die can count as free when the illness invoked to justify it may be the thing producing the request.

The argument from autonomy

The starting point is the principle behind Spain’s law and most assisted-death laws: a competent adult may decide that her suffering is no longer bearable, and that judgment is hers to make.

Argument

The Autonomy Argument

1. A competent adult has the right to make decisions about her own body and life, including how much suffering she will endure.
2. Noelia Castillo was a competent adult enduring constant, irreversible suffering with

no relief she found tolerable.

3. Therefore, the decision to end that suffering through euthanasia was hers to make, and the law was right to honor it.

Premise 1 is widely accepted in the abstract. So, as in many euthanasia cases, the whole weight of the argument falls on **premise 2** — the claim that Castillo was genuinely competent. That is exactly where her father attacked.

The capacity challenge: symptom or choice?

Her father did not argue that suffering people should be forced to live. He argued that Castillo's particular suffering came wrapped in a mental illness that made her wish to die unreliable. On this view, a borderline personality disorder marked by suicidal urges does not just *accompany* the request to die — it may *produce* it. The wish is a symptom, not a settled preference.

Argument

The Capacity Challenge

1. A choice counts as autonomous only if it reflects the person's stable values rather than a symptom of illness.
2. Castillo's wish to die may be a symptom of her psychiatric disorders, which involve suicidality.
3. The state has a duty to protect vulnerable people from acting on such symptoms.
4. Therefore, her request should not have been honored without resolving whether it was a stable choice or a symptom.

This challenge raises a problem psychiatrists have not solved. With a terminal cancer, doctors can say with confidence that the illness is **irremediable** — beyond cure. With a psychiatric condition, predicting that suffering will never lift is far harder. The line between *preventing suicide* and *assisting death* can blur (Grassi et al. 2022). The same health system offers suicide prevention to one patient and euthanasia to another. What decides which door a person walks through?

Background

Capacity and irremediability. **Decisional capacity** is the ability to understand relevant information, appreciate how it applies to oneself, reason about options, and communicate a choice. It is decision-specific, not global: a person can have a diagnosis and still have capacity for a given decision. **Irremediable** suffering is suffering that cannot be relieved by any means the patient finds acceptable. Both concepts are easier to apply to a failing body than to a suffering mind — which is why psychiatric cases sit at the frontier of these laws.

The reply: a double standard

Castillo's defenders — including a group of philosophers writing in *The Lancet Psychiatry* after her death — argue that the capacity challenge hides a double standard. They identify three moves against psychiatric euthanasia that they regard as fallacies (Rueda et al. 2026).

- **Branding the wish as a symptom.** Treating any death wish rooted in mental illness as automatically incompetent applies a test we never apply to physical illness. When a cancer patient in agony asks to die, we take it seriously; we do not dismiss it as the disease talking. Capacity must be judged case by case.
- **The “better support” objection.** Saying that stronger family or social support might have changed her mind does not show that her actual decision was invalid. Almost any choice could be waved away by imagining kinder circumstances that never existed.
- **The “time heals” objection.** Saying trauma *might* ease with more years and more treatment imposes an open-ended waiting period on the patient — a standard of proof that no other irremediable condition has to meet.

Argument

The Anti-Double-Standard Reply

1. We accept a suffering patient's request to die when their condition is physical and judged irremediable.
2. Refusing the same request when the suffering is psychiatric, solely because it is psychiatric, treats like cases differently without justification.
3. Castillo was found to have capacity by every court and committee that examined her.
4. Therefore, refusing her request would have been unjust discrimination against the mentally ill, not protection of them.

The reply is powerful precisely because it turns the protective instinct around: shielding the mentally ill from their own choices can itself be a way of *not* taking them seriously as persons.

The vulnerability counter: a free choice, or a failed society?

But the strongest objection to Castillo's euthanasia does not deny her capacity at all. It accepts that she chose freely — and asks what she was choosing *between*.

The philosopher Loughran Butcher argues that defenders of psychiatric euthanasia answer the wrong worry. They point to safeguards: independent reviews, capacity assessments, committee approval. But safeguards only check whether a person *qualifies*. They do nothing about *why* a person reaches that point (Butcher 2025). Castillo's life held a lost home, years in institutional care, untreated trauma, poverty, and assaults that were never investigated. Each of those is a social failure, not a medical fact. A safeguard confirms she is eligible. It does not ask whether better support would have left her wanting to live.

This is also where the **disability-rights** critique enters. Castillo was not dying. She was paralyzed and in pain. Many disabled people live with such conditions for decades and do not regard them as a reason to die. Critics worry that approving her death sends a message: that a paraplegic life is a life not worth living. The British pianist James Rhodes, himself an abuse survivor, offered to pay for her physical and mental healthcare. His gesture dramatized the objection. Was death offered to her more readily than a real chance at a livable life?

Argument

The Vulnerability / Disability Argument

1. A choice is only as free as the alternatives a person actually has.
2. Castillo's alternatives were narrowed by social failures — poverty, lost housing, institutional care, untreated trauma, and inadequate support.
3. When death becomes more available than the support that would make life bearable, “choosing” death is not a clean exercise of autonomy.
4. Therefore, honoring her request may have ratified a social failure rather than respected an autonomous person.

Notice that this argument does not call Castillo incompetent. It grants everything the double-standard reply demands — and still finds something wrong. The Spanish bishops, in their own register, made a parallel point: the case showed “an accumulation of personal suffering and institutional shortcomings that calls the whole of society into question.”

The unresolved frontier

So the case ends where the law itself is unsettled. Only a handful of countries — the Netherlands, Belgium, Spain — permit assisted death for suffering that is not terminal and is partly or wholly psychiatric. Canada voted to extend its system to people whose sole condition is a mental illness, then **postponed** the change repeatedly — most recently to March 2027. In June 2026 a parliamentary committee went further and recommended excluding psychiatric-only cases **indefinitely**. Clinicians could not agree on how to tell irremediable mental illness from the treatable kind ([Grassi et al. 2022](#)). The very question Castillo's father raised is one that a whole national health system has so far refused to answer.

Her case also inverts the older end-of-life disputes. In the landmark American cases, families fought to *stop* treatment while the state insisted on preserving life. Here a competent woman fought for death while her father fought to keep her alive. The courts sided with the woman against the family. That reversal is not a footnote. It shows how far the center of gravity has shifted toward individual autonomy. It also shows why the people most worried are not only religious conservatives, but disability advocates and psychiatrists.

What remains unresolved is whether Castillo's death was autonomy honored or vulnerability abandoned. Both descriptions fit the same facts. The courts answered the narrow legal question — did she have capacity? They answered yes. They did not, and perhaps could not, answer the

larger one. Would a just society have given her something to live for before it gave her a way to die?

Discussion Questions

1. Every court that examined Castillo found she had the capacity to choose. Is a finding of capacity enough to settle whether honoring her request was *ethical*, or can a fully capable choice still be the wrong one to grant? Explain the gap, if there is one.
2. The *Lancet Psychiatry* authors argue that treating a psychiatric death-wish as automatically incompetent applies a double standard we never apply to physical illness. Is the difference between physical and psychiatric suffering a real moral difference, or just a prejudice? Defend your answer.
3. Butcher argues that safeguards check whether someone *qualifies* for euthanasia but ignore *why* they reached that point. If a person's wish to die grows out of poverty and lack of support, is their choice still autonomous? What would have to be true for it to count as free?
4. Castillo's father argued the state should have protected her from her own decision; the courts ruled that protecting her meant respecting it. When a competent adult and her family disagree this completely, whose judgment should the law privilege, and why?
5. Canada has twice postponed allowing euthanasia for mental illness alone because doctors cannot reliably judge when such suffering is irremediable. Given that uncertainty, should a country permit psychiatric euthanasia now, wait until the judgment is more reliable, or refuse it permanently? What does your answer cost?

Singapore's Health Care System: Cost, Access, Outcomes, and What the United States Can (and Can't) Learn

The model everyone admires — and reads selectively

At a Glance

- **What:** Singapore covers everyone, lives longer than almost anyone, and spends about a quarter of what the US spends — through an unusual mix of forced savings, catastrophic insurance, subsidies, and heavy government control.
- **The numbers:** Singapore spends ~5% of GDP on health; the US spends ~17.6%. Life expectancy is 83.5 years vs. 79.0 in the US (2024). 100% of Singaporeans have public coverage.
- **The shadow:** roughly a third of Singapore's population are migrant workers, who are **excluded** from government subsidies. COVID-19 exposed this sharply.
- **Why it matters:** Singapore is the system US reformers on the **left and right** both point to. This case asks whether the US can learn from it — and whether its success can be separated from the trade-offs that produce it.
- **Pair with:** Why US Health Care Costs So Much

The Case

Every American who studies health policy eventually runs into Singapore, and has the same reaction: *how?*

The United States spends close to **18% of its entire economy** on health care — about **\$14,900 per person** (2024) — and gets middling results. Singapore spends roughly **5% of GDP**, a quarter of the US share. Yet Singaporeans live to **83.5** on average, about four and a half years longer than Americans, and **everyone** is covered ([The Commonwealth Fund 2026](#)). On paper, Singapore looks like it solved the puzzle the US can't.

It did not do so by building a British-style national health service, or a Canadian single-payer plan, or a German system of insurance funds. It built something stranger.

Background

The “S+3Ms” framework. Singapore funds health care through four layers ([The Commonwealth Fund 2026](#)).

- **Subsidies:** the government pays up to 80% of the bill in public hospital wards, with the biggest subsidies for those who choose the most basic accommodations.
- **MediSave (1984):** a **compulsory savings** account. Workers must set aside **8–10.5% of every paycheck** in a personal medical account they can later spend on care.
- **MediShield Life (2015):** mandatory national insurance for **catastrophic** bills — big hospital stays and costly treatments. Everyone is in it, including people with pre-existing conditions.
- **MediFund:** a government endowment that catches those who still can't pay after the other three layers.

The design has a philosophy. Singapore wants you to feel the cost of routine care, so you spend your own MediSave dollars on small things. It pools money only for the **big, unpredictable** bills that would bankrupt a family. And it keeps a safety net under everyone who falls through.

This system was not built in one stroke. It grew piece by piece over forty years — MediSave in 1984, MediShield in 1990, MediFund in 1993, and finally MediShield Life in 2015, which closed the gaps by making coverage truly universal and lifelong. Each step added a layer rather than replacing the system, much as the US did. The difference is that Singapore was building toward a **plan**, while the US was patching an accident.

The money still comes from a mix of sources. In 2022 the government paid about **57%** of all health spending, and individuals paid roughly **a quarter out of pocket** — a higher share than in most rich countries ([The Commonwealth Fund 2026](#)). Singapore is comfortable making patients pay at the point of care, on the theory that a price you can feel is a price you think about. Whether that restraint is wise or harsh is part of what the case is about.

The system also nudges behavior in ways Americans would find startling. Public hospital wards are **tiered**: a “C” ward holds many beds in one room and is heavily subsidized; an “A” ward is a private room with little or no subsidy. You can have comfort, or you can have a big subsidy, but not both. The choice quietly sorts people by what they're willing to pay for — a form of rationing that patients impose on themselves.

Background

The part that's easy to miss. Behind the savings accounts sits a powerful government hand. Singapore's state **owns most of the acute hospitals**, controls how many are built, and **negotiates or sets the prices** of drugs and procedures. This is **supply-side control** — managing the cost and amount of care from the top, not just the demand for it. Much of Singapore's low spending comes from this control, not from the savings accounts alone ([Haseltine 2013](#)).

Then came the test no one plans for.

When COVID-19 arrived, Singapore looked like a global model. It moved early and hard. National apps — **TraceTogether** and **SafeEntry** — let officials trace contacts within hours. When community spread grew, the country imposed a “circuit breaker” lockdown. Later it ran one of the world’s most successful vaccination drives, reaching more than **96%** of eligible people. Its **case-fatality rate settled at about 0.08% — among the very lowest on Earth.**

But the worst outbreak did not happen among citizens. It tore through the **dormitories of migrant workers** — the men who build Singapore’s towers and clean its streets. Hundreds of thousands of them live packed into shared rooms, often a dozen or more to a unit. The virus moved through that crowding fast. At the peak, well over **47% of dormitory workers** had been infected, and for months they made up the overwhelming majority of Singapore’s cases ([Koh 2020](#)).

Very few of these workers **died** — most are young men, so the dormitory outbreak was enormous in cases but small in deaths. Yet they had been living in conditions that made an outbreak almost certain, in a country famous for control and order. These workers are about a third of Singapore’s population, and they are **not** eligible for the government subsidies that protect citizens.

So Singapore’s celebrated COVID numbers told a true story and hid one at the same time. The country protected its citizens superbly. It had also built a second, lower track for the people it depended on most — and the pandemic shoved that track into view. The low *national* death rate was partly a fact about Singapore’s system, and partly a fact about who was counted as fully inside it.

The Argument Singapore Started

Singapore is not really one argument. It is a screen onto which people project the system they already wanted. The exchange below tries to separate what Singapore actually does from what its admirers wish it did.

The case for copying it

Start with the appeal, because it is real and bipartisan.

Argument

The Emulation Argument

1. A just and efficient health system should deliver good health outcomes for everyone at a sustainable cost.
2. Singapore delivers near-top outcomes, universal coverage, and long lives at about a quarter of the US cost.
3. Therefore, the US should adopt the key features of Singapore’s system.

What makes this argument unusual is **who** makes it. American conservatives love **MediSave**: people spend their own money, feel the cost, and avoid waste — personal responsibility built into the system. It is the dream behind US health savings accounts. American progressives love that coverage is **universal**, that the **government sets prices** and owns the hospitals, and that no one goes bankrupt from a hospital bill. It is the dream behind a public system.

Both camps look at Singapore and see their own ideals confirmed. A system claimed as proof by people who agree on nothing is usually doing something more complicated than either story admits — and each side is describing only the part that flatters its own view.

Which model is it?

Background

The four classic models. Most countries build their system on one of four templates (Reid 2009).

- **Beveridge:** the government owns the hospitals and pays for care from taxes (the UK's NHS).
- **National Health Insurance:** care is delivered privately, but a single government insurer pays the bills (Canada).
- **Bismarck:** private providers and private nonprofit insurance “funds,” tightly regulated so everyone is covered (Germany).
- **Out-of-pocket:** there is no system; you pay for what you can afford (much of the developing world).

Singapore fits **none** of them cleanly.

- Its forced **savings accounts** look like a regulated version of *out-of-pocket* — you pay, but from a mandated account.
- Its **catastrophic insurance** (MediShield Life) looks like *National Health Insurance* for big bills.
- Its **government-owned hospitals and price-setting** look like *Beveridge*.

It is, in effect, a **fifth model** — a deliberate hybrid. In the US debate, each faction grabs the one piece that resembles its preferred model and ignores the rest. The conservative sees the out-of-pocket gear; the progressive sees the Beveridge gear. Singapore only works because it bolts them together.

The hidden engine

This is where the most common American lesson goes wrong.

The popular story is that Singapore is cheap because patients spend their own money and shop carefully. There is something to that. But it is not the main engine, and Haseltine's history of

the system makes the point plainly: Singapore's low costs come mostly from the **government's heavy hand on supply and prices** (Haseltine 2013).

Argument

The Whole-System Argument

1. Singapore's low costs depend on the government owning hospitals, limiting their number, and setting prices — not mainly on patient cost-sharing.
2. A reform that imports the savings accounts (MediSave) **without** the government's supply-side control would keep US prices high.
3. The features of Singapore's system work as an interlocking whole, not as parts you can take one at a time.
4. Therefore, "copy MediSave" is not a coherent lesson from Singapore; the price control is doing the heavy lifting.

What does that control look like in practice? The government decides how many hospitals get built and where, so supply does not balloon. It publishes the prices hospitals charge and pressures them to stay in line. It buys drugs in bulk for the whole country and negotiates hard, the way Canada or the UK does — leverage the fragmented US system never has. None of this is the savings accounts. It is the state acting as the dominant buyer and planner of care.

There is also a cultural engine Americans tend to skip. Singapore's government is openly **paternalistic**. It tells citizens how much to save, steers them toward basic wards, and ran national tracking systems during COVID that most Americans would reject outright. The savings accounts only restrain spending because a strong state stands behind them, shaping what things cost in the first place. Remove the state, and the savings accounts restrain nothing: a patient shopping carefully at American prices still gets American bills.

The shadow: who's left out

Now the hardest move, and the moral heart of the case.

A defender says: *Singapore proves you can cover everyone well and cheaply*. But "everyone" hides a choice. Singapore's superb averages rest partly on a population it does **not** fully include — the migrant workers who are barred from subsidies and were left in crowded dormitories when the virus came.

Background

Why this is a justice problem, not just a footnote. Philosophers who study health justice argue that a society's duties run to **all** the people whose labor sustains it, and that health is shaped by **social determinants** — housing, work, and income, not just clinics (MacKay and Sreenivasan 2021). On a Rawlsian view, a system should be judged by how it treats its **worst-off** members. By that test, Singapore's dormitory residents are exactly the people the numbers leave out.

Argument

The Exclusion Argument

1. A health system should be judged by how it treats its worst-off members, not only by its averages.
2. Singapore achieves part of its low cost and high averages by excluding migrant workers — about a third of its population — from full coverage.
3. Excellent averages bought partly through exclusion do not show that a system is **just**.
4. Therefore, Singapore's headline success cannot be read, on its own, as a model of justice for others to copy.

Singapore's achievement is real; the exclusion changes what it shows. The efficiency Americans admire may not be **separable** from it — tidy national numbers and a hidden second track may simply come together.

The US is in no position to feel superior. It has its own excluded groups: undocumented immigrants who get only emergency care, and the millions who fall into the coverage gap between Medicaid and the exchanges. Every health system draws a line around who fully counts. Singapore's line is just unusually visible — drawn between citizens and the workers who build the city. The US draws lines too, and its own are not obviously easier to defend.

Can it transfer?

Suppose the US wanted to learn the real lessons anyway. Could it?

Several features that make Singapore work may not cross the ocean.

- **Scale.** Singapore is a wealthy **city-state** of under six million. The US is a continent of 340 million, with fifty states and no single health authority.
- **Governance.** Singapore is run by one dominant party that can impose forced savings, set prices, and track citizens with little pushback. US politics blocks far smaller moves.
- **Culture.** Singapore's system assumes a level of trust in government, and acceptance of paternalism, that the US does not share.
- **Membership.** Singapore's model depends on a sharp line between citizens, who are protected, and non-citizens, who are not. The US draws lines too, but a model built on that exclusion sits badly with its stated ideals.

Some pieces might still travel. The US could borrow Singapore's **supply-side price control** — public price-setting and bulk drug purchasing — without forcing anyone to open a savings account. It could add **catastrophic-only** public insurance, so that no illness bankrupts a family, while leaving routine care to the market. It could fund a real safety net like MediFund for those who exhaust everything else.

But notice that each of these “lessons” is really a **value choice** dressed as a technical fix. Price control means deciding that the government, not the market, sets what care is worth. Catastrophic-only insurance means deciding that people *should* feel the cost of routine care — a stance that

looks like prudence to one person and cruelty to another. Singapore did not stumble into these answers. It chose them, and accepted the trade-offs. The US cannot borrow the answers without making the same choices.

What Singapore really offers the US is not a blueprint. It is a mirror. Each side sees the system it wanted and calls it proof. The honest lesson may be the uncomfortable one: that Singapore's enviable numbers come bundled with state control and a two-track membership most Americans say they would refuse — and that you cannot reliably buy the outcomes without buying the trade-offs.

Discussion Questions

1. Both American conservatives and progressives cite Singapore as proof of their own approach. What does that tell you about using a single foreign country as evidence in a policy debate? How should you reason about “what works elsewhere”?
2. Singapore's low costs come largely from government control of hospitals and prices, not from patient savings accounts. If the US adopted MediSave-style accounts *without* price control, what would you predict — and why?
3. Is excluding migrant workers from subsidies a **flaw** in Singapore's system, or a **feature** that helps produce its low costs and good averages? Can a system be both highly efficient and unjust?
4. Singapore's model depends on a paternalistic government, forced savings, and national health tracking. Are these acceptable prices for universal, low-cost care? Defend your line using a theory of justice.
5. Pick one element of Singapore's system you think the US *should* adopt and one it *should not*. For each, name the value (from Part A) that drives your choice.

Andrew Wakefield and the MMR-Autism Myth: Misinformation, Vaccine Hesitancy, and Public Trust

“The facts were settled, and the belief grew anyway” — a bioethics case study

At a Glance

- **Who:** Andrew Wakefield (b. 1957), a British gastroenterologist; and the millions of parents who came to fear the MMR vaccine.
- **What:** A 1998 study falsely linked the measles-mumps-rubella vaccine to autism. The claim was wrong, and later shown to be fraud. The fear it started has outlived every correction.
- **Where / When:** London, 1998; then the US, Europe, and the world, up to today.
- **Why it matters:** The science here is not in doubt. The vaccine does not cause autism. The live questions are about **vaccine hesitancy**, **manufactured doubt**, and public trust — and how a clinic or a government should respond when facts alone do not work.
- **Pair with:** Public Health Ethics — False Beliefs and the Ethics of Correction.

The Case

A pediatrician has fifteen minutes. Across the exam table sits a mother with a healthy toddler and a folder of printouts. She is kind, college-educated, and certain: she will not allow the MMR shot. She has read that it causes autism. She loves her child, and she is not careless. The doctor has met this conversation a hundred times. Where did it come from?

It came from one paper. In February 1998, the British medical journal *The Lancet* published a study by Andrew Wakefield and twelve co-authors. It described **twelve children** with autism and bowel problems. It suggested their symptoms began soon after the MMR vaccine. The paper itself was cautious. At a press conference, Wakefield was not. He told reporters that parents should consider single shots instead of the combined vaccine. The story went around the world.

What the public did not learn for years was what sat behind it. The journalist **Brian Deer** spent years taking the study apart. He tracked down the families. He compared what the parents remembered to what the records said. The two did not match. Records had been altered to fit the story (Deer 2011). Some children had shown signs of autism *before* the shot. Some had no

autism at all. One child described as healthy before the vaccine had, in fact, been delayed from birth.

Deer found the money, too. Wakefield had been paid more than **£435,000** by lawyers who were preparing to sue vaccine makers. The lawyers wanted a study that pointed at the vaccine. They got one. Wakefield had also filed a patent on a rival, single-measles vaccine — a product that could profit if the combined shot fell out of favor. None of this appeared in the paper. The study was not just wrong. It was, in Deer's word, a fraud.

The damage did not wait for the correction. In Britain, MMR vaccination fell from about 92% to below 80%. In some London boroughs it dropped further. Measles, once nearly gone, came back. In 2013, an outbreak in Swansea, Wales, infected more than 1,200 people and killed a young man. Outbreaks followed across Europe and the United States. Children who could not be vaccinated — newborns, kids with cancer — lost the shield that high vaccination had given them. The fear was abstract. The harm was not.



Figure 1: The measles rash. Before the vaccine, nearly every American child caught measles; the disease still kills more than 100,000 people a year worldwide.

US Centers for Disease Control and Prevention. Public domain.

Background

Herd immunity and R_0

A disease's **basic reproduction number**, or R_0 , is how many people one sick person infects in a group with no immunity. Measles is among the most contagious diseases known, with an R_0 near **12 to 18**. Because it spreads so easily, a community needs about **95%** of people vaccinated to stop it. That shared protection is called **herd immunity**. When vaccination dips even a little, measles is the first disease to return.

The official response came slowly but fully. In 2004, ten of Wakefield's co-authors withdrew their support for the paper's claims. In 2010, *The Lancet* retracted it. The same year, Britain's medical board found Wakefield guilty of dishonesty and struck him off the register. He could no longer practice medicine.

He did not disappear. He moved to the United States. He became a leader of the anti-vaccine movement and directed a 2016 film, *Vaxxed*. He drew large, devoted crowds. The retraction had ended his career. It had not ended his influence. If anything, being cast out by the establishment made him, to his followers, more believable.

Meanwhile the science only grew stronger. This is the part the printouts in the mother's folder leave out.

- A Danish study in 2002 followed **537,303 children**. It found no link between MMR and autism ([Madsen et al. 2002](#)).
- A larger Danish study in 2019 followed **657,461 children**, including those with autism risk factors. Again, no link ([Hviid et al. 2019](#)).
- Dozens of studies on several continents agree. The combined evidence covers millions of children.

There is also a simple reason the myth felt true. The MMR shot is given around **12 to 15 months**. The first clear signs of autism often appear around the same age. One follows the other in time, so the mind links them. But timing is not cause. When researchers compared vaccinated and unvaccinated children, autism rates were the same.

So we reach a strange place. The evidence is overwhelming. The original study was retracted and exposed as fraud. And yet, more than twenty-five years later, the fear is stronger and broader than ever. The honest question is no longer *whether* the science is settled. It is. The question is why the belief grew anyway — and what a clinician or a government may rightly do about it.

The Argument Wakefield Started

The science here was settled years ago: the study was retracted, the fraud exposed, the supposed link disproven many times over. The argument Wakefield left behind is the one that won't settle, and it sits one level below the facts. Scholars and clinicians who fully accept the science still split over two questions — why the false belief spread anyway, and what, if anything, can be said to undo it.

Just tell them the facts

The first response is the natural one. People believe a falsehood because they lack the truth. So give them the truth — clearly, often, and from trusted experts. Correct the record. Label the lies. Teach the science in school. This is the instinct behind most fact-checking and public health messaging. It is also the instinct of the tired clinician who simply hands the mother a fact sheet. The logic feels airtight.

Argument

The Correction Argument

1. People reject the MMR vaccine because they hold a false belief.
2. False beliefs are caused by missing or wrong information.
3. Giving people correct information removes the false belief.
4. Therefore, the cure for vaccine fear is better facts, clearly delivered.

The weak link is premise 3. It assumes that facts, once delivered, do their work. Sometimes they do. But the past twenty-five years are a hard test of that premise, and it often fails. The retraction, the fraud finding, the millions of children studied — all of it was delivered. The fear grew anyway. So the next question is: why doesn't correction work?

Why correction can backfire

The philosopher C. Thi Nguyen offers an answer. He separates two things that look alike. In an **epistemic bubble**, people simply miss the other side; show them the evidence and the bubble can pop. An **echo chamber** is different and far more stubborn. It is a place where members are actively taught that outside sources — doctors, the CDC, “the media” — are liars. The chamber does not hide the experts. It discredits them in advance.

Once someone believes that, a fact-check is not neutral. It is proof of the plot. Hand such a parent the retraction, and she does not see a correction. She sees the establishment closing ranks. Wakefield's own story feeds this. He was cast out by the medical board, so to his followers his exile is not disgrace. It is evidence that he was silenced for telling the truth.

Argument

The Backfire Reply

1. Many vaccine-fearful people belong to communities that distrust mainstream experts on principle.
2. To such a person, a correction from those experts looks like more evidence of a cover-up.
3. So official corrections can deepen the false belief instead of removing it.
4. Therefore, premise 3 of the Correction Argument is false: facts alone can make things worse.

This reply is strong, but it can sound smug. It risks treating every doubter as a brainwashed dupe. That picture is both unkind and, often, untrue. Many people who distrust the vaccine system are not trapped in a chamber. They are reading a real history. That history is the next move.

Sometimes the distrust is earned

Heidi Larson, who studies vaccine confidence around the world, argues that we misread hesitancy when we treat it only as ignorance. Trust is not refused at random. It is withheld from institutions that have broken it (Larson 2020). A parent who distrusts the drug industry is not making a wild leap. She is generalizing from a record. And medicine has handed her that record, more than once.

- Drug companies have hidden harms. Purdue Pharma marketed OxyContin as barely addictive and helped start the opioid crisis. Its makers paid billions in settlements.
- The Tuskegee study let Black men go untreated for syphilis for forty years. Its memory still shapes trust in Black communities today.
- During COVID, officials sometimes spoke with more certainty than the evidence allowed, then reversed course. Masks, school closures, and transmission all saw public flip-flops. Each reversal spent trust that was hard to earn back.

Argument

The Earned-Distrust Argument

1. People reasonably lower trust in institutions that have deceived them before.
2. Medical and government institutions have, in real cases, deceived the public.
3. So some vaccine distrust is a rational response to a real record, not mere ignorance.
4. Therefore, the cure is not louder facts but more trustworthy institutions.

This reframes the whole problem. If distrust is partly earned, then “just trust the experts” repeats the insult. The work moves from the patient’s head to the institution’s conduct. But this raises a harder political question. What happens when the distrust is bundled with a broad movement — one that mixes fair complaints with false ones?

Health freedom, and the trouble with bundles

By the 2020s, vaccine doubt had become more than a private worry. It had become an identity and a politics. The “Make America Healthy Again” (MAHA) movement is the clearest example. Its appeal is that it is not only about vaccines. It also raises concerns many experts share. Americans do eat too much ultra-processed food. Chronic disease is rising. Regulators and the companies they oversee are often too close. On these points, the movement is not crankery. It is closer to common sense.

This is exactly what makes it hard to answer. Its true claims and its false ones travel in the same bag. A parent arrives worried about food dyes and added sugar — fair worries — and leaves doubting the measles vaccine. The good points open the door. The dangerous one walks in behind them. A movement that was only false would be easy to refute. One that is half right is not.

Scholars who fully accept the vaccine science split sharply on what to do about this. The split is not about facts. It is about strategy, and about what counts as a clean hand.

Argument

The Engagement Argument

1. A movement gains trust by being heard on the points where it is right.
2. MAHA is right that chronic disease and processed food are neglected problems.
3. Engaging those true concerns could rebuild the trust that moves people on vaccines.
4. Therefore, public health should work with such movements where they are right, not just fight them.

The vaccine scientists Paul Offit and Peter Hotez push back hard ([Offit 2011](#); [Hotez 2021](#)). To engage, they argue, is to lend your credibility to the false parts. Shaking hands on diet can legitimize a leader who also spreads vaccine lies. Stand next to such a figure on a stage, and you have endorsed the whole stage. Hotez goes further. He calls organized anti-vaccine activism a lethal force, one that has killed people who trusted it. You do not negotiate with that, he argues. You name it.

The disputed premise is 3: that engagement builds trust rather than laundering falsehood. Larson's side answers that refusing to engage has its own cost. Treat every doubter as an enemy, and you push the merely worried into the arms of the truly committed. The clinician feels this directly. Lecture the mother, and she may never come back. Listen to her, and you keep the door open — but you may also seem to grant that her fears are reasonable. Which risk is worse is not a question of science. It is a question of strategy and ethics, and it is unresolved.

The move that hasn't ended

The argument is no longer academic. It now runs the government. In 2025, **Robert F. Kennedy Jr.**, a long-time vaccine skeptic and a face of the MAHA movement, became US Secretary of Health and Human Services. Supporters saw a long-overdue challenge to a captured system. Critics saw the fox handed the keys to the henhouse. Both were watching the same appointment.

What followed was not rhetoric but machinery:

- **June 2025:** Kennedy dismissed every sitting member of the **ACIP** — the federal advisory committee that recommends vaccines — and reconstituted it with his own appointees, several with records of vaccine skepticism.
- **January 2026:** he directed the CDC to rewrite the **childhood immunization schedule**, cutting routine recommendations from 17 diseases to 11. Flu, rotavirus, hepatitis A, hepatitis B, RSV, and some meningococcal vaccines were moved off the routine schedule and recommended only for “high-risk” groups. Notably, the Secretary can direct this change without an ACIP vote ([Congressional Research Service 2026](#)).

- **March 16, 2026:** a federal court in Massachusetts **stayed** the new schedule in a challenge led by the American Academy of Pediatrics, reverting it to the May 2025 version. The legal fight continues.

Meanwhile the disease came back. **2026 has been the worst US measles year since 1991** — more than **2,100 confirmed cases across 41 jurisdictions** by mid-year, with months still to run, and **three deaths** since 2025: two unvaccinated children in Texas and one adult in New Mexico. The United States has held **measles elimination status** since 2000. The Pan American Health Organization is scheduled to review that status in **November 2026** — during our term — and sustained transmission makes losing it a real possibility.

Sit with what that would mean. Elimination was one of the clearest public-health achievements of the twentieth century, and it can be reversed not by a new pathogen but by a change in belief. For the doubters, each outbreak is a tragedy to be explained away. For public health, it is the cost of the argument made real, in a way no study could be.

At the same time, the courts narrowed one tool. In *Murthy v. Missouri* (2024), the Supreme Court took up claims that federal officials had pressured social media platforms to remove “misinformation” ([Supreme Court of the United States 2024](#)). The case intensified a worry the Backfire Reply had already raised: when the government leans on speech it calls false, it can confirm the very story that officials are hiding the truth.

So the clinician’s fifteen minutes have become a national problem, and every move in the dialogue is now policy. Offit and Hotez would hold the line and refuse to legitimize the doubt. Larson would rebuild trust and engage the fair complaints. Nguyen would warn that the wrong response feeds the chamber. None of them disputes the science. They dispute the cure. And here the honest case must stop, not on an answer, but on the open question: when the facts are settled and the fear still spreads, what does a free society owe the people it is trying to protect — and what may it never do to them?

Discussion Questions

1. Reconstruct the Correction Argument and the Backfire Reply in your own words. They share one disputed premise. Name it, and say which side you find more convincing and why.
2. The Earned-Distrust Argument says some vaccine refusal is rational. Does calling it “rational” let people off the hook for a dangerous choice? Defend your view, and use at least one of the real betrayals named in the case.
3. A premise in the Engagement Argument is that working with a movement on its true claims builds trust on its false ones. Argue either that this is wise or that it is a trap, using the MAHA example.
4. You are the pediatric nurse with the mother and her folder of printouts. You have fifteen minutes and you know the facts may backfire. Describe exactly what you would do first, and explain why it is honest persuasion and not manipulation.

5. After *Murthy v. Missouri*, should the government ask platforms to remove false vaccine claims? Argue for one clear policy, and address the risk that removal proves the doubters' point.

Further Reading

- Deer, Brian. *The Doctor Who Fooled the World: Science, Deception, and the War on Vaccines*. Johns Hopkins University Press, 2020.
- Biss, Eula. *On Immunity: An Inoculation*. Graywolf Press, 2014.
- Larson, Heidi J. *Stuck: How Vaccine Rumors Start — and Why They Don't Go Away*. Oxford University Press, 2020.

The Optum Algorithm: Racial Bias, Proxy Variables, and Justice in Medical AI

It was measuring spending, not sickness — a bioethics case study

At a Glance

- **Who:** Ziad Obermeyer, an emergency physician turned researcher, and his colleagues — plus the roughly 200 million Americans scored each year by tools like the one they audited.
- **What:** A widely used health algorithm ranked patients for extra care using predicted *cost* as its measure of *need*. At the same risk score, Black patients were far sicker than white patients. The tool routed help away from the people who needed it most.
- **Where / When:** A large U.S. academic hospital; published in *Science*, October 2019; the repair and the argument continue today.
- **Why it matters:** The case is the clearest example on record of **algorithmic bias** — a system that harms one group without anyone intending it to. It shows how a **proxy variable** (a stand-in measure for the thing you really want) can carry decades of inequality into a brand-new machine. And it forces a hard question: can a tool be *accurate* and *unjust* at the same time?
- **Pair with:** AI in Medicine

The Case

Start with a number in a chart. At hospitals and insurers across the United States, every patient in certain health plans gets a **risk score** — a number from 1 to 100, recalculated as claims roll in. The score predicts how much trouble a patient's health will bring next year. Patients above the 97th percentile are flagged for automatic enrollment in a care-management program. Patients above the 55th get a closer look.

Nobody in the building picks the number. Nobody in the building can quite say how it is picked. The score arrives with the software.

Background

What care management is. Some patients are far sicker than others, and a small share of patients account for most health spending. *Care-management programs* give those patients extra support: a nurse who calls to check in, help juggling many prescriptions, fast access when something goes wrong. The programs work — they keep fragile patients out of the emergency room. But the slots are scarce and expensive. So the whole system turns on one question: *which patients should get them?*

That question — who gets the extra help — is what the algorithm was built to answer. Tools of this kind are applied to roughly 200 million Americans every year ([Obermeyer et al. 2019](#)). The one in this story was among the most widely used in the country.

The people who finally checked the number came at it from two directions. Ziad Obermeyer was an emergency physician before he was a researcher. Years of ER shifts had left him with a working obsession: medicine’s judgments — human or machine — are tested against messy reality, and they fail more often than anyone records. By the late 2010s he was a professor at Berkeley, studying what algorithms in medicine actually do. Sendhil Mullainathan, his longtime collaborator, was an economist — a MacArthur “genius” award winner who had spent a career studying human bias, and who saw algorithms as a new kind of evidence about it. With them were the physicians Brian Powers and Christine Vogeli.

What the team got, at one large academic hospital, was a dataset that almost never exists. Companies guard their algorithms’ scores as trade secrets. Hospitals hold the medical records. The two are almost never in the same room — which is why claims about algorithmic bias usually end in a shrug, with no way to check. Here, for once, both halves sat side by side: the machine’s ranking, *and* the truth it was supposed to track.

So they asked a simple question. Take a Black patient and a white patient with the *same* risk score. The algorithm says they have the same level of need. Are they, in fact, equally sick?

They were not. At the same score, Black patients were carrying about **26 percent more chronic conditions** than white patients ([Obermeyer et al. 2019](#)). And the gap was not just a count of diagnoses. It showed up in the lab values themselves: at the same score, Black patients had higher blood sugar, higher blood pressure, worse kidney function, more severe anemia. These were not borderline cases sorted slightly differently. These were sicker bodies, scored as if they were healthier ones. The algorithm was telling care managers the two groups were equally needy. The blood work said otherwise.

The team then ran the repair as a thought experiment. Keep the program the same size. Rank patients by *actual sickness* instead of by the algorithm’s score. The share of Black patients selected for extra help would jump from **17.7 percent to 46.5 percent**. Nearly two-thirds of the Black patients who should have been getting help were not getting it.

How does a machine do that? Not the way most people first guess. The developers had not fed the algorithm race at all. They had excluded it on purpose, in the belief that leaving race out made the tool fair. The problem was hiding somewhere else — in a single design choice that had seemed purely technical.

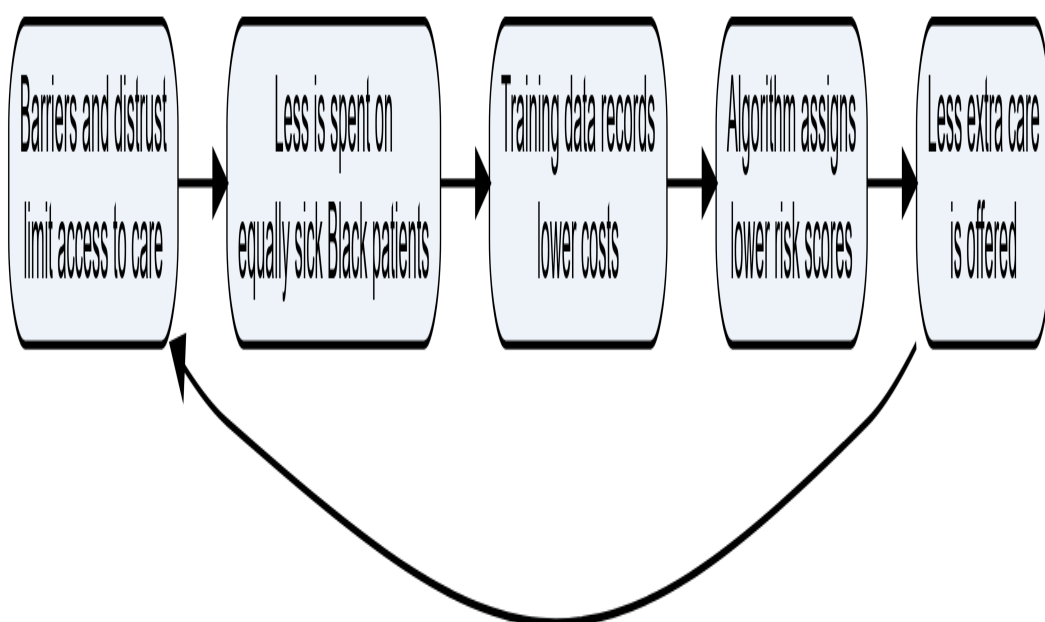
The algorithm needed a target: a number, in the data, meaning “how much will this patient’s health need next year?” No such number exists in any database. *Sickness* is not a field you can sort by. So the designers chose something close by that *is* in the data, down to the penny: **next year’s health-care costs**. Predict the spending, the logic ran, and you have found the need.

Background

What a proxy variable is. A *proxy variable* is a stand-in: something you can measure, used in place of something you actually care about but cannot measure directly. Proxies are everywhere and often harmless — a thermometer reading is a proxy for how feverish you feel. The danger comes when the proxy and the real target drift apart for *some* people more than others. Then a system built on the proxy will be quietly wrong in a patterned way.

Cost is a proxy for sickness — and it is a warped one. On average, sicker patients do cost more. But cost does not measure sickness alone. It measures sickness *times* everything that stands between a patient and care. Less money is spent on Black patients who are exactly as sick as white patients — the study measured the gap directly. The reasons are layered. Some are practical: distance from clinics, jobs without sick leave, thinner insurance. Some live inside the exam room, where research has long shown that Black patients’ pain and symptoms are taken less seriously, and fewer tests and referrals follow. And some of it is earned distrust. A community that remembers Tuskegee comes to the doctor later, and less often.

Every one of those forces leaves the same fingerprint in the data: **lower spending**. The algorithm read that fingerprint exactly as designed. Less money spent meant a lower predicted cost. A lower predicted cost meant a lower score. A lower score meant no care manager, no extra support — and another year in which less was spent. The bias did not just persist. It looped.



The proxy feedback loop. Each step follows from the last — and the loop closes on itself. No

step requires anyone to intend harm.

The study appeared in *Science* on October 25, 2019 (Obermeyer et al. 2019). The response was unusually fast for a research paper. Within days, New York’s financial-services and health regulators sent a joint letter to the algorithm’s corporate parent, demanding proof that the tool did not discriminate — “These discriminatory results, whether intentional or not, are unacceptable and are unlawful in New York,” it warned. Newspapers around the world covered the finding.

Background

A note on the name. The *Science* paper never names the product — that was part of the deal that got the researchers access. Reporters quickly identified it as Impact Pro, sold by Optum, a subsidiary of UnitedHealth Group. The company disputed parts of the coverage but engaged with the researchers on a fix. Because the underlying design choice — cost as the label — was industry-standard, the authors stressed that the problem was bigger than any one product.

Then came the part that makes this case unusual among scandals: a third act. The researchers did not stop at exposure. Working with the manufacturer, they rebuilt the target — predicting a measure closer to actual health instead of cost alone. That one change removed about **84 percent** of the measured bias (Obermeyer et al. 2019). The fix was that close at hand, once someone could see the problem.

And that is what makes the case hard, not easy. The algorithm had done exactly what it was built to do, and done it well. The result was unjust anyway. So where, exactly, did the wrong live — in the math, in the label, or in the world the data described? The argument over that question outlived the algorithm, and it is not finished.

The Argument the Audit Started

Algorithmic fairness already had a central puzzle before Optum: a model can use no forbidden variable — not race, nor anything close to it — and still reproduce the bias of the world it learned from. The Optum audit turned that abstraction into a measured number, on roughly 200 million patients. What the people who studied it then disagreed about was *where the wrong actually lived*: in the data, the proxy, the way the score was used, or the assumption that a clean fix was available at all.

“Accurate and colorblind”: the designers’ defense

Begin with the position the developers — and much of the industry — actually held. It deserves its strongest form, because most of us walk in believing some version of it. The algorithm used no race data. And at its assigned task, it was *accurate*: it predicted future costs about equally well for Black and white patients alike. By the standard tests the field used, the tool passed.

Argument

The Colorblind Accuracy Argument

1. A prediction tool is fair if it ignores race and predicts its target equally well for every group.
2. This algorithm excluded race and predicted costs equally well for Black and white patients.
3. Therefore, the algorithm was fair.

Premise 2 is true — the study itself confirmed it. The whole dispute lives in premise 1, and specifically in two words: *its target*. The premise assumes that hitting the target is what fairness means. It never asks whether the target was the right one. That is exactly where the study's authors aimed their reply.

“The label is where the values hide”: the auditors’ reply

Obermeyer’s team answered that fairness is not a property of the math. It is a property of the match between the math and the *job*. The job was to route scarce help to the sickest patients. The tool was never asked to do that job. It was asked to predict spending — and spending, in America, tracks race for reasons that have nothing to do with health.

Argument

The Warped Proxy Argument

1. A tool that allocates care is fair only if it serves the allocation’s real goal: helping the sickest.
2. The tool predicted cost, and cost understates sickness for Black patients, because less is spent on them at every level of illness.
3. A tool that accurately predicts a warped proxy reproduces the warp in every decision built on it.
4. Therefore, the algorithm was unfair — *because* it was accurate, not in spite of it.

On this view, choosing the label was never a technical decision. It was an ethical decision wearing technical clothes, made in a room where no one thought ethics was on the agenda. The most chilling detail is how the bias survived inspection: every internal quality check compared predictions to *cost*. The yardstick and the flaw were the same object. You cannot audit your way out of a proxy problem with the proxy.

But the reply has a soft spot, and the defenders of algorithms pressed on it at once. The warp in the proxy came from the world, not from the code. If the data record an unequal system, is the *algorithm* really the thing to blame — or is it just the messenger that finally got caught carrying the message?

“Biased algorithms are easier to fix than biased people”: the optimists’ counter

The most forceful version of that counter came from inside the study itself. Two months after the paper, Sendhil Mullainathan — one of its authors — published an essay whose title carried the whole argument: “Biased Algorithms Are Easier to Fix Than Biased People” (Mullainathan 2019). Yes, the algorithm absorbed the world’s bias. But notice what happened next. The bias was *found*, because an algorithm’s behavior can be tested on thousands of cases. It was *measured*, down to the percentage point. And it was mostly *repaired*, by changing one label — within months.

Argument

The Fixable Machine Argument

1. Bias in human judgment is diffuse, deniable, and very hard to change.
2. Bias in an algorithm lives in specific, inspectable choices — above all, the label.
3. This algorithm’s bias was found, measured, and 84 percent repaired by changing the label.
4. Therefore, audited and regulated algorithms can *reduce* injustice, compared with the human status quo.

Consider the human baseline honestly. Decades of studies show clinicians undertreating Black patients’ pain. No one has ever cut that bias by 84 percent with any intervention, let alone in one quarter. The machine, on this view, did medicine a favor: it gathered a thousand scattered human biases into one place where they could finally be seen and operated on.

The soft spot here is the word *audited*. The fix happened because outside researchers got something the industry almost never grants: access to the scores and the truth, side by side. Premise 3 quietly assumes someone gets to look. That assumption is where the next speaker took the argument.

“The bug-fix is not the lesson”: Benjamin’s reframing

Ruha Benjamin, a Princeton sociologist, wrote the commentary that ran beside the study in the same issue of *Science* (Benjamin 2019). She did not dispute the finding or oppose the fix. She challenged the frame around them. Treating the case as a bug that engineers patched, she argued, hides the questions that produced the bug in the first place. Who decided that *cost* was the objective? Who was in the room? Who was not? Colorblind design was not a neutral default that happened to fail — it was a choice to *not look*, and the harm it concealed fell exactly where American health care’s harms have always fallen.

Benjamin’s reframe moves the problem from engineering to governance. The lesson of the case is not “pick better labels.” It is that label-picking is an exercise of power, and power needs structure: mandatory audits, impact assessments before deployment, regulators with authority, and seats at the design table for the communities the tool will score. On her view, Mullainathan’s optimism is true but conditional — machines are easier to fix *if* anyone with power is obliged to look for

the flaw. The Optum case had volunteer auditors and a cooperative company. Justice should not have to depend on that kind of luck.

So the exchange narrowed to a real question: is the fixable machine the *normal* case, or was it a lucky one? The years since 2019 have been running that experiment.

The move that hasn't ended: what the years since have shown

The case did not close. It split into three live threads, and they pull in different directions.

The lesson spread — into medicine's oldest tools. The audit taught a generation to ask what a number actually measures, and the question escaped AI entirely. In 2020, physicians writing in the *New England Journal of Medicine* catalogued the “race corrections” built into everyday clinical formulas — kidney function, lung capacity, birth risk — adjustments that quietly moved Black patients away from treatment (Vyas et al. 2020). The kidney equation was rebuilt without race in 2021. Federal rules followed: in 2024, health regulators made hospitals responsible for discrimination risk in the decision-support tools they use, effective mid-2025. Score one for the optimists: visibility produced repair, at scale, beyond anything the authors predicted.

But the technology changed underneath the debate. The audit method that caught Optum needs two things: one label to inspect, and ground truth to compare it against. The generative AI now entering medicine has neither. When Stanford researchers tested large language models in 2023, the chatbots repeated debunked race-based medicine — false claims about Black patients' kidneys, skin, and pain thresholds — woven into fluent clinical answers (Omiye et al. 2023). There is no label to relabel. The bias is not *in* a column of the spreadsheet; it is dissolved into the model's whole way of talking.

Argument

The Vanishing Label Argument

1. The Optum bias was findable and fixable because one label could be checked against the truth and replaced.
2. Generative models have no single label, and often no ground truth to check their answers against.
3. An injustice that cannot be *located* cannot be repaired by relabeling.
4. Therefore, the “easier to fix” defense weakens just as medicine adopts the technology it fits least.

And the politics moved — backwards. The governance Benjamin called for advanced, then collapsed. Colorado passed the country's most ambitious algorithmic-discrimination law in 2024, covering health care by name. It never took effect. Lawmakers delayed it from February to June 2026, and then, in **May 2026**, repealed and replaced it outright (SB 26-189) (Skadden, Arps, Slate, Meagher and Flom 2026). The replacement drops the three provisions that gave the original its teeth: risk-management programs, impact assessments, and — most pointedly for this case — **the duty to use reasonable care to prevent algorithmic discrimination**. What

remains is a transparency law: tell people an automated system was used, disclose adverse outcomes within thirty days, correct bad data, offer human review. Compliance begins January 1, 2027.

Read that against the Optum facts. A pure disclosure regime would have required UnitedHealth to say *an algorithm helped decide this*. It would not have required anyone to notice that the algorithm's target variable was cost rather than illness — the very thing Obermeyer's team had to obtain proprietary data to discover. Meanwhile federal oversight of clinical AI loosened through 2025–26 in the name of innovation, and the original sore spot remains untouched: no law yet guarantees the next Obermeyer the data access that made the first audit possible. The fix was real. The system that produced the flaw — and the luck the fix required — look much the same.

So the question the case leaves open has widened rather than settled. The Optum story had an address for its injustice: one label, one fix, 84 percent. If the same injustice now lives everywhere and nowhere in a model no one can relabel — was the 2019 case the *model* for governing medical AI, or merely the last one easy enough to solve?

Discussion Questions

1. Reconstruct the Colorblind Accuracy Argument and the Warped Proxy Argument. The two sides agree on the facts. Identify the single premise they actually dispute, and say which side you find more convincing — and why.
2. The developers excluded race from the algorithm *in order to be fair* — and that choice helped hide the bias for years. Is colorblindness in a decision system a moral virtue, a neutral default, or a refusal to look? Defend your answer using the specifics of this case.
3. You are a nurse care-coordinator. Your daily call list comes from a risk score, and you notice it rarely includes the patients you know are struggling most. What do you do that week — and what would your hospital need to provide for your concern to actually be checkable?
4. Mullainathan says biased algorithms are easier to fix than biased people. Benjamin says the fix depended on power and luck that the system does not guarantee. Whose position is better supported by what has happened *since* 2019? Use at least one documented development from the final section.
5. A hospital's new chatbot gives clinically biased advice, but there is no risk score, no label, and no ground-truth file to audit. Design the 2026 version of the Obermeyer audit — what would you compare against what? — or argue that no such audit is possible and say what should replace it.

Further Reading

- Benjamin, Ruha. *Race After Technology: Abolitionist Tools for the New Jim Code*. Polity, 2019. (The book-length version of the reframing move.)

- Mullainathan, Sendhil. “Biased Algorithms Are Easier to Fix Than Biased People.” *The New York Times*, December 6, 2019.
- Obermeyer, Ziad, et al. *Algorithmic Bias Playbook*. Chicago Booth Center for Applied Artificial Intelligence, 2021. (The audit, turned into a how-to guide for health systems.)

United States v. Skrmetti: Gender-Affirming Care, Equal Protection, and Evidence Under Law

“We leave questions regarding its policy to the democratic process” — a bioethics case study

At a Glance

- **Who:** L.W., a sixteen-year-old transgender girl from Nashville, and her parents Brian and Samantha Williams; Jonathan Skrmetti, the Attorney General of Tennessee; Chase Strangio, the ACLU lawyer who argued for the families.
- **What:** In June 2025, the Supreme Court upheld a Tennessee law that bans puberty blockers and hormones for minors’ gender transition — while leaving the same drugs legal for other uses. The vote was 6–3.
- **Where / When:** A federal trial court in Nashville, the Sixth Circuit, and the Supreme Court, 2023–2025.
- **Why it matters:** The case turned on **equal protection** and on which **standard of review** a court must use. It asked whether a ban that sorts patients by *purpose* really sorts them by *sex* — and how a court should rule when the medical evidence is thin and the experts disagree.
- **Pair with:** Gender, Identity & Disability

The Case

L.W. was fourteen when she first put words to what she had felt for years. She told her parents she was a girl. Brian and Samantha Williams, of Nashville, did not panic and did not rush. They took her to doctors. After careful assessment, L.W. was diagnosed with **gender dysphoria** and, in time, began hormone therapy. She later described the years before treatment as “real-life body horror.” The medicine, she said, let her “feel normal.”

Background

Gender dysphoria. This is a clinical diagnosis, not a feeling alone. In the DSM-5-TR, it names the marked distress that can come from a mismatch between a person’s felt gender and their sex recorded at birth. The distress must last at least six months and must cause real impairment. Being transgender is not itself a disorder; the diagnosis treats the suffering,

not the identity.

Then the law changed. In March 2023, Tennessee passed Senate Bill 1, often called SB1. The law bans doctors from giving minors puberty blockers, hormones, or surgery *when the goal is gender transition*. One feature of the law would become the heart of the case. The very same drugs stay legal for other reasons. A boy with **precocious puberty** can still take a blocker to slow it down. A boy whose body makes too little testosterone can still take testosterone. What SB1 forbids is using those drugs to help a minor live as a gender that differs from their birth sex.

Background

Precocious puberty. This is puberty that starts far too early, sometimes at age six or seven. Doctors have paused it for decades using the same class of drugs — GnRH agonists — now debated in gender care. The drug is identical. Under SB1, only the purpose decides whether it is legal.

For the Williams family, the new law meant the road. To keep L.W. on the care her doctors had prescribed, the family now had to drive out of state. Samantha Williams was bitter about what she saw as a double standard. Tennessee lawmakers had loudly defended “parental rights” during the COVID-19 school fights, she noted. Now those same lawmakers were overruling her family’s medical choices. “They made this medical decision for our child,” she said.

The Williamses did not stay quiet. Together with two other transgender teens and their families, and a Tennessee doctor, they sued. The Biden administration joined them. A federal law lets the United States enter a private suit when it claims a state has denied **equal protection** of the laws.

Background

The Equal Protection Clause and tiers of review. The Fourteenth Amendment says no state may “deny to any person the equal protection of the laws.” Courts do not treat every law the same. Most laws get **rational basis review**: the law survives if the state had any reasonable reason for it. This test is easy to pass. But laws that sort people by race get **strict scrutiny**, and laws that sort by sex get **intermediate scrutiny** — a much harder test the state usually loses. So the whole fight often becomes a single question: which test applies?

The case moved up the ladder. The trial judge in Nashville blocked the law, finding it likely violated parents’ rights. Then the Sixth Circuit Court of Appeals reversed. Chief Judge Jeffrey Sutton wrote that the right of parents to direct a child’s medical care is not “deeply rooted in this Nation’s history and tradition,” the test the Supreme Court had set in an earlier case. He held that SB1 needed only rational basis review — and that it passed. Judge Helene White dissented in part. She thought the law “likely unconstitutional” as sex discrimination. The split set the stage for the Supreme Court.

The argument came on December 4, 2024. It drew a crowd, in part because of who stood at the lectern. Chase Strangio, a lawyer for the ACLU, argued for the families. He is the first openly

transgender person known to argue a case before the Supreme Court. Solicitor General Elizabeth Prelogar argued for the United States; she told the justices the case was about whether L.W. could simply “be a girl” in Tennessee. Matt Rice, Tennessee’s solicitor general, argued for the state.

There was a strange twist before the ruling came down. In February 2025, after the election, the Trump administration switched the government’s side. It now told the Court that SB1 did *not* violate equal protection. But it asked the Court to rule anyway. Because oral argument was already over, the switch changed little.

On June 18, 2025, the Court ruled 6–3 to uphold the law. Chief Justice John Roberts wrote for the majority. He insisted the ruling was narrow and not a verdict on transgender people. “Our role is not ‘to judge the wisdom, fairness, or logic’ of the law before us,” he wrote, “but only to ensure that it does not violate the equal protection guarantee.” Having found no violation, he closed: “We leave questions regarding its policy to the people, their elected representatives, and the democratic process.”

Justice Sonia Sotomayor read part of her dissent aloud, a rare move that signals deep disagreement. The Court, she wrote, “invites legislatures to engage in discrimination by hiding blatant sex classifications in plain sight.” The question the case raised is easy to state and hard to answer. When a law turns the same drug on or off depending on the patient, is that a line about *purpose* — or a line about *sex*?

The Argument *Skrmetti* Started

The legal fight in *Skrmetti* did not begin in 2025. It inherited a fifty-year-old machinery for deciding when a law that treats people differently must clear a higher constitutional bar. The whole case turns on one move inside that machinery: whether a ban that sorts patients by the *purpose* of a drug is, underneath, sorting them by *sex* — and so whether courts owe it skeptical review or near-total deference.

The families’ argument: this is a sex line

Strangio and Prelogar built their case on a simple test. Change one fact about the patient — her sex — and see if the law’s answer changes. Under SB1, a patient born male may take testosterone to match a male identity. A patient born female may not take testosterone to match a male identity. Same drug, same dose, same goal of looking more masculine. The only thing that flipped the result was the sex on the birth certificate.

Argument

The Sex-Classification Argument

1. A law classifies by sex if changing only the patient’s sex changes what the law allows.
2. Under SB1, whether a minor may receive testosterone or estrogen depends on the sex they were assigned at birth.

3. Any law that classifies by sex must pass intermediate scrutiny, which SB1 cannot.
4. Therefore, SB1 violates the Equal Protection Clause.

The families leaned on *Bostock v. Clayton County* (2020), where the Court held that firing someone for being transgender is firing them “because of sex” under federal job law ([Supreme Court of the United States 2020](#)). You cannot define transgender status, they argued, without referring to sex. The disputed premise is the second one. Does the law really sort by sex — or does it sort by the *reason* the drug is given? Tennessee staked everything on saying it sorts by reason.

Tennessee’s reply: this is a purpose line

The state, and then the majority, denied that sex was doing the work. SB1, they said, draws two lines, and neither is sex. The first line is age: the law restricts minors, not adults. The second line is medical purpose: it allows these drugs for conditions like precocious puberty and bars them for gender transition. A transgender boy and a transgender girl are treated exactly the same — neither may transition as a minor. So, the state argued, no one is treated worse for being male or female.

Argument

The Medical-Purpose Reply

1. A law classifies by sex only if it treats males and females differently.
2. SB1 bans transition care for every minor, of either sex, while allowing the same drugs for other conditions for every minor, of either sex.
3. So SB1 classifies by age and medical purpose, not by sex.
4. Therefore, only rational basis review applies, and SB1 passes it.

The soft spot here is premise 2’s claim that the line is truly sex-neutral. The dissent pressed exactly there. It argued that “transition care” cannot even be described without naming sex, because the treatment a patient seeks is defined against the sex they were born with.

The dissent’s counter-reply: hidden lines and a warning from history

Sotomayor, joined by Justice Ketanji Brown Jackson, answered that the “purpose” framing was a magic trick. A sex line does not stop being a sex line just because a state describes it in other words. Her most telling point came from history, and Jackson had raised it at oral argument. In *Loving v. Virginia* (1967), Virginia defended its ban on interracial marriage by saying it was not racist, because it punished the white and the Black partner equally ([Supreme Court of the United States 1967](#)). The Court rejected that reasoning. A law can rest on race even when it falls on every race “equally.”

Argument

The *Loving* Parallel

1. In *Loving*, a law that applied to all races was still struck down as race discrimination, because the forbidden act was defined by race.
2. SB1 applies to all sexes, but the forbidden treatment is defined by the patient's sex.
3. Laws that hide a protected classification inside an "even-handed" rule still trigger heightened scrutiny.
4. Therefore, SB1 should face intermediate scrutiny, which it cannot survive.

This is where a concept from constitutional ethics earns its place. The point of **heightened scrutiny** is to protect groups that the ordinary political process tends to overlook or mistreat. To Sotomayor, retreating to rational basis "exactly where it matters most" abandoned the children the clause exists to guard. The majority's answer was not that trans youth deserve no concern. It was that courts are the wrong body to weigh contested medicine — which sets up the next move.

The reframing move: who decides a question of weak evidence?

Roberts did not rest only on logic about classification. He reached for the medical record, and this is where the case meets the rest of the gender-care debate. The majority cited the **Cass Review**, the four-year independent study commissioned by England's National Health Service.

Background

The Cass Review. Led by senior pediatrician Hilary Cass, this 2024 review used systematic evidence methods from the University of York. It concluded that the evidence behind puberty blockers and hormones for minors is "remarkably weak" — small studies, short follow-up, and low certainty about long-term benefit. It did not call the treatment harmful. It called the evidence thin, and urged caution. England, Sweden, and Finland all moved toward restriction on similar grounds.

Roberts drew a lesson from that uncertainty: "Recent developments only underscore the need for legislative flexibility in this area." Where evidence is contested, he argued, elected lawmakers — not judges — should set the rule. This reframes the whole case. The fight is no longer only about sex. It is about *who* decides a medical question that science has not settled.

The evidence fight had a fierce subplot in the briefs. The American Academy of Pediatrics, the Endocrine Society, and other major medical groups filed briefs against the ban, calling the care necessary. On the other side, Tennessee's supporting experts drew scrutiny. *The Guardian* reported that four doctors backing the ban had been criticized by various courts as biased, and that three of them had never treated a transgender youth. Alabama, defending its own ban, filed a brief claiming that WPATH, the group that writes transgender care standards, had shaped its guidelines with an eye on lawsuits rather than pure science. Each side, in short, accused the other of dressing up values as evidence. That is the deeper puzzle the case leaves open: when

the studies are weak, does the benefit of the doubt belong to the patient who wants care, or to the legislature that wants to wait?

The move that hasn't ended: the squeeze after the ruling

Skrametti settled one narrow question and reopened many. The Court did not rule that the bans are wise, that the treatments are harmful, or that transgender people can never be a protected class. It also left untouched the *parental rights* claim — whether the Constitution protects a parent's choice to get this care — because that question was not before it. At oral argument, Justice Amy Coney Barrett seemed open to that issue, which means the next lawsuit may look different.

One concurring opinion shows how live that next question is. Justice Barrett wrote separately to argue that transgender people should *not* count as a protected class at all. A protected class, she reasoned, is usually marked by an “obvious, immutable, or distinguishing” trait like race or sex, and has faced formal, written-in-law discrimination over time. Transgender people, she argued, do not fit that mold. Sotomayor answered her directly. Anyone “searching for more evidence of de jure discrimination,” she wrote, “need look no further than the present” — pointing to the expulsion of transgender service members from the military and threats to pull funding from schools that support transgender students. The two justices were not just reading history differently. They were arguing about whether the harm is past or happening now.

What followed the ruling moved faster than the ruling itself. *Skrametti* concerned state bans on care for *minors*. But Executive Order 14187, signed in January 2025, directed federally funded institutions to stop the care for patients under nineteen — past the usual age of adulthood. Federal health agencies sent hospitals warning letters. Major children's hospitals paused or ended services, including some in states where the care is fully legal. Minnesota is a refuge state, with laws meant to protect patients and providers. Yet even there, hospitals that depend on federal money felt the pressure.

The order's own legal status is the strangest part of the story. Courts **preliminarily enjoined** EO 14187, so it is not currently enforceable; sixteen state attorneys general sued in August 2025 arguing it violates the Tenth Amendment and exceeds the President's authority, and that case was still being litigated through 2026. Hospitals stopped the care anyway. When the government asked the court to throw the suit out, the judge pointed to precisely that — providers announcing they had ended services — as proof of real harm. A blocked order had already achieved much of what it directed, because no hospital will bet its federal funding on winning a lawsuit later.

As of mid-2026 the picture is roughly: **27 states** restricting or prohibiting the care for minors, **17 states and D.C.** with shield laws protecting it, and a federal executive order that is legally suspended but practically potent. (Check these counts before you rely on them; they move.)

So the legal map is no longer a simple split between “ban states” and “shield states.” Federal funding reaches across that line. *Skrametti* handed the policy to “the democratic process” — but the democratic process now runs through courts, statehouses, and federal budget offices all at once. The live question is no longer just *what should we do for a child like L.W.?* It is *who now holds the pen — and can anyone overrule them?*

Discussion Questions

1. Reconstruct the **Sex-Classification Argument** and the **Medical-Purpose Reply** in your own words. They share one disputed premise: whether SB1's line is drawn by sex or by purpose. Take a side and defend it, using the actual text of the law.
2. Justice Jackson compared SB1 to the interracial-marriage ban struck down in *Loving v. Virginia*. Tennessee replied that the cases are not alike. Where is the analogy strong, and where does it break down? Does the comparison persuade you?
3. The majority argued that when medical evidence is weak, *legislatures* — not courts — should set the rule. Pressure-test that claim. Are there cases where weak evidence should make a court *more* protective of an individual, not less? When, and why?
4. You are the nurse caring for a teen like L.W. in a Minnesota clinic. The care is legal in your state, but your hospital has paused it under federal funding pressure. The family asks what you can still do for them. What do you say, and what do you refuse to promise?
5. After *Skrmetti*, federal orders reached into shield states like Minnesota and into the care of patients up to age nineteen. Was the Court right that this dispute belongs to “the democratic process,” or does that phrase hide who actually holds the power now? Defend your answer using the events after the ruling.

Further Reading

- Hilary Cass, *The Cass Review: Final Report* (NHS England, 2024). Read the foreword and recommendations directly; both sides quote it, and few read it.
- ABC News, “Transgender Teen Implores Supreme Court to Strike Tennessee Gender-Affirming Care Ban” (2024) — a profile of L.W. and the Williams family in their own words.
- The majority and dissenting opinions in *United States v. Skrmetti* (2025), which are written for non-lawyers more than most and repay a direct read.

David Bennett and the First Pig-Heart Transplant: Xenotransplantation, Consent, and the Rule of Rescue

“It was either die or do this transplant” — a bioethics case study

At a Glance

- **Who:** David Bennett Sr. (1964–2022), a 57-year-old Maryland handyman dying of heart failure; surgeons Bartley Griffith and Muhammad Mohiuddin of the University of Maryland.
- **What:** On January 7, 2022, Bennett received the heart of a genetically modified pig — the first such transplant into a living human. He lived about two months.
- **Where / When:** University of Maryland Medical Center, Baltimore, 2022.
- **Why it matters:** The case tests whether a dying patient can give real **informed consent**, whether the **rule of rescue** justified an untried procedure, and whether a risk to the wider public — **xenozoonosis** — can ever be consented to by one patient alone.
- **Pair with:** Animal Ethics

The Case

By the fall of 2021, David Bennett’s heart was failing for the last time. He was 57, a father of two, a former handyman who liked motorcycles and the beach. He had lived for years with a faulty heart rhythm. Now he lay in a Baltimore hospital bed, hooked to a machine called **ECMO** that pumped and oxygenated his blood because his own heart no longer could. He could not leave the bed. Without something drastic, he would die within weeks.

Background

ECMO (extracorporeal membrane oxygenation). A machine that takes over for the heart and lungs, pumping a patient’s blood out of the body, adding oxygen, and returning it. It buys time during a crisis. It is not a long-term fix; patients cannot live on it indefinitely, and it carries its own risks of clots and bleeding.

Bennett was out of ordinary options. He was not eligible for a human heart transplant. His doctors said he had not always followed medical advice in the past — he had missed appointments and skipped medications — and transplant programs weigh such history, because a donated heart is

scarce and must not be wasted. He was also turned down for an implanted mechanical pump, because his irregular heartbeat made it too risky. Every standard door had closed.

Then his doctors offered a door no one had ever walked through alive. Bartley Griffith, the surgeon who would lead the operation, put it to him plainly: they could not give him a human heart, but they might be able to give him a pig's. Bennett, by the hospital's account, understood the gamble. "It was either die or do this transplant," he said before the surgery. "I know it's a shot in the dark, but it's my last choice."

What made the offer possible was decades of genetic engineering. The pig heart did not come from an ordinary farm animal. It came from a pig built, gene by gene, to survive inside a human body.

Background

Xenotransplantation and the gene-edited pig. Xenotransplantation means transplanting organs or tissue from one species into another. The human body destroys ordinary pig tissue within minutes. The donor pig was modified at ten genes: three pig genes that trigger instant rejection were switched off, six human genes were added to calm the body's attack, and one growth gene was disabled so the heart would not outgrow its new chest. The pig was supplied by a company, Revivicor, that breeds animals for this purpose.

The surgery, on January 7, 2022, was not a normal, approved treatment. No such operation had ever been formally tested in people. Instead, the U.S. Food and Drug Administration granted a one-time emergency clearance — a path called **compassionate use** — that let a single dying patient try an experimental therapy outside of a clinical trial.

Background

Compassionate use (expanded access). A route by which a patient with a serious illness and no good alternatives may receive an experimental treatment that has not finished testing. It is meant as mercy for an individual, not as a study. Because it treats one person at a time, it produces little of the careful, controlled data that a real trial is designed to gather.

The operation itself was the easy part to imagine and the hard part to reach. Muhammad Mohiuddin, who ran the Maryland program's xenotransplant lab, had spent some thirty years trying to keep pig hearts alive in animals long enough to matter. Griffith had practiced the surgery on the bench for years. On the day, the team removed Bennett's failing heart and sewed in the pig's, then watched to see whether the human body would tear it apart. It did not. The heart took.

For a while, it worked. The pig heart beat on its own. Bennett came off ECMO. He watched the Super Bowl, did physical therapy, and joked with his nurses about going home to see his dog. For several weeks he was, by every visible sign, a man saved by an animal's heart. It was a genuine medical first, reported around the world.

The decline came in March. The new heart thickened and stiffened. Fluid built up in his body, and the organ slowly lost its strength. On March 8, 2022 — about sixty days after the surgery —

David Bennett died.

Months later came a detail that reframed the whole story. In analyzing the heart, the Maryland team found traces of a pig virus, **porcine cytomegalovirus (PCMV)**, that screening was supposed to have caught and missed ([Mohiuddin et al. 2023](#)). The virus may have contributed to the heart's failure. It also pointed at the deepest worry about the entire enterprise: a germ carried in an animal organ, into a human body, that no one had agreed to invite.

His family chose to read the sixty days as a gift. His son thanked the doctors for “every extra moment” and for the chance to help patients who would come later. Griffith, the surgeon, said the team had learned a great deal and grieved the man they had come to know. Even framed as a triumph of nerve and science, though, the case left a set of questions that the obituaries mostly skipped.

There was one more fact, and it sat uneasily with the celebration. Decades earlier, in 1988, Bennett had stabbed a young man named Edward Shumaker, leaving him paralyzed; Shumaker died years later after a life of disability. When Bennett received his historic heart, Shumaker's sister asked aloud why a man who had ruined her brother's life should be the one to make medical history ([Associated Press 2022](#)). Bennett's family asked that he be judged by his final act of courage, not his worst one. The hospital said, as medicine generally does, that it does not rank patients by their pasts.

So the case leaves a hard question on the table. When a desperate patient, a designed animal, and an unconsenting public all meet in one operating room, what made this permissible — and was it?

The Argument the Bennett Case Started

Every time medicine tries something for the first time, the same two questions return: where is the line between treating a patient and experimenting on one, and can a dying person freely consent to either? Bennett's case poses both as starkly as the first heart transplants did in 1967 — and adds a third those cases never faced. A pig's organ can carry a pig's viruses, so the risk reached past Bennett to a public that never sat in the consent conversation.

The rescue argument: he had nothing to lose

The simplest defense is also the most powerful. Bennett was going to die in days. He understood that the pig heart had never kept a human alive. He chose it anyway. To deny him that choice would be to deny a dying man his last act of self-determination — and might cost him the only chance he had.

Argument

The Rule-of-Rescue Argument

1. A competent patient may choose a risky treatment when the alternative is near-certain

- death.
2. Bennett was competent, fully informed of the risks, and faced near-certain death.
3. The transplant offered a real, if small, chance of benefit and could not leave him worse off than dying.
4. Therefore, performing the transplant respected Bennett's autonomy and served his good.

This appeals to the **rule of rescue** — the deep pull we feel to save an identifiable person in front of us, whatever the odds. The disputed premise is the second one: that Bennett was *fully informed* and freely choosing, and that the benefit was real rather than imagined. That is exactly where the reply lands.

The reply: this was an experiment dressed as treatment

A treatment is meant to help the patient. An experiment is meant to produce knowledge. The Bennett operation, critics argue, was mostly the second while being sold as the first — and that confusion poisons the consent.

Argument

The Therapeutic-Misconception Reply

1. Valid consent to treatment requires a realistic expectation of personal benefit.
2. No pig heart had ever kept a human alive, so the realistic aim was knowledge for future patients, not benefit to Bennett.
3. A patient who believes a research procedure is really treatment is under a **therapeutic misconception** and cannot give valid consent to it.
4. Therefore, Bennett's "choice" was not the free, informed consent the rescue argument assumes.

Background

Therapeutic misconception. The well-documented tendency of patients in research to believe that an experimental procedure is designed to help *them*, when its real purpose is to gather data. The more desperate the patient, the stronger the misconception — and the harder it is to correct.

Add desperation to the mix and the worry grows. A man hours from death, offered his "last choice," is in no position to weigh odds coolly. Consent given at the edge of the grave may be sincere and still not be free.

The counter-reply: desperate patients can still choose

Defenders of the surgery answer that the reply proves too much. If desperation canceled consent, no dying patient could ever volunteer for anything — and medicine would be forbidden from trying the new thing that someday becomes standard. The first human heart transplant patients, in 1967, also died within weeks; we do not now call them dupes.

But this answer concedes the key point: if the real goal was knowledge, then Bennett was, at least in part, a **research subject** — and research subjects are owed protections that a one-off compassionate-use case cannot provide. There is no control group. There is no plan to test the result against an alternative. There is no committee weighing risks to a whole class of future patients. The data from a single dramatic case are thin, and a death teaches less than a trial would.

Here Kant's warning bites. To use Bennett *partly* as a means to medical knowledge is not automatically wrong; Kant forbids using a person *merely* as a means, without regard to their own ends. Bennett wanted to live, and he also wanted his ordeal to help others — he said as much. So the surgeons could honor both aims at once. The hard question is which aim was really steering the ship. If the chance of saving him was close to zero, then "treatment" was the story told at the bedside while "experiment" was the work being done. A dying man's hope can be honored as his own end, or quietly borrowed to serve science, and from the outside the two can look identical.

The reframing move: it was never only about the patient

The whole debate so far assumes the only person at stake is Bennett. The most distinctive feature of xenotransplantation is that this assumption is false.

A pig organ can carry pig viruses. If one of them adapts to humans inside a recipient, it could spread to other people — a **xenozoonosis**, an animal-to-human infection with the recipient as the doorway. The discovery of PCMV in Bennett's heart was a live rehearsal of that fear. The HIV pandemic almost certainly began as an animal virus that crossed into humans; the worry is not science fiction.

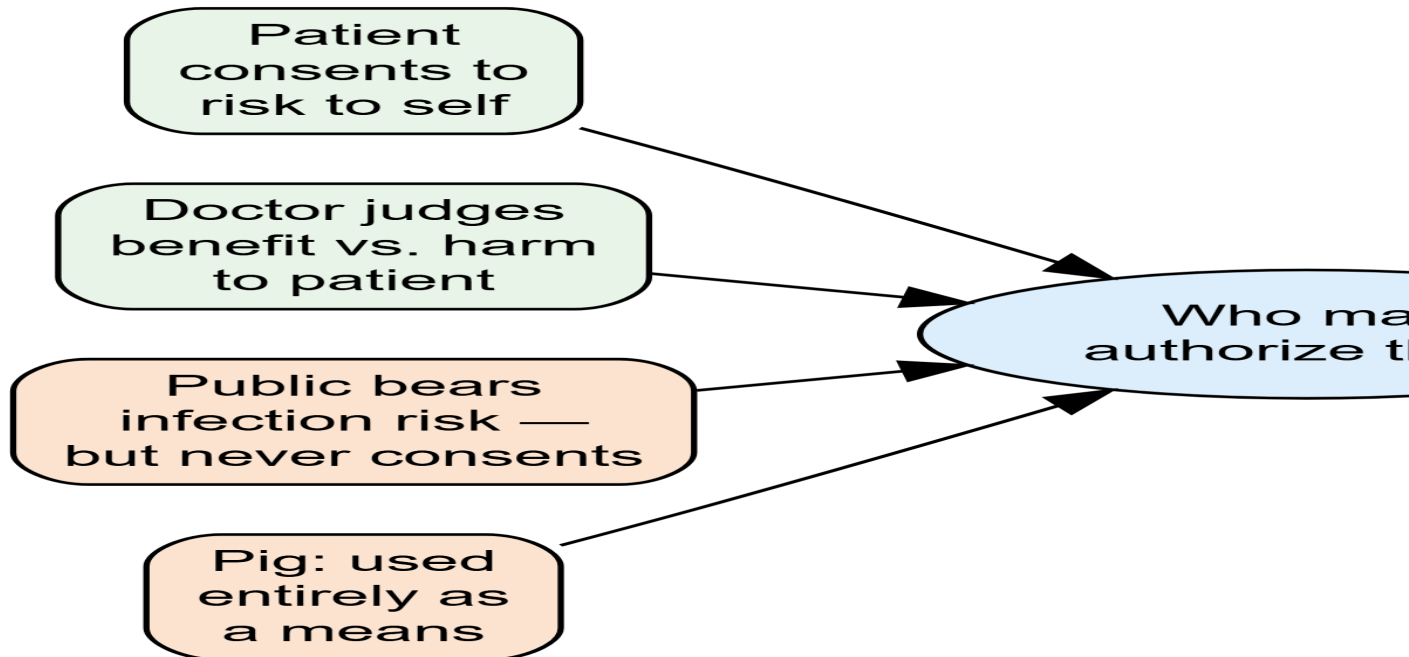
And here consent runs out.

Argument

The Public-Risk Argument

1. A person can consent to a risk that falls only on themselves.
2. Xenotransplantation creates a risk of a new infection that could spread to other people.
3. That shared risk falls on family, clinicians, and strangers who never agreed to it.
4. Therefore, the patient's consent alone cannot make the procedure permissible; the public has a stake no individual can sign away.

Bennett could accept a risk to *himself*. He could not consent on behalf of the nurses who cared for him or the strangers who would bear the cost of an outbreak. There is also the pig, engineered from conception to be killed for its parts and raised in sterile isolation — a being used wholly as a means, with no say at all.



This reframes the decision from a private one — patient and doctor — into a public one. If the risk is shared by people who never agreed to it, then perhaps patient and surgeon alone should never have held the whole decision. Some body speaking for the community arguably belongs in the room.

The move that hasn't ended: from rescue to routine

Bennett's case was a single act of rescue. The field has since tried to become a discipline. In 2024, surgeons gave gene-edited pig kidneys to living patients — Rick Slayman and Towana Looney — who, like Bennett, were out of other options. And in 2025, the FDA cleared the first true **clinical trials** of pig-organ transplantation, moving the practice out of one-off exceptions and into planned, regulated studies with rules, oversight, and consent designed in advance ([United Therapeutics Corporation 2025](#)).

Slayman lived about two months after his pig kidney; Looney's organ was removed after 130 days. Then Tim Andrews, a retired engineer from New Hampshire, lived **271 days** with a 69-gene-edited pig kidney before it failed in October 2025 — and in January 2026 he received a **human** kidney, becoming the first person to cross from an animal organ to a human one. That possibility, xenotransplant as a temporary **bridge** rather than a permanent replacement, may end up being the field's most consequential result so far: a bridge only has to hold for months, not decades. Each result, good or bad, now feeds a planned study rather than a single news cycle. This is meant to answer the objections. A trial replaces the therapeutic misconception with

honest research consent. It builds in screening for viruses like PCMV. It submits the public-risk question to regulators rather than to a single operating team. Yet the deepest tensions do not vanish. Trial subjects will still be desperate people choosing under pressure. The infection risk still falls on a public that cannot sit in the consent conversation. The pigs are still made to die. The move from rescue to routine makes the practice more careful — but careful is not the same as settled. The question Bennett’s case opened is now being asked at scale: not *may we save this one dying man*, but on *what terms may we build a whole supply of human organs out of animals*?

Discussion Questions

1. Reconstruct the **Rule-of-Rescue Argument** and the **Therapeutic-Misconception Reply**. They turn on one shared premise — whether Bennett’s consent was genuinely informed and free. Take a side and defend it, using the facts of his condition.
2. Bennett had been turned down for a conventional transplant partly because of his medical history, yet was chosen for a world-first surgery. Was that consistent? Should a patient’s past — including the 1988 stabbing — ever factor into who receives scarce or experimental care? Defend your answer.
3. You are the nurse responsible for walking Bennett through the consent form the night before surgery. He says, “So this is going to fix me, right?” What do you say, and what would you refuse to let him believe before you signed off?
4. The “nothing to lose” defense assumes the transplant could not leave Bennett worse off than dying. Pressure-test that premise. Are there outcomes — prolonged suffering, dying as a research subject, a public infection — that count as *worse*? Who should weigh them?
5. In 2025 the FDA approved the first regulated clinical trials of pig-organ transplantation. Does moving from one-off compassionate use to planned trials actually resolve the consent and public-risk objections raised by Bennett’s case, or only manage them? Defend your view.

Further Reading

- Roni Caryn Rabin, “In a First, Man Receives a Heart from a Genetically Altered Pig,” *The New York Times*, January 10, 2022 — the contemporary account of the surgery and the patient.
- Muhammad M. Mohiuddin et al., “Graft Dysfunction in Compassionate Use of Genetically Engineered Pig-to-Human Cardiac Xenotransplantation,” *The Lancet*, 2023 — the surgical team’s own analysis of what went wrong, including the PCMV finding.

Works Cited

Works Cited

The references below collect every source cited across the introduction and the case studies.

Alsan, Marcella, Owen Garrick, and Grant Graziani. 2019. "Does Diversity Matter for Health? Experimental Evidence from Oakland." *American Economic Review* 109 (12): 4071–111. <https://doi.org/10.1257/aer.20181446>.

Alsan, Marcella, and Marianne Wanamaker. 2018. "Tuskegee and the Health of Black Men." *The Quarterly Journal of Economics* 133 (1): 407–55. <https://doi.org/10.1093/qje/qjx029>.

Appelbaum, Paul S. 2007. "Assessment of Patients' Competence to Consent to Treatment." *New England Journal of Medicine* 357 (18): 1834–40. <https://doi.org/10.1056/NEJMc074045>.

Aristotle. 340 BC. *Nicomachean Ethics*.

Arnold, Denis G., and Paul T. Menzel. 1998. "When Comes 'the End of the Day'? A Comment on the Dialogue Between Dax Cowart and Robert Burt." *Hastings Center Report* 28 (1): 25–27. <https://doi.org/10.2307/3527970>.

Associated Press. 2022. *Pig Heart Recipient Stabbed Man in 1988, Victim's Family Says*. Associated Press. <https://apnews.com/article/pig-heart-transplant-stabbing>.

Bajaj, Simar Singh, and Fatima Cody Stanford. 2021. "Beyond Tuskegee — Vaccine Distrust and Everyday Racism." *New England Journal of Medicine* 384 (5): e12. <https://doi.org/10.1056/NEJMp2035827>.

Beauchamp, Tom L., and James F. Childress. 2019. *Principles of Biomedical Ethics*. 8th ed. Oxford University Press.

Benjamin, Ruha. 2019. "Assessing Risk, Automating Racism." *Science* 366 (6464): 421–22. <https://doi.org/10.1126/science.aaz3873>.

Bentham, Jeremy. 1789. *An Introduction to the Principles of Morals and Legislation*. T. Payne; Son.

Brewster, Luke P., Barbara K. Bennett, and Richard L. Gamelli. 2006. "Application of Rehabilitation Ethics to a Selected Burn Patient Population's Perspective." *Journal of the American College of Surgeons* 203 (5): 766–71. <https://doi.org/10.1016/j.jamcollsurg.2006.06.024>.

Works Cited

- Brock, Dan W. 2008. "Conscientious Refusal by Physicians and Pharmacists: Who Is Obligated to Do What, and Why?" *Theoretical Medicine and Bioethics* 29 (3): 187–200. <https://doi.org/10.1007/s11017-008-9076-y>.
- Burton, Keith. 1989. "A Chronicle: Dax's Case as It Happened." In *Dax's Case: Essays in Medical Ethics and Human Meaning*, edited by Lonnie D. Kliever. Southern Methodist University Press.
- Butcher, Loughran. 2025. "MAiD, Mental Disorders, and Vulnerability: How Common Responses to Vulnerability Concerns Are Inadequate." *Canadian Journal of Bioethics / Revue Canadienne de Bioéthique* 8 (4): 56–63. <https://doi.org/10.7202/1121335ar>.
- Caplan, Arthur. 2025. *The Adriana Smith Case Unfolding in Atlanta Raises Many Questions*. Bioethics Today (American Journal of Bioethics). <https://bioethicstoday.org/blog/the-adriana-smith-case-unfolding-in-atlanta-raises-many-questions/>.
- Congressional Research Service. 2026. *The 2026 Childhood Immunization Schedule*. No. R48982. CRS Report. <https://www.congress.gov/crs-product/R48982>.
- Cortes Generales (Spain). 2021. *Ley Orgánica 3/2021, de 24 de Marzo, de Regulación de La Eutanasia*. Boletín Oficial del Estado, no. 72. <https://www.boe.es/eli/es/lo/2021/03/24/3>.
- Cowart, Dax, and Robert Burt. 1998. "Confronting Death: Who Chooses, Who Controls?" *Hastings Center Report* 28 (1): 14–24. <https://doi.org/10.2307/3528337>.
- Cyranoski, David. 2020. "What CRISPR-Baby Prison Sentences Mean for Research." *Nature* 577 (7789): 154–55. <https://doi.org/10.1038/d41586-020-00001-y>.
- Dada, Damilola, Joseph Nguemo Djiometio, SaraJane M. McFadden, et al. 2022. "Strategies That Promote Equity in COVID-19 Vaccine Uptake for Black Communities: A Review." *Journal of Urban Health* 99 (1): 15–27. <https://doi.org/10.1007/s11524-021-00594-3>.
- Deer, Brian. 2011. "How the Case Against the MMR Vaccine Was Fixed." *BMJ* 342: c5347. <https://doi.org/10.1136/bmj.c5347>.
- Dukes, Deidra. 2025. *Court Grants Sole Custody to Father After Mother Declared Brain-Dead*. FOX 5 Atlanta. <https://www.fox5atlanta.com/news/court-grants-sole-custody-father-after-mother-declared-brain-dead>.
- Elliott, Carl. 2018. "Tuskegee Truth Teller." *The American Scholar* 87 (1): 44–56. <https://www.jstor.org/stable/26756050>.
- Emanuel, Ezekiel J., and Linda L. Emanuel. 1992. "Four Models of the Physician-Patient Relationship." *JAMA* 267 (16): 2221–26. <https://doi.org/10.1001/jama.1992.03480160079038>.

- Faden, Ruth R., and Tom L. Beauchamp. 1986. *A History and Theory of Informed Consent*. Oxford University Press.
- Gamble, Vanessa Northington. 1997. "Under the Shadow of Tuskegee: African Americans and Health Care." *American Journal of Public Health* 87 (11): 1773–78. <https://doi.org/10.2105/AJPH.87.11.1773>.
- Gerrek, Monica L. 2018. "Getting Past Dax." *AMA Journal of Ethics* 20 (6): 581–88. <https://doi.org/10.1001/journalofethics.2018.20.6.mhst1-1806>.
- Gilligan, Carol. 1982. *In a Different Voice: Psychological Theory and Women's Development*. Harvard University Press.
- Grassi, Luigi, Federica Folesani, Marco Marella, et al. 2022. "Debating Euthanasia and Physician-Assisted Death in People with Psychiatric Disorders." *Current Psychiatry Reports* 24 (6): 325–35. <https://doi.org/10.1007/s11920-022-01339-y>.
- Greely, Henry T. 2019. "CRISPR'd Babies: Human Germline Genome Editing in the 'He Jiankui Affair'." *Journal of Law and the Biosciences* 6 (1): 111–83. <https://doi.org/10.1093/jlb/lasz010>.
- Habermas, Jürgen. 2003. *The Future of Human Nature*. Polity Press.
- Harris, Lisa H., Daniel Grossman, et al. 2024. "Beneath the Sword of Damocles: Moral Obligations of Physicians in a Post-Dobbs Landscape." *Hastings Center Report* 54 (3): 15–25. <https://doi.org/10.1002/hast.1583>.
- Haseltine, William A. 2013. *Affordable Excellence: The Singapore Healthcare Story*. Brookings Institution Press.
- Hotez, Peter J. 2021. *Preventing the Next Pandemic: Vaccine Diplomacy in a Time of Anti-Science*. Johns Hopkins University Press.
- Hviid, Anders, Jorgen Vinslov Hansen, Morten Frisch, and Mads Melbye. 2019. "Measles, Mumps, Rubella Vaccination and Autism: A Nationwide Cohort Study." *Annals of Internal Medicine* 170 (8): 513–20.
- Jones, James H. 1981. *Bad Blood: The Tuskegee Syphilis Experiment*. The Free Press.
- Jones, Sam. 2026. *Spanish Woman Who Won Legal Battle for Right to Euthanasia Has Assisted Death*. The Guardian. <https://www.theguardian.com/society/2026/mar/26/spanish-woman-noelia-castillo-euthanasia-assisted-death>.
- Jonsen, Albert R. 1998. *The Birth of Bioethics*. Oxford University Press.

Works Cited

- Kant, Immanuel. 1785. *Groundwork of the Metaphysics of Morals*.
- Kipnis, Kenneth. 2001. "Vulnerability in Research Subjects: A Bioethical Taxonomy." In *Ethical and Policy Issues in Research Involving Human Participants, Volume II*. National Bioethics Advisory Commission.
- Kliever, Lonnie D., ed. 1989. *Dax's Case: Essays in Medical Ethics and Human Meaning*. Southern Methodist University Press.
- Koh, David. 2020. "Migrant Workers and COVID-19." *Occupational and Environmental Medicine* 77 (9): 634–36. <https://doi.org/10.1136/oemed-2020-106626>.
- Larson, Heidi J. 2020. *Stuck: How Vaccine Rumors Start—and Why They Don't Go Away*. Oxford University Press.
- Lesné, Sylvain, Ming Teng Koh, Linda Kotilinek, et al. 2006. "A Specific Amyloid-Beta Protein Assembly in the Brain Impairs Memory." *Nature* 440 (7082): 352–57. <https://doi.org/10.1038/nature04533>.
- Lewis, Ariane, Gwendolyn Quinn, and Kimberly Mutcherson. 2025. "Ethical Controversies in the Adriana Smith Case in Georgia: Brain Death/Death by Neurologic Criteria in Pregnancy." *The American Journal of Bioethics*, ahead of print. <https://doi.org/10.1080/15265161.2025.2533359>.
- MacKay, Douglas, and Gopal Sreenivasan. 2021. *Justice, Inequality, and Health*. The Stanford Encyclopedia of Philosophy (Winter 2021 Edition), Edward N. Zalta (ed.). <https://plato.stanford.edu/archives/win2021/entries/justice-inequality-health/>.
- Madsen, Kreesten M., Anders Hviid, Mogens Vestergaard, et al. 2002. "A Population-Based Study of Measles, Mumps, and Rubella Vaccination and Autism." *New England Journal of Medicine* 347 (19): 1477–82.
- Mill, John Stuart. 1863. *Utilitarianism*. Parker, Son,; Bourn.
- Mohiuddin, Muhammad M., Avneesh K. Singh, Linda Scobie, Corbin E. Goerlich, et al. 2023. "Graft Dysfunction in Compassionate Use of Genetically Engineered Pig-to-Human Cardiac Xenotransplantation: A Case Report." *The Lancet* 402 (10399): 397–410. [https://doi.org/10.1016/S0140-6736\(23\)00775-4](https://doi.org/10.1016/S0140-6736(23)00775-4).
- Mullainathan, Sendhil. 2019. "Biased Algorithms Are Easier to Fix Than Biased People." *The New York Times*, December. <https://www.nytimes.com/2019/12/06/business/algorithm-bias-fix.html>.
- National Academies of Sciences, Engineering, and Medicine. 2017. *Human Genome Editing: Science, Ethics, and Governance*. The National Academies Press. <https://doi.org/10.17226/246>

23.

- National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. 1979. *The Belmont Report: Ethical Principles and Guidelines for the Protection of Human Subjects of Research*. U.S. Department of Health, Education, and Welfare. <https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/index.html>.
- Nuremberg Military Tribunals. 1947. *The Nuremberg Code*. Trials of War Criminals before the Nuremberg Military Tribunals under Control Council Law No. 10.
- Obermeyer, Ziad, Brian Powers, Christine Vogeli, and Sendhil Mullainathan. 2019. "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations." *Science* 366 (6464): 447–53. <https://doi.org/10.1126/science.aax2342>.
- Office of the Attorney General of Georgia. 2025. *Response to State Senator Nabilah Islam Parkes Regarding the LIFE Act and Brain-Dead Pregnant Patients*. Office of the Georgia Attorney General.
- Offit, Paul A. 2011. *Deadly Choices: How the Anti-Vaccine Movement Threatens Us All*. Basic Books.
- Omiye, Jesutofunmi A., Jenna C. Lester, Simon Spichak, Veronica Rotemberg, and Roxana Daneshjou. 2023. "Large Language Models Propagate Race-Based Medicine." *Npj Digital Medicine* 6: 195. <https://doi.org/10.1038/s41746-023-00939-z>.
- Parsi, Kayhan, and William J. Winslade. 2019. "Why Dax's Case Still Matters." *The American Journal of Bioethics* 19 (9): 8–10. <https://doi.org/10.1080/15265161.2019.1643944>.
- Piller, Charles. 2022. "Blots on a Field?" *Science* 377 (6604): 358–63. <https://doi.org/10.1126/science.add9993>.
- Piller, Charles. 2025. *Doctored: Fraud, Arrogance, and Tragedy in the Quest to Cure Alzheimer's*. Atria/One Signal Publishers.
- Presidential Commission for the Study of Bioethical Issues. 2011. "Ethically Impossible": *STD Research in Guatemala from 1946 to 1948*. Washington, D.C., September 2011. <https://bioethicsarchive.georgetown.edu/pcsbi/node/654.html>.
- Reid, T. R. 2009. *The Healing of America: A Global Quest for Better, Cheaper, and Fairer Health Care*. Penguin Press.
- Reverby, Susan M. 2009. *Examining Tuskegee: The Infamous Syphilis Study and Its Legacy*. University of North Carolina Press.
- Reverby, Susan M. 2011. "'Normal Exposure' and Inoculation Syphilis: A PHS 'Tuskegee' Doctor in Guatemala, 1946–1948." *Journal of Policy History* 23 (1): 6–28. <https://doi.org/10.1017/S0>

Works Cited

898030610000291.

- Roberts, Dorothy E. 1997. *Killing the Black Body: Race, Reproduction, and the Meaning of Liberty*. Pantheon Books.
- Rueda, Jon, Rosana Triviño-Caballero, Iris Parra Jounou, Stuart Youngner, and David Rodríguez-Arias. 2026. "Three Fallacies in Arguments Made Against Euthanasia for People with Psychiatric Disorders." *The Lancet Psychiatry*, ahead of print. [https://doi.org/10.1016/S2215-0366\(26\)00154-9](https://doi.org/10.1016/S2215-0366(26)00154-9).
- Ryan, Isa, Ashish Premkumar, and Katie Watson. 2022. "Why the Post-Roe Era Requires Protecting Conscientious Provision as We Protect Conscientious Refusal in Health Care." *AMA Journal of Ethics* 24 (9): E906–912. <https://doi.org/10.1001/amajethics.2022.906>.
- Singer, Peter. 1975. *Animal Liberation*. HarperCollins.
- SisterSong Women of Color Reproductive Justice Collective. 2025. *Statement on the Case of Adriana Smith*. SisterSong.
- Skadden, Arps, Slate, Meagher and Flom. 2026. *Colorado Repeals and Replaces Its AI Act*. Client alert, June 2026. <https://www.skadden.com/insights/publications/2026/06/colorado-repeals-and-replaces-its-ai-act>.
- Slotnik, Daniel E. 2019. *Dax Cowart, Who Fought for Patients' Rights, Dies at 71*. *The New York Times*, May 15, 2019. <https://www.nytimes.com/2019/05/15/obituaries/dax-cowart-dead.html>.
- Stahl, Ronit Y., and Ezekiel J. Emanuel. 2017. "Physicians, Not Conscripts — Conscientious Objection in Health Care." *New England Journal of Medicine* 376 (14): 1380–85. <https://doi.org/10.1056/NEJMs1612472>.
- State of Georgia. 2019. *Living Infants Fairness and Equality (LIFE) Act, o.c.g.a. § 16-12-141*. Georgia General Assembly (House Bill 481).
- Stegenga, Jacob. 2018. *Medical Nihilism*. Oxford University Press.
- Supreme Court of New Jersey. 1976. *In Re Quinlan*, 70 n.j. 10, 355 a.2d 647. Supreme Court of New Jersey.
- Supreme Court of Texas. 2023. *In re State of Texas*, No. 23-0994 (*per curiam*). Supreme Court of Texas. <https://www.txcourts.gov/supreme/>.
- Supreme Court of the United States. 1967. *Loving v. Virginia*, 388 u.s. 1.
- Supreme Court of the United States. 1990. *Cruzan v. Director, Missouri Department of Health*, 497

- u.s.* 261. Supreme Court of the United States.
- Supreme Court of the United States. 2020. *Bostock v. Clayton County*, 590 *u.s.* 644.
- Supreme Court of the United States. 2024. *Murthy v. Missouri*, 603 *u.s.* ____.
- The Commonwealth Fund. 2026. *International Health Care System Profiles: Singapore*. The Commonwealth Fund. <https://www.commonwealthfund.org/international-health-policy-center/countries/singapore>.
- Tobin, Martin J. 2022. "Fiftieth Anniversary of Uncovering the Tuskegee Syphilis Study: The Story and Timeless Lessons." *American Journal of Respiratory and Critical Care Medicine* 205 (10): 1145–58. <https://doi.org/10.1164/rccm.202201-0136SO>.
- United States Court of Appeals, District of Columbia Circuit. 1972. *Canterbury v. Spence*, 464 *f.2d* 772. United States Court of Appeals, D.C. Circuit.
- United Therapeutics Corporation. 2025. *FDA Clears First Clinical Trial of Genetically Modified Pig Kidney Xenotransplantation*. United Therapeutics press release. <https://ir.unither.com/>.
- Vyas, Darshali A., Leo G. Eisenstein, and David S. Jones. 2020. "Hidden in Plain Sight — Reconsidering the Use of Race Correction in Clinical Algorithms." *New England Journal of Medicine* 383 (9): 874–82. <https://doi.org/10.1056/NEJMms2004740>.
- Washington, Harriet A. 2006. *Medical Apartheid: The Dark History of Medical Experimentation on Black Americans from Colonial Times to the Present*. Doubleday.
- Watson, Katie. 2025. "Brain Death in Pregnancy—Abortion, Advance-Directive, or End-of-Life Law?" In *New England Journal of Medicine*, No. 4, vol. 393. <https://doi.org/10.1056/NEJMP2507544>.
- White, Robert B. 1974. *Please Let Me Die [Film]*. University of Texas Medical Branch, Galveston.
- Wicclair, Mark R. 2000. "Conscientious Objection in Medicine." *Bioethics* 14 (3): 205–27. <https://doi.org/10.1111/1467-8519.00191>.
- World Medical Association. 2024. *WMA Declaration of Helsinki – Ethical Principles for Medical Research Involving Human Participants*. Adopted by the 75th WMA General Assembly, Helsinki, Finland. <https://www.wma.net/policies-post/wma-declaration-of-helsinki/>.

